APPENDIX B

Detailed User Stories and Acceptance Flows

VitaMate — Senior Project 2

Academic Appendix Derived from the Validated User Stories and Acceptance Criteria

Academic Year 2025–2026

B.1 Purpose and Scope

This appendix provides an academic, test-oriented elaboration of the validated VitaMate user-story baseline. The source document contains 194 user stories organized into 16 epics and records the implementation status, priority, and concise acceptance criteria for each story. The present appendix preserves those identifiers and statuses while expressing every story in English and expanding it into a structured acceptance flow suitable for requirements review, acceptance testing, system demonstration, and traceability.

The elaborated flows are derived from the source user stories and acceptance criteria. They are intended to clarify actor interaction, validation, system response, and observable results without changing the documented implementation state or introducing unsupported functionality.

B.2 User-Story and Acceptance Format

Each user story follows the standard agile form "As a [role], I want [capability], so that [value]." For academic reporting, each story is accompanied by implementation metadata, testable acceptance criteria, and a stepwise operational flow. Stories marked as device-dependent, partially implemented, or backlog retain that limitation explicitly.

Element	Definition
User Story	A concise statement of actor, requested capability, and intended value.
Acceptance Criteria	Observable, testable conditions that define acceptable behavior for the story.
Operational Flow	A stepwise interaction between the actor and VitaMate showing initiation, validation, execution, and observable result.
Implementation Status	The documented current state: implemented, device/permission dependent, partially implemented, or backlog.
Traceability	The original story identifier and epic are preserved to support RTM and test-case mapping.

B.3 Epic Index

Epic	Title	Story Count	Primary Scope
EP-01	Account, Authentication, and Onboarding	12	Secure account access and an initial health profile that establishes configurable baseline targets.
EP-02	Navigation, Persistent Layout, and Home Dashboard	11	A concise daily entry point with persistent navigation that does not obscure content and directs the user to the most relevant health action.
EP-03	My VitaMate: Profile, Goals, Security, and Privacy	14	A centralized management area for identity, health profile, goals, notifications, security, and privacy controls.
EP-04	Nutrition and Meal Management	16	Manual meal recording from the food library with stable nutrient snapshots and coordinated effects across related trackers.
EP-05	AI-Assisted Meal Analysis	12	A review-assisted image-analysis workflow that proposes meal components and weights before the user confirms the final nutrition record.
EP-06	Vitamins, Minerals, and Focused Tracking	9	Focused micronutrient monitoring that can combine food snapshots, beverages, taken supplements, laboratory context, and effective health constraints.
EP-07	Hydration and Beverages	10	Hydration logging that distinguishes effective hydration contribution from raw beverage volume and maintains bidirectional coordination with nutrition and habits.
EP-08	Activity, Workouts, and Steps	13	Unified movement tracking that combines manual activity, live sessions, and automatic step data without duplicate counting.

Epic	Title	Story Count	Primary Scope
EP-09	Sleep and Sleep Coach	9	Sleep recording and planning that uses behavior and caffeine context as explanatory factors rather than diagnostic conclusions.
EP-10	Medication and Adherence	15	A unified medication-plan model for manual medications, condition-linked medications, and supplements, with actionable dose states.
EP-11	Unhealthy Habits and Bidirectional Integration	11	Reduction plans for smoking, caffeine, and fast food using explicit check-ins and cross-tracker projections without duplicate logging.
EP-12	Chronic Conditions, Readings, and Health Constraints	14	Management of supported chronic conditions, readings, goals, alerts, and constraints that safely influence related trackers.
EP-13	Unified Health State, Progress, and History	10	A single consistent daily health-state view across relevant trackers and constraints, separate from motivational XP.
EP-14	Points, Missions, Badges, and Motivation	12	A motivation subsystem that rewards verified user actions through an auditable point ledger, missions, streaks, badges, levels, and contextual feedback.
EP-15	Notification Hub and In-App Alerts	14	Centralized notification decision-making in Django with synchronized local Android delivery, health priority, quiet hours, and localization.
EP-16	Language, Accessibility, Reliability, and Cross-Cutting Security	12	Cross-cutting requirements for bilingual operation, accessibility, responsive layouts, error recovery, data isolation, idempotency, and medical safety.

B.4 Detailed User Stories

The following sections document every validated user story. Each story begins on a new page to preserve table integrity and make the appendix suitable for direct academic referencing.

B.4 EP-01 — Account, Authentication, and Onboarding

Secure account access and an initial health profile that establishes configurable baseline targets.

Epic Attribute	Academic Description
Scope	Secure account access and an initial health profile that establishes configurable baseline targets.
Story Count	12
Baseline Rule 1	JWT access and refresh tokens are stored using secure client storage.
Baseline Rule 2	Passwords are never stored as plain text, and duplicate email addresses are rejected case-insensitively.
Baseline Rule 3	Initial calorie, hydration, step, and activity targets are derived from the completed health profile.

US-01-01 — Create Account

Epic	EP-01 — Account, Authentication, and Onboarding	**Primary Actor**	Visitor
Priority	High	**Implementation Status**	Implemented
User Story	As a visitor, I want to create account, so that I can access VitaMate securely and establish a valid initial health profile.		
Traceability	Source Story ID: US-01-01 | Source Epic: EP-01 | Appendix Story 1 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Create Account' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Relevant inputs and preconditions shall be validated before the operation is accepted.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Visitor	Opens the relevant authentication or onboarding function for create account.	The requested screen or onboarding step becomes available.
2	Visitor	Provides or selects the information required to create account.	The application captures only the fields relevant to the story.
3	VitaMate System	Validates the request against authentication, profile, and input rules.	Invalid or incomplete input is rejected with an explicit message.
4	VitaMate System	Completes the authorized operation and persists any required account/profile state.	The stored state and derived onboarding information remain internally consistent.
5	Visitor	Receives the resulting authenticated, onboarding, or error state.	The next valid navigation step is available without duplicate submission.

US-01-02 — Sign In

Epic	EP-01 — Account, Authentication, and Onboarding	**Primary Actor**	Visitor
Priority	High	**Implementation Status**	Implemented
User Story	As a visitor, I want to sign in, so that I can access VitaMate securely and establish a valid initial health profile.		
Traceability	Source Story ID: US-01-02 | Source Epic: EP-01 | Appendix Story 2 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Sign In' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-3	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Visitor	Opens the relevant authentication or onboarding function for sign in.	The requested screen or onboarding step becomes available.
2	Visitor	Provides or selects the information required to sign in.	The application captures only the fields relevant to the story.
3	VitaMate System	Validates the request against authentication, profile, and input rules.	Invalid or incomplete input is rejected with an explicit message.
4	VitaMate System	Completes the authorized operation and persists any required account/profile state.	The stored state and derived onboarding information remain internally consistent.
5	Visitor	Receives the resulting authenticated, onboarding, or error state.	The next valid navigation step is available without duplicate submission.

US-01-03 — Restore Session

Epic	EP-01 — Account, Authentication, and Onboarding	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to restore session, so that I can access VitaMate securely and establish a valid initial health profile.		
Traceability	Source Story ID: US-01-03 | Source Epic: EP-01 | Appendix Story 3 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Restore Session' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens the relevant authentication or onboarding function for restore session.	The requested screen or onboarding step becomes available.
2	Registered User	Provides or selects the information required to restore session.	The application captures only the fields relevant to the story.
3	VitaMate System	Validates the request against authentication, profile, and input rules.	Invalid or incomplete input is rejected with an explicit message.
4	VitaMate System	Completes the authorized operation and persists any required account/profile state.	The stored state and derived onboarding information remain internally consistent.
5	Registered User	Receives the resulting authenticated, onboarding, or error state.	The next valid navigation step is available without duplicate submission.

US-01-04 — Select Language Before Sign-In

Epic	EP-01 — Account, Authentication, and Onboarding	**Primary Actor**	Visitor
Priority	High	**Implementation Status**	Implemented
User Story	As a visitor, I want to select language before sign-in, so that I can access VitaMate securely and establish a valid initial health profile.		
Traceability	Source Story ID: US-01-04 | Source Epic: EP-01 | Appendix Story 4 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Select Language Before Sign-In' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Visitor	Opens the relevant authentication or onboarding function for select language before sign-in.	The requested screen or onboarding step becomes available.
2	Visitor	Provides or selects the information required to select language before sign-in.	The application captures only the fields relevant to the story.
3	VitaMate System	Validates the request against authentication, profile, and input rules.	Invalid or incomplete input is rejected with an explicit message.
4	VitaMate System	Completes the authorized operation and persists any required account/profile state.	The stored state and derived onboarding information remain internally consistent.
5	Visitor	Receives the resulting authenticated, onboarding, or error state.	The next valid navigation step is available without duplicate submission.

US-01-05 — Select Language During Registration

Epic	EP-01 — Account, Authentication, and Onboarding	**Primary Actor**	Visitor
Priority	High	**Implementation Status**	Implemented
User Story	As a visitor, I want to select language during registration, so that I can access VitaMate securely and establish a valid initial health profile.		
Traceability	Source Story ID: US-01-05 | Source Epic: EP-01 | Appendix Story 5 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Select Language During Registration' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The visible result shall remain consistent with the selected locale, supported directionality, and accessibility semantics.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Visitor	Opens the relevant authentication or onboarding function for select language during registration.	The requested screen or onboarding step becomes available.
2	Visitor	Provides or selects the information required to select language during registration.	The application captures only the fields relevant to the story.
3	VitaMate System	Validates the request against authentication, profile, and input rules.	Invalid or incomplete input is rejected with an explicit message.
4	VitaMate System	Completes the authorized operation and persists any required account/profile state.	The stored state and derived onboarding information remain internally consistent.
5	Visitor	Receives the resulting authenticated, onboarding, or error state.	The next valid navigation step is available without duplicate submission.

US-01-06 — Enter Basic Health Data

Epic	EP-01 — Account, Authentication, and Onboarding	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to enter basic health data, so that I can access VitaMate securely and establish a valid initial health profile.		
Traceability	Source Story ID: US-01-06 | Source Epic: EP-01 | Appendix Story 6 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Enter Basic Health Data' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Relevant inputs and preconditions shall be validated before the operation is accepted.
AC-3	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-4	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens the relevant authentication or onboarding function for enter basic health data.	The requested screen or onboarding step becomes available.
2	Registered User	Provides or selects the information required to enter basic health data.	The application captures only the fields relevant to the story.
3	VitaMate System	Validates the request against authentication, profile, and input rules.	Invalid or incomplete input is rejected with an explicit message.
4	VitaMate System	Completes the authorized operation and persists any required account/profile state.	The stored state and derived onboarding information remain internally consistent.
5	Registered User	Receives the resulting authenticated, onboarding, or error state.	The next valid navigation step is available without duplicate submission.

US-01-07 — Set Activity Level

Epic	EP-01 — Account, Authentication, and Onboarding	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to set activity level, so that I can access VitaMate securely and establish a valid initial health profile.		
Traceability	Source Story ID: US-01-07 | Source Epic: EP-01 | Appendix Story 7 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Set Activity Level' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens the relevant authentication or onboarding function for set activity level.	The requested screen or onboarding step becomes available.
2	Registered User	Provides or selects the information required to set activity level.	The application captures only the fields relevant to the story.
3	VitaMate System	Validates the request against authentication, profile, and input rules.	Invalid or incomplete input is rejected with an explicit message.
4	VitaMate System	Completes the authorized operation and persists any required account/profile state.	The stored state and derived onboarding information remain internally consistent.
5	Registered User	Receives the resulting authenticated, onboarding, or error state.	The next valid navigation step is available without duplicate submission.

US-01-08 — Set Weight Goal

Epic	EP-01 — Account, Authentication, and Onboarding	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to set weight goal, so that I can access VitaMate securely and establish a valid initial health profile.		
Traceability	Source Story ID: US-01-08 | Source Epic: EP-01 | Appendix Story 8 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Set Weight Goal' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-4	Any motivation effect shall follow the defined point, mission, streak, badge, or level rules and shall not be duplicated.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens the relevant authentication or onboarding function for set weight goal.	The requested screen or onboarding step becomes available.
2	Registered User	Provides or selects the information required to set weight goal.	The application captures only the fields relevant to the story.
3	VitaMate System	Validates the request against authentication, profile, and input rules.	Invalid or incomplete input is rejected with an explicit message.
4	VitaMate System	Completes the authorized operation and persists any required account/profile state.	The stored state and derived onboarding information remain internally consistent.
5	Registered User	Receives the resulting authenticated, onboarding, or error state.	The next valid navigation step is available without duplicate submission.

US-01-09 — Review Onboarding

Epic	EP-01 — Account, Authentication, and Onboarding	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to review onboarding, so that I can access VitaMate securely and establish a valid initial health profile.		
Traceability	Source Story ID: US-01-09 | Source Epic: EP-01 | Appendix Story 9 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Review Onboarding' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Relevant inputs and preconditions shall be validated before the operation is accepted.
AC-3	The prohibited record or side effect identified by the source story shall not be created.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens the relevant authentication or onboarding function for review onboarding.	The requested screen or onboarding step becomes available.
2	Registered User	Provides or selects the information required to review onboarding.	The application captures only the fields relevant to the story.
3	VitaMate System	Validates the request against authentication, profile, and input rules.	Invalid or incomplete input is rejected with an explicit message.
4	VitaMate System	Completes the authorized operation and persists any required account/profile state.	The stored state and derived onboarding information remain internally consistent.
5	Registered User	Receives the resulting authenticated, onboarding, or error state.	The next valid navigation step is available without duplicate submission.

US-01-10 — Access Home Dashboard

Epic	EP-01 — Account, Authentication, and Onboarding	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to access home dashboard, so that I can access VitaMate securely and establish a valid initial health profile.		
Traceability	Source Story ID: US-01-10 | Source Epic: EP-01 | Appendix Story 10 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Access Home Dashboard' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	Duplicate records, rewards, plans, and side effects shall be prevented using idempotent or uniqueness-aware behavior.
AC-4	Navigation shall resolve to the intended route or detail view while preserving valid session context.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens the relevant authentication or onboarding function for access home dashboard.	The requested screen or onboarding step becomes available.
2	Registered User	Provides or selects the information required to access home dashboard.	The application captures only the fields relevant to the story.
3	VitaMate System	Validates the request against authentication, profile, and input rules.	Invalid or incomplete input is rejected with an explicit message.
4	VitaMate System	Completes the authorized operation and persists any required account/profile state.	The stored state and derived onboarding information remain internally consistent.
5	Registered User	Receives the resulting authenticated, onboarding, or error state.	The next valid navigation step is available without duplicate submission.

US-01-11 — Sign Out

Epic	EP-01 — Account, Authentication, and Onboarding	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to sign out, so that I can access VitaMate securely and establish a valid initial health profile.		
Traceability	Source Story ID: US-01-11 | Source Epic: EP-01 | Appendix Story 11 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Sign Out' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The state-changing action shall preserve required historical information and linked consistency according to the source story.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens the relevant authentication or onboarding function for sign out.	The requested screen or onboarding step becomes available.
2	Registered User	Provides or selects the information required to sign out.	The application captures only the fields relevant to the story.
3	VitaMate System	Validates the request against authentication, profile, and input rules.	Invalid or incomplete input is rejected with an explicit message.
4	VitaMate System	Completes the authorized operation and persists any required account/profile state.	The stored state and derived onboarding information remain internally consistent.
5	Registered User	Receives the resulting authenticated, onboarding, or error state.	The next valid navigation step is available without duplicate submission.

US-01-12 — Forgot Password

Epic	EP-01 — Account, Authentication, and Onboarding	**Primary Actor**	Visitor
Priority	Medium	**Implementation Status**	Backlog / Not Implemented
User Story	As a visitor, I want to use forgot password, so that I can access VitaMate securely and establish a valid initial health profile.		
Traceability	Source Story ID: US-01-12 | Source Epic: EP-01 | Appendix Story 12 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Forgot Password' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Navigation shall resolve to the intended route or detail view while preserving valid session context.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Visitor	Opens the relevant authentication or onboarding function for forgot password.	The requested screen or onboarding step becomes available.
2	Visitor	Provides or selects the information required to use forgot password.	The application captures only the fields relevant to the story.
3	VitaMate System	Validates the request against authentication, profile, and input rules.	Invalid or incomplete input is rejected with an explicit message.
4	VitaMate System	Completes the authorized operation and persists any required account/profile state.	The stored state and derived onboarding information remain internally consistent.
5	VitaMate System	The intended acceptance flow is documented but not represented as completed functionality.	The report and appendix must continue to identify this story as backlog until implementation and verification are completed.

Implementation Note: This flow documents the intended acceptance behavior only. It must not be represented in the main report as completed functionality.

B.5 EP-02 — Navigation, Persistent Layout, and Home Dashboard

A concise daily entry point with persistent navigation that does not obscure content and directs the user to the most relevant health action.

Epic Attribute	Academic Description
Scope	A concise daily entry point with persistent navigation that does not obscure content and directs the user to the most relevant health action.
Story Count	11
Baseline Rule 1	Top and bottom navigation areas reserve their own layout space instead of covering page content.
Baseline Rule 2	The motivation status area is persistent in authenticated application views but is excluded from authentication and onboarding screens.
Baseline Rule 3	The Home dashboard treats activity as a complete tracker rather than using steps as a substitute for the whole activity domain.

US-02-01 — Unified Top Bar

Epic	EP-02 — Navigation, Persistent Layout, and Home Dashboard	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to use unified top bar, so that I can navigate the application efficiently and understand my current daily health state.		
Traceability	Source Story ID: US-02-01 | Source Epic: EP-02 | Appendix Story 13 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Unified Top Bar' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The interface shall preserve readable layout and shall not obscure or overflow required content.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens the Home or navigation context related to unified top bar.	The persistent application shell loads without obscuring content.
2	VitaMate System	Retrieves the current authorized overview and applicable tracker state.	Unavailable data are not replaced with fabricated values.
3	VitaMate System	Applies the presentation rule for unified top bar.	The displayed information is consistent with the current route and health state.
4	Registered User	Reviews the information or uses the presented navigation/action control.	The intended destination or action is reachable directly.
5	VitaMate System	Handles refresh or failure conditions when required.	The user receives an explicit retry or stable fallback state.

US-02-02 — Unified Bottom Navigation

Epic	EP-02 — Navigation, Persistent Layout, and Home Dashboard	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to use unified bottom navigation, so that I can navigate the application efficiently and understand my current daily health state.		
Traceability	Source Story ID: US-02-02 | Source Epic: EP-02 | Appendix Story 14 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Unified Bottom Navigation' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-3	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens the Home or navigation context related to unified bottom navigation.	The persistent application shell loads without obscuring content.
2	VitaMate System	Retrieves the current authorized overview and applicable tracker state.	Unavailable data are not replaced with fabricated values.
3	VitaMate System	Applies the presentation rule for unified bottom navigation.	The displayed information is consistent with the current route and health state.
4	Registered User	Reviews the information or uses the presented navigation/action control.	The intended destination or action is reachable directly.
5	VitaMate System	Handles refresh or failure conditions when required.	The user receives an explicit retry or stable fallback state.

US-02-03 — Highlight Current Destination

Epic	EP-02 — Navigation, Persistent Layout, and Home Dashboard	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to highlight current destination, so that I can navigate the application efficiently and understand my current daily health state.		
Traceability	Source Story ID: US-02-03 | Source Epic: EP-02 | Appendix Story 15 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Highlight Current Destination' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Navigation shall resolve to the intended route or detail view while preserving valid session context.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens the Home or navigation context related to highlight current destination.	The persistent application shell loads without obscuring content.
2	VitaMate System	Retrieves the current authorized overview and applicable tracker state.	Unavailable data are not replaced with fabricated values.
3	VitaMate System	Applies the presentation rule for highlight current destination.	The displayed information is consistent with the current route and health state.
4	Registered User	Reviews the information or uses the presented navigation/action control.	The intended destination or action is reachable directly.
5	VitaMate System	Handles refresh or failure conditions when required.	The user receives an explicit retry or stable fallback state.

US-02-04 — Display User Identity

Epic	EP-02 — Navigation, Persistent Layout, and Home Dashboard	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to display user identity, so that I can navigate the application efficiently and understand my current daily health state.		
Traceability	Source Story ID: US-02-04 | Source Epic: EP-02 | Appendix Story 16 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Display User Identity' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-3	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens the Home or navigation context related to display user identity.	The persistent application shell loads without obscuring content.
2	VitaMate System	Retrieves the current authorized overview and applicable tracker state.	Unavailable data are not replaced with fabricated values.
3	VitaMate System	Applies the presentation rule for display user identity.	The displayed information is consistent with the current route and health state.
4	Registered User	Reviews the information or uses the presented navigation/action control.	The intended destination or action is reachable directly.
5	VitaMate System	Handles refresh or failure conditions when required.	The user receives an explicit retry or stable fallback state.

US-02-05 — Daily Summary

Epic	EP-02 — Navigation, Persistent Layout, and Home Dashboard	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view daily summary, so that I can navigate the application efficiently and understand my current daily health state.		
Traceability	Source Story ID: US-02-05 | Source Epic: EP-02 | Appendix Story 17 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Daily Summary' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any motivation effect shall follow the defined point, mission, streak, badge, or level rules and shall not be duplicated.
AC-3	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens the Home or navigation context related to daily summary.	The persistent application shell loads without obscuring content.
2	VitaMate System	Retrieves the current authorized overview and applicable tracker state.	Unavailable data are not replaced with fabricated values.
3	VitaMate System	Applies the presentation rule for daily summary.	The displayed information is consistent with the current route and health state.
4	Registered User	Reviews the information or uses the presented navigation/action control.	The intended destination or action is reachable directly.
5	VitaMate System	Handles refresh or failure conditions when required.	The user receives an explicit retry or stable fallback state.

US-02-06 — Today's Focus

Epic	EP-02 — Navigation, Persistent Layout, and Home Dashboard	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to use today's focus, so that I can navigate the application efficiently and understand my current daily health state.		
Traceability	Source Story ID: US-02-06 | Source Epic: EP-02 | Appendix Story 18 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Today's Focus' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Navigation shall resolve to the intended route or detail view while preserving valid session context.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens the Home or navigation context related to today's focus.	The persistent application shell loads without obscuring content.
2	VitaMate System	Retrieves the current authorized overview and applicable tracker state.	Unavailable data are not replaced with fabricated values.
3	VitaMate System	Applies the presentation rule for today's focus.	The displayed information is consistent with the current route and health state.
4	Registered User	Reviews the information or uses the presented navigation/action control.	The intended destination or action is reachable directly.
5	VitaMate System	Handles refresh or failure conditions when required.	The user receives an explicit retry or stable fallback state.

US-02-07 — Core Tracker Cards

Epic	EP-02 — Navigation, Persistent Layout, and Home Dashboard	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view core tracker cards, so that I can navigate the application efficiently and understand my current daily health state.		
Traceability	Source Story ID: US-02-07 | Source Epic: EP-02 | Appendix Story 19 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Core Tracker Cards' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens the Home or navigation context related to core tracker cards.	The persistent application shell loads without obscuring content.
2	VitaMate System	Retrieves the current authorized overview and applicable tracker state.	Unavailable data are not replaced with fabricated values.
3	VitaMate System	Applies the presentation rule for core tracker cards.	The displayed information is consistent with the current route and health state.
4	Registered User	Reviews the information or uses the presented navigation/action control.	The intended destination or action is reachable directly.
5	VitaMate System	Handles refresh or failure conditions when required.	The user receives an explicit retry or stable fallback state.

US-02-08 — Today's Plan

Epic	EP-02 — Navigation, Persistent Layout, and Home Dashboard	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view today's plan, so that I can navigate the application efficiently and understand my current daily health state.		
Traceability	Source Story ID: US-02-08 | Source Epic: EP-02 | Appendix Story 20 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Today's Plan' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Navigation shall resolve to the intended route or detail view while preserving valid session context.
AC-3	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens the Home or navigation context related to today's plan.	The persistent application shell loads without obscuring content.
2	VitaMate System	Retrieves the current authorized overview and applicable tracker state.	Unavailable data are not replaced with fabricated values.
3	VitaMate System	Applies the presentation rule for today's plan.	The displayed information is consistent with the current route and health state.
4	Registered User	Reviews the information or uses the presented navigation/action control.	The intended destination or action is reachable directly.
5	VitaMate System	Handles refresh or failure conditions when required.	The user receives an explicit retry or stable fallback state.

US-02-09 — Health Conditions Summary

Epic	EP-02 — Navigation, Persistent Layout, and Home Dashboard	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view health conditions summary, so that I can navigate the application efficiently and understand my current daily health state.		
Traceability	Source Story ID: US-02-09 | Source Epic: EP-02 | Appendix Story 21 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Health Conditions Summary' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Duplicate records, rewards, plans, and side effects shall be prevented using idempotent or uniqueness-aware behavior.
AC-3	Navigation shall resolve to the intended route or detail view while preserving valid session context.
AC-4	The requested information shall be presented from current authorized data without fabricating unavailable values.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens the Home or navigation context related to health conditions summary.	The persistent application shell loads without obscuring content.
2	VitaMate System	Retrieves the current authorized overview and applicable tracker state.	Unavailable data are not replaced with fabricated values.
3	VitaMate System	Applies the presentation rule for health conditions summary.	The displayed information is consistent with the current route and health state.
4	Registered User	Reviews the information or uses the presented navigation/action control.	The intended destination or action is reachable directly.
5	VitaMate System	Handles refresh or failure conditions when required.	The user receives an explicit retry or stable fallback state.

US-02-10 — Notification Button

Epic	EP-02 — Navigation, Persistent Layout, and Home Dashboard	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to use notification button, so that I can navigate the application efficiently and understand my current daily health state.		
Traceability	Source Story ID: US-02-10 | Source Epic: EP-02 | Appendix Story 22 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Notification Button' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Navigation shall resolve to the intended route or detail view while preserving valid session context.
AC-3	Notification-related changes shall remain synchronized with current preferences, schedule, priority, and device state.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens the Home or navigation context related to notification button.	The persistent application shell loads without obscuring content.
2	VitaMate System	Retrieves the current authorized overview and applicable tracker state.	Unavailable data are not replaced with fabricated values.
3	VitaMate System	Applies the presentation rule for notification button.	The displayed information is consistent with the current route and health state.
4	Registered User	Reviews the information or uses the presented navigation/action control.	The intended destination or action is reachable directly.
5	VitaMate System	Handles refresh or failure conditions when required.	The user receives an explicit retry or stable fallback state.

US-02-11 — Refresh and Error Handling

Epic	EP-02 — Navigation, Persistent Layout, and Home Dashboard	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to use refresh and error handling, so that I can navigate the application efficiently and understand my current daily health state.		
Traceability	Source Story ID: US-02-11 | Source Epic: EP-02 | Appendix Story 23 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Refresh and Error Handling' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-3	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens the Home or navigation context related to refresh and error handling.	The persistent application shell loads without obscuring content.
2	VitaMate System	Retrieves the current authorized overview and applicable tracker state.	Unavailable data are not replaced with fabricated values.
3	VitaMate System	Applies the presentation rule for refresh and error handling.	The displayed information is consistent with the current route and health state.
4	Registered User	Reviews the information or uses the presented navigation/action control.	The intended destination or action is reachable directly.
5	VitaMate System	Handles refresh or failure conditions when required.	The user receives an explicit retry or stable fallback state.

B.6 EP-03 — My VitaMate: Profile, Goals, Security, and Privacy

A centralized management area for identity, health profile, goals, notifications, security, and privacy controls.

Epic Attribute	Academic Description
Scope	A centralized management area for identity, health profile, goals, notifications, security, and privacy controls.
Story Count	14
Baseline Rule 1	Visible management actions must be functional or explicitly identified as limited.
Baseline Rule 2	An email change is represented as a pending email until verification is complete.
Baseline Rule 3	A successful password change terminates the current session and requires authentication with the new password.

US-03-01 — Account Summary

Epic	EP-03 — My VitaMate: Profile, Goals, Security, and Privacy	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view account summary, so that I can manage my profile, goals, security settings, and personal data from one place.		
Traceability	Source Story ID: US-03-01 | Source Epic: EP-03 | Appendix Story 24 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Account Summary' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Failure states shall expose a recoverable retry or error path without reporting fabricated success.
AC-3	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens My VitaMate and selects account summary.	The current authorized profile or setting is loaded.
2	Registered User	Reviews or changes the information required to view account summary.	The application captures the requested management action.
3	VitaMate System	Validates ownership, security, and field-level constraints.	Invalid or unauthorized changes are rejected.
4	VitaMate System	Persists the valid change and triggers any required goal, constraint, locale, session, or notification update.	Dependent state is synchronized after the successful write.
5	Registered User	Receives confirmation or the documented limitation state.	The updated value is reflected consistently across subsequent views.

US-03-02 — Edit Name

Epic	EP-03 — My VitaMate: Profile, Goals, Security, and Privacy	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to edit name, so that I can manage my profile, goals, security settings, and personal data from one place.		
Traceability	Source Story ID: US-03-02 | Source Epic: EP-03 | Appendix Story 25 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Edit Name' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens My VitaMate and selects edit name.	The current authorized profile or setting is loaded.
2	Registered User	Reviews or changes the information required to edit name.	The application captures the requested management action.
3	VitaMate System	Validates ownership, security, and field-level constraints.	Invalid or unauthorized changes are rejected.
4	VitaMate System	Persists the valid change and triggers any required goal, constraint, locale, session, or notification update.	Dependent state is synchronized after the successful write.
5	Registered User	Receives confirmation or the documented limitation state.	The updated value is reflected consistently across subsequent views.

US-03-03 — Change Region and Language

Epic	EP-03 — My VitaMate: Profile, Goals, Security, and Privacy	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to change region and language, so that I can manage my profile, goals, security settings, and personal data from one place.		
Traceability	Source Story ID: US-03-03 | Source Epic: EP-03 | Appendix Story 26 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Change Region and Language' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	The visible result shall remain consistent with the selected locale, supported directionality, and accessibility semantics.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens My VitaMate and selects change region and language.	The current authorized profile or setting is loaded.
2	Registered User	Reviews or changes the information required to change region and language.	The application captures the requested management action.
3	VitaMate System	Validates ownership, security, and field-level constraints.	Invalid or unauthorized changes are rejected.
4	VitaMate System	Persists the valid change and triggers any required goal, constraint, locale, session, or notification update.	Dependent state is synchronized after the successful write.
5	Registered User	Receives confirmation or the documented limitation state.	The updated value is reflected consistently across subsequent views.

US-03-04 — Request Email Change

Epic	EP-03 — My VitaMate: Profile, Goals, Security, and Privacy	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Partially Implemented / Operationally Limited
User Story	As a registered user, I want to request email change, so that I can manage my profile, goals, security settings, and personal data from one place.		
Traceability	Source Story ID: US-03-04 | Source Epic: EP-03 | Appendix Story 27 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Request Email Change' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens My VitaMate and selects request email change.	The current authorized profile or setting is loaded.
2	Registered User	Reviews or changes the information required to request email change.	The application captures the requested management action.
3	VitaMate System	Validates ownership, security, and field-level constraints.	Invalid or unauthorized changes are rejected.
4	VitaMate System	Persists the valid change and triggers any required goal, constraint, locale, session, or notification update.	Dependent state is synchronized after the successful write.
5	Registered User	Receives confirmation or the documented limitation state.	The updated value is reflected consistently across subsequent views.

Implementation Note: The core flow exists, but complete acceptance depends on the device, permission, external service, or operational limitation identified by the source story.

US-03-05 — Upload Profile Photo

Epic	EP-03 — My VitaMate: Profile, Goals, Security, and Privacy	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to upload profile photo, so that I can manage my profile, goals, security settings, and personal data from one place.		
Traceability	Source Story ID: US-03-05 | Source Epic: EP-03 | Appendix Story 28 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Upload Profile Photo' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Relevant inputs and preconditions shall be validated before the operation is accepted.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens My VitaMate and selects upload profile photo.	The current authorized profile or setting is loaded.
2	Registered User	Reviews or changes the information required to upload profile photo.	The application captures the requested management action.
3	VitaMate System	Validates ownership, security, and field-level constraints.	Invalid or unauthorized changes are rejected.
4	VitaMate System	Persists the valid change and triggers any required goal, constraint, locale, session, or notification update.	Dependent state is synchronized after the successful write.
5	Registered User	Receives confirmation or the documented limitation state.	The updated value is reflected consistently across subsequent views.

US-03-06 — Delete Profile Photo

Epic	EP-03 — My VitaMate: Profile, Goals, Security, and Privacy	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to delete profile photo, so that I can manage my profile, goals, security settings, and personal data from one place.		
Traceability	Source Story ID: US-03-06 | Source Epic: EP-03 | Appendix Story 29 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Delete Profile Photo' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The state-changing action shall preserve required historical information and linked consistency according to the source story.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens My VitaMate and selects delete profile photo.	The current authorized profile or setting is loaded.
2	Registered User	Reviews or changes the information required to delete profile photo.	The application captures the requested management action.
3	VitaMate System	Validates ownership, security, and field-level constraints.	Invalid or unauthorized changes are rejected.
4	VitaMate System	Persists the valid change and triggers any required goal, constraint, locale, session, or notification update.	Dependent state is synchronized after the successful write.
5	Registered User	Receives confirmation or the documented limitation state.	The updated value is reflected consistently across subsequent views.

US-03-07 — Edit Health Profile

Epic	EP-03 — My VitaMate: Profile, Goals, Security, and Privacy	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to edit health profile, so that I can manage my profile, goals, security settings, and personal data from one place.		
Traceability	Source Story ID: US-03-07 | Source Epic: EP-03 | Appendix Story 30 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Edit Health Profile' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Relevant inputs and preconditions shall be validated before the operation is accepted.
AC-3	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens My VitaMate and selects edit health profile.	The current authorized profile or setting is loaded.
2	Registered User	Reviews or changes the information required to edit health profile.	The application captures the requested management action.
3	VitaMate System	Validates ownership, security, and field-level constraints.	Invalid or unauthorized changes are rejected.
4	VitaMate System	Persists the valid change and triggers any required goal, constraint, locale, session, or notification update.	Dependent state is synchronized after the successful write.
5	Registered User	Receives confirmation or the documented limitation state.	The updated value is reflected consistently across subsequent views.

US-03-08 — Customize Goal

Epic	EP-03 — My VitaMate: Profile, Goals, Security, and Privacy	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to customize goal, so that I can manage my profile, goals, security settings, and personal data from one place.		
Traceability	Source Story ID: US-03-08 | Source Epic: EP-03 | Appendix Story 31 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Customize Goal' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	Compared and displayed values shall use the applicable units consistently.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens My VitaMate and selects customize goal.	The current authorized profile or setting is loaded.
2	Registered User	Reviews or changes the information required to customize goal.	The application captures the requested management action.
3	VitaMate System	Validates ownership, security, and field-level constraints.	Invalid or unauthorized changes are rejected.
4	VitaMate System	Persists the valid change and triggers any required goal, constraint, locale, session, or notification update.	Dependent state is synchronized after the successful write.
5	Registered User	Receives confirmation or the documented limitation state.	The updated value is reflected consistently across subsequent views.

US-03-09 — Restore Default Goal

Epic	EP-03 — My VitaMate: Profile, Goals, Security, and Privacy	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to restore default goal, so that I can manage my profile, goals, security settings, and personal data from one place.		
Traceability	Source Story ID: US-03-09 | Source Epic: EP-03 | Appendix Story 32 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Restore Default Goal' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens My VitaMate and selects restore default goal.	The current authorized profile or setting is loaded.
2	Registered User	Reviews or changes the information required to restore default goal.	The application captures the requested management action.
3	VitaMate System	Validates ownership, security, and field-level constraints.	Invalid or unauthorized changes are rejected.
4	VitaMate System	Persists the valid change and triggers any required goal, constraint, locale, session, or notification update.	Dependent state is synchronized after the successful write.
5	Registered User	Receives confirmation or the documented limitation state.	The updated value is reflected consistently across subsequent views.

US-03-10 — Notification Settings

Epic	EP-03 — My VitaMate: Profile, Goals, Security, and Privacy	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to use notification settings, so that I can manage my profile, goals, security settings, and personal data from one place.		
Traceability	Source Story ID: US-03-10 | Source Epic: EP-03 | Appendix Story 33 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Notification Settings' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-4	Notification-related changes shall remain synchronized with current preferences, schedule, priority, and device state.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens My VitaMate and selects notification settings.	The current authorized profile or setting is loaded.
2	Registered User	Reviews or changes the information required to use notification settings.	The application captures the requested management action.
3	VitaMate System	Validates ownership, security, and field-level constraints.	Invalid or unauthorized changes are rejected.
4	VitaMate System	Persists the valid change and triggers any required goal, constraint, locale, session, or notification update.	Dependent state is synchronized after the successful write.
5	Registered User	Receives confirmation or the documented limitation state.	The updated value is reflected consistently across subsequent views.

US-03-11 — Change Password

Epic	EP-03 — My VitaMate: Profile, Goals, Security, and Privacy	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to change password, so that I can manage my profile, goals, security settings, and personal data from one place.		
Traceability	Source Story ID: US-03-11 | Source Epic: EP-03 | Appendix Story 34 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Change Password' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Relevant inputs and preconditions shall be validated before the operation is accepted.
AC-3	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens My VitaMate and selects change password.	The current authorized profile or setting is loaded.
2	Registered User	Reviews or changes the information required to change password.	The application captures the requested management action.
3	VitaMate System	Validates ownership, security, and field-level constraints.	Invalid or unauthorized changes are rejected.
4	VitaMate System	Persists the valid change and triggers any required goal, constraint, locale, session, or notification update.	Dependent state is synchronized after the successful write.
5	Registered User	Receives confirmation or the documented limitation state.	The updated value is reflected consistently across subsequent views.

US-03-12 — Medical Data Center

Epic	EP-03 — My VitaMate: Profile, Goals, Security, and Privacy	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to use medical data center, so that I can manage my profile, goals, security settings, and personal data from one place.		
Traceability	Source Story ID: US-03-12 | Source Epic: EP-03 | Appendix Story 35 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Medical Data Center' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Duplicate records, rewards, plans, and side effects shall be prevented using idempotent or uniqueness-aware behavior.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens My VitaMate and selects medical data center.	The current authorized profile or setting is loaded.
2	Registered User	Reviews or changes the information required to use medical data center.	The application captures the requested management action.
3	VitaMate System	Validates ownership, security, and field-level constraints.	Invalid or unauthorized changes are rejected.
4	VitaMate System	Persists the valid change and triggers any required goal, constraint, locale, session, or notification update.	Dependent state is synchronized after the successful write.
5	Registered User	Receives confirmation or the documented limitation state.	The updated value is reflected consistently across subsequent views.

US-03-13 — Export Data

Epic	EP-03 — My VitaMate: Profile, Goals, Security, and Privacy	**Primary Actor**	Registered User
Priority	Medium	**Implementation Status**	Partially Implemented / Operationally Limited
User Story	As a registered user, I want to use export data, so that I can manage my profile, goals, security settings, and personal data from one place.		
Traceability	Source Story ID: US-03-13 | Source Epic: EP-03 | Appendix Story 36 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Export Data' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Owner-scoped authorization shall prevent cross-user access to protected records.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens My VitaMate and selects export data.	The current authorized profile or setting is loaded.
2	Registered User	Reviews or changes the information required to use export data.	The application captures the requested management action.
3	VitaMate System	Validates ownership, security, and field-level constraints.	Invalid or unauthorized changes are rejected.
4	VitaMate System	Persists the valid change and triggers any required goal, constraint, locale, session, or notification update.	Dependent state is synchronized after the successful write.
5	Registered User	Receives confirmation or the documented limitation state.	The updated value is reflected consistently across subsequent views.

Implementation Note: The core flow exists, but complete acceptance depends on the device, permission, external service, or operational limitation identified by the source story.

US-03-14 — Request Account Deletion

Epic	EP-03 — My VitaMate: Profile, Goals, Security, and Privacy	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to request account deletion, so that I can manage my profile, goals, security settings, and personal data from one place.		
Traceability	Source Story ID: US-03-14 | Source Epic: EP-03 | Appendix Story 37 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Request Account Deletion' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens My VitaMate and selects request account deletion.	The current authorized profile or setting is loaded.
2	Registered User	Reviews or changes the information required to request account deletion.	The application captures the requested management action.
3	VitaMate System	Validates ownership, security, and field-level constraints.	Invalid or unauthorized changes are rejected.
4	VitaMate System	Persists the valid change and triggers any required goal, constraint, locale, session, or notification update.	Dependent state is synchronized after the successful write.
5	Registered User	Receives confirmation or the documented limitation state.	The updated value is reflected consistently across subsequent views.

B.7 EP-04 — Nutrition and Meal Management

Manual meal recording from the food library with stable nutrient snapshots and coordinated effects across related trackers.

Epic Attribute	Academic Description
Scope	Manual meal recording from the food library with stable nutrient snapshots and coordinated effects across related trackers.
Story Count	16
Baseline Rule 1	Meal nutrient values are stored as snapshots so that historical records do not change when the food catalog is updated.
Baseline Rule 2	Beverages may synchronize with hydration but are not counted as genuine meals for three-meal mission logic.
Baseline Rule 3	Meal records are owner-scoped and support the user's local health date.

US-04-01 — Calorie Summary

Epic	EP-04 — Nutrition and Meal Management	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view calorie summary, so that I can record and review nutrition information accurately.		
Traceability	Source Story ID: US-04-01 | Source Epic: EP-04 | Appendix Story 38 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Calorie Summary' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Nutrition and initiates calorie summary.	The relevant meal, catalog, or nutrition context is loaded.
2	Registered User	Selects, enters, or reviews the data required to view calorie summary.	The application captures the intended food, meal, quantity, time, or filter information.
3	VitaMate System	Validates ownership, quantity, meal classification, and applicable nutrition rules.	Invalid or unauthorized input is rejected.
4	VitaMate System	Applies the nutrition-domain operation and preserves a stable nutrient snapshot when a meal write occurs.	Nutrition totals and linked beverage/hydration effects remain consistent.
5	Registered User	Reviews the updated meal or nutrition result.	The latest nutrition state and eligible motivation effects are visible without duplicate side effects.

US-04-02 — Macronutrient Summary

Epic	EP-04 — Nutrition and Meal Management	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view macronutrient summary, so that I can record and review nutrition information accurately.		
Traceability	Source Story ID: US-04-02 | Source Epic: EP-04 | Appendix Story 39 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Macronutrient Summary' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-3	Navigation shall resolve to the intended route or detail view while preserving valid session context.
AC-4	The requested information shall be presented from current authorized data without fabricating unavailable values.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Nutrition and initiates macronutrient summary.	The relevant meal, catalog, or nutrition context is loaded.
2	Registered User	Selects, enters, or reviews the data required to view macronutrient summary.	The application captures the intended food, meal, quantity, time, or filter information.
3	VitaMate System	Validates ownership, quantity, meal classification, and applicable nutrition rules.	Invalid or unauthorized input is rejected.
4	VitaMate System	Applies the nutrition-domain operation and preserves a stable nutrient snapshot when a meal write occurs.	Nutrition totals and linked beverage/hydration effects remain consistent.
5	Registered User	Reviews the updated meal or nutrition result.	The latest nutrition state and eligible motivation effects are visible without duplicate side effects.

US-04-03 — Nutrition Details

Epic	EP-04 — Nutrition and Meal Management	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view nutrition details, so that I can record and review nutrition information accurately.		
Traceability	Source Story ID: US-04-03 | Source Epic: EP-04 | Appendix Story 40 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Nutrition Details' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Nutrition and initiates nutrition details.	The relevant meal, catalog, or nutrition context is loaded.
2	Registered User	Selects, enters, or reviews the data required to view nutrition details.	The application captures the intended food, meal, quantity, time, or filter information.
3	VitaMate System	Validates ownership, quantity, meal classification, and applicable nutrition rules.	Invalid or unauthorized input is rejected.
4	VitaMate System	Applies the nutrition-domain operation and preserves a stable nutrient snapshot when a meal write occurs.	Nutrition totals and linked beverage/hydration effects remain consistent.
5	Registered User	Reviews the updated meal or nutrition result.	The latest nutrition state and eligible motivation effects are visible without duplicate side effects.

US-04-04 — Today's Meals

Epic	EP-04 — Nutrition and Meal Management	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to use today's meals, so that I can record and review nutrition information accurately.		
Traceability	Source Story ID: US-04-04 | Source Epic: EP-04 | Appendix Story 41 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Today's Meals' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-3	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Nutrition and initiates today's meals.	The relevant meal, catalog, or nutrition context is loaded.
2	Registered User	Selects, enters, or reviews the data required to use today's meals.	The application captures the intended food, meal, quantity, time, or filter information.
3	VitaMate System	Validates ownership, quantity, meal classification, and applicable nutrition rules.	Invalid or unauthorized input is rejected.
4	VitaMate System	Applies the nutrition-domain operation and preserves a stable nutrient snapshot when a meal write occurs.	Nutrition totals and linked beverage/hydration effects remain consistent.
5	Registered User	Reviews the updated meal or nutrition result.	The latest nutrition state and eligible motivation effects are visible without duplicate side effects.

US-04-05 — Search Food Library

Epic	EP-04 — Nutrition and Meal Management	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to search food library, so that I can record and review nutrition information accurately.		
Traceability	Source Story ID: US-04-05 | Source Epic: EP-04 | Appendix Story 42 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Search Food Library' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-3	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Nutrition and initiates search food library.	The relevant meal, catalog, or nutrition context is loaded.
2	Registered User	Selects, enters, or reviews the data required to search food library.	The application captures the intended food, meal, quantity, time, or filter information.
3	VitaMate System	Validates ownership, quantity, meal classification, and applicable nutrition rules.	Invalid or unauthorized input is rejected.
4	VitaMate System	Applies the nutrition-domain operation and preserves a stable nutrient snapshot when a meal write occurs.	Nutrition totals and linked beverage/hydration effects remain consistent.
5	Registered User	Reviews the updated meal or nutrition result.	The latest nutrition state and eligible motivation effects are visible without duplicate side effects.

US-04-06 — Filter Food Library

Epic	EP-04 — Nutrition and Meal Management	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to filter food library, so that I can record and review nutrition information accurately.		
Traceability	Source Story ID: US-04-06 | Source Epic: EP-04 | Appendix Story 43 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Filter Food Library' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The visible result shall remain consistent with the selected locale, supported directionality, and accessibility semantics.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Nutrition and initiates filter food library.	The relevant meal, catalog, or nutrition context is loaded.
2	Registered User	Selects, enters, or reviews the data required to filter food library.	The application captures the intended food, meal, quantity, time, or filter information.
3	VitaMate System	Validates ownership, quantity, meal classification, and applicable nutrition rules.	Invalid or unauthorized input is rejected.
4	VitaMate System	Applies the nutrition-domain operation and preserves a stable nutrient snapshot when a meal write occurs.	Nutrition totals and linked beverage/hydration effects remain consistent.
5	Registered User	Reviews the updated meal or nutrition result.	The latest nutrition state and eligible motivation effects are visible without duplicate side effects.

US-04-07 — Favorites

Epic	EP-04 — Nutrition and Meal Management	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view favorites, so that I can record and review nutrition information accurately.		
Traceability	Source Story ID: US-04-07 | Source Epic: EP-04 | Appendix Story 44 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Favorites' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Nutrition and initiates favorites.	The relevant meal, catalog, or nutrition context is loaded.
2	Registered User	Selects, enters, or reviews the data required to view favorites.	The application captures the intended food, meal, quantity, time, or filter information.
3	VitaMate System	Validates ownership, quantity, meal classification, and applicable nutrition rules.	Invalid or unauthorized input is rejected.
4	VitaMate System	Applies the nutrition-domain operation and preserves a stable nutrient snapshot when a meal write occurs.	Nutrition totals and linked beverage/hydration effects remain consistent.
5	Registered User	Reviews the updated meal or nutrition result.	The latest nutrition state and eligible motivation effects are visible without duplicate side effects.

US-04-08 — Recent Foods

Epic	EP-04 — Nutrition and Meal Management	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view recent foods, so that I can record and review nutrition information accurately.		
Traceability	Source Story ID: US-04-08 | Source Epic: EP-04 | Appendix Story 45 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Recent Foods' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Owner-scoped authorization shall prevent cross-user access to protected records.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Nutrition and initiates recent foods.	The relevant meal, catalog, or nutrition context is loaded.
2	Registered User	Selects, enters, or reviews the data required to view recent foods.	The application captures the intended food, meal, quantity, time, or filter information.
3	VitaMate System	Validates ownership, quantity, meal classification, and applicable nutrition rules.	Invalid or unauthorized input is rejected.
4	VitaMate System	Applies the nutrition-domain operation and preserves a stable nutrient snapshot when a meal write occurs.	Nutrition totals and linked beverage/hydration effects remain consistent.
5	Registered User	Reviews the updated meal or nutrition result.	The latest nutrition state and eligible motivation effects are visible without duplicate side effects.

US-04-09 — Select Meal Type

Epic	EP-04 — Nutrition and Meal Management	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to select meal type, so that I can record and review nutrition information accurately.		
Traceability	Source Story ID: US-04-09 | Source Epic: EP-04 | Appendix Story 46 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Select Meal Type' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Relevant inputs and preconditions shall be validated before the operation is accepted.
AC-3	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-4	Explicitly excluded records shall not contribute to the affected metric, completion rule, or mission.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Nutrition and initiates select meal type.	The relevant meal, catalog, or nutrition context is loaded.
2	Registered User	Selects, enters, or reviews the data required to select meal type.	The application captures the intended food, meal, quantity, time, or filter information.
3	VitaMate System	Validates ownership, quantity, meal classification, and applicable nutrition rules.	Invalid or unauthorized input is rejected.
4	VitaMate System	Applies the nutrition-domain operation and preserves a stable nutrient snapshot when a meal write occurs.	Nutrition totals and linked beverage/hydration effects remain consistent.
5	Registered User	Reviews the updated meal or nutrition result.	The latest nutrition state and eligible motivation effects are visible without duplicate side effects.

US-04-10 — Select Quantity

Epic	EP-04 — Nutrition and Meal Management	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to select quantity, so that I can record and review nutrition information accurately.		
Traceability	Source Story ID: US-04-10 | Source Epic: EP-04 | Appendix Story 47 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Select Quantity' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Relevant inputs and preconditions shall be validated before the operation is accepted.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Nutrition and initiates select quantity.	The relevant meal, catalog, or nutrition context is loaded.
2	Registered User	Selects, enters, or reviews the data required to select quantity.	The application captures the intended food, meal, quantity, time, or filter information.
3	VitaMate System	Validates ownership, quantity, meal classification, and applicable nutrition rules.	Invalid or unauthorized input is rejected.
4	VitaMate System	Applies the nutrition-domain operation and preserves a stable nutrient snapshot when a meal write occurs.	Nutrition totals and linked beverage/hydration effects remain consistent.
5	Registered User	Reviews the updated meal or nutrition result.	The latest nutrition state and eligible motivation effects are visible without duplicate side effects.

US-04-11 — Set Consumption Time

Epic	EP-04 — Nutrition and Meal Management	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to set consumption time, so that I can record and review nutrition information accurately.		
Traceability	Source Story ID: US-04-11 | Source Epic: EP-04 | Appendix Story 48 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Set Consumption Time' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	Time-sensitive values shall preserve the validated UTC/local-date or timezone behavior.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Nutrition and initiates set consumption time.	The relevant meal, catalog, or nutrition context is loaded.
2	Registered User	Selects, enters, or reviews the data required to set consumption time.	The application captures the intended food, meal, quantity, time, or filter information.
3	VitaMate System	Validates ownership, quantity, meal classification, and applicable nutrition rules.	Invalid or unauthorized input is rejected.
4	VitaMate System	Applies the nutrition-domain operation and preserves a stable nutrient snapshot when a meal write occurs.	Nutrition totals and linked beverage/hydration effects remain consistent.
5	Registered User	Reviews the updated meal or nutrition result.	The latest nutrition state and eligible motivation effects are visible without duplicate side effects.

US-04-12 — Save Manual Meal

Epic	EP-04 — Nutrition and Meal Management	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to save manual meal, so that I can record and review nutrition information accurately.		
Traceability	Source Story ID: US-04-12 | Source Epic: EP-04 | Appendix Story 49 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Save Manual Meal' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Nutrition and initiates save manual meal.	The relevant meal, catalog, or nutrition context is loaded.
2	Registered User	Selects, enters, or reviews the data required to save manual meal.	The application captures the intended food, meal, quantity, time, or filter information.
3	VitaMate System	Validates ownership, quantity, meal classification, and applicable nutrition rules.	Invalid or unauthorized input is rejected.
4	VitaMate System	Applies the nutrition-domain operation and preserves a stable nutrient snapshot when a meal write occurs.	Nutrition totals and linked beverage/hydration effects remain consistent.
5	Registered User	Reviews the updated meal or nutrition result.	The latest nutrition state and eligible motivation effects are visible without duplicate side effects.

US-04-13 — Meal Details

Epic	EP-04 — Nutrition and Meal Management	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view meal details, so that I can record and review nutrition information accurately.		
Traceability	Source Story ID: US-04-13 | Source Epic: EP-04 | Appendix Story 50 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Meal Details' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	Navigation shall resolve to the intended route or detail view while preserving valid session context.
AC-4	The requested information shall be presented from current authorized data without fabricating unavailable values.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Nutrition and initiates meal details.	The relevant meal, catalog, or nutrition context is loaded.
2	Registered User	Selects, enters, or reviews the data required to view meal details.	The application captures the intended food, meal, quantity, time, or filter information.
3	VitaMate System	Validates ownership, quantity, meal classification, and applicable nutrition rules.	Invalid or unauthorized input is rejected.
4	VitaMate System	Applies the nutrition-domain operation and preserves a stable nutrient snapshot when a meal write occurs.	Nutrition totals and linked beverage/hydration effects remain consistent.
5	Registered User	Reviews the updated meal or nutrition result.	The latest nutrition state and eligible motivation effects are visible without duplicate side effects.

US-04-14 — Edit Meal

Epic	EP-04 — Nutrition and Meal Management	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to edit meal, so that I can record and review nutrition information accurately.		
Traceability	Source Story ID: US-04-14 | Source Epic: EP-04 | Appendix Story 51 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Edit Meal' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Nutrition and initiates edit meal.	The relevant meal, catalog, or nutrition context is loaded.
2	Registered User	Selects, enters, or reviews the data required to edit meal.	The application captures the intended food, meal, quantity, time, or filter information.
3	VitaMate System	Validates ownership, quantity, meal classification, and applicable nutrition rules.	Invalid or unauthorized input is rejected.
4	VitaMate System	Applies the nutrition-domain operation and preserves a stable nutrient snapshot when a meal write occurs.	Nutrition totals and linked beverage/hydration effects remain consistent.
5	Registered User	Reviews the updated meal or nutrition result.	The latest nutrition state and eligible motivation effects are visible without duplicate side effects.

US-04-15 — Delete Meal

Epic	EP-04 — Nutrition and Meal Management	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to delete meal, so that I can record and review nutrition information accurately.		
Traceability	Source Story ID: US-04-15 | Source Epic: EP-04 | Appendix Story 52 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Delete Meal' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any motivation effect shall follow the defined point, mission, streak, badge, or level rules and shall not be duplicated.
AC-3	The state-changing action shall preserve required historical information and linked consistency according to the source story.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Nutrition and initiates delete meal.	The relevant meal, catalog, or nutrition context is loaded.
2	Registered User	Selects, enters, or reviews the data required to delete meal.	The application captures the intended food, meal, quantity, time, or filter information.
3	VitaMate System	Validates ownership, quantity, meal classification, and applicable nutrition rules.	Invalid or unauthorized input is rejected.
4	VitaMate System	Applies the nutrition-domain operation and preserves a stable nutrient snapshot when a meal write occurs.	Nutrition totals and linked beverage/hydration effects remain consistent.
5	Registered User	Reviews the updated meal or nutrition result.	The latest nutrition state and eligible motivation effects are visible without duplicate side effects.

US-04-16 — Weight Goal and Points

Epic	EP-04 — Nutrition and Meal Management	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to align nutrition-related motivation rules with the configured weight goal, so that I can record and review nutrition information accurately.		
Traceability	Source Story ID: US-04-16 | Source Epic: EP-04 | Appendix Story 53 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Weight Goal and Points' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Duplicate records, rewards, plans, and side effects shall be prevented using idempotent or uniqueness-aware behavior.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Nutrition and initiates weight goal and points.	The relevant meal, catalog, or nutrition context is loaded.
2	Registered User	Selects, enters, or reviews the data required to align nutrition-related motivation rules with the configured weight goal.	The application captures the intended food, meal, quantity, time, or filter information.
3	VitaMate System	Validates ownership, quantity, meal classification, and applicable nutrition rules.	Invalid or unauthorized input is rejected.
4	VitaMate System	Applies the nutrition-domain operation and preserves a stable nutrient snapshot when a meal write occurs.	Nutrition totals and linked beverage/hydration effects remain consistent.
5	Registered User	Reviews the updated meal or nutrition result.	The latest nutrition state and eligible motivation effects are visible without duplicate side effects.

B.8 EP-05 — AI-Assisted Meal Analysis

A review-assisted image-analysis workflow that proposes meal components and weights before the user confirms the final nutrition record.

Epic Attribute	Academic Description
Scope	A review-assisted image-analysis workflow that proposes meal components and weights before the user confirms the final nutrition record.
Story Count	12
Baseline Rule 1	The mobile client communicates with the AI runtime through Django REST rather than calling the FastAPI service directly.
Baseline Rule 2	AI output is provisional; detected components and weights require user review before a MealLog is finalized.
Baseline Rule 3	A complete weight-estimation failure requires manual input and must not be represented as a valid zero-weight result.

US-05-01 — Capture Meal Image

Epic	EP-05 — AI-Assisted Meal Analysis	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to capture meal image, so that I can use AI assistance while retaining control over the final meal record.		
Traceability	Source Story ID: US-05-01 | Source Epic: EP-05 | Appendix Story 54 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Capture Meal Image' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Required device capabilities or permissions shall be checked, and unavailable dependencies shall be communicated clearly.
AC-3	AI-derived information shall remain provisional until the user completes the required review or confirmation step.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Enters the AI meal-analysis workflow and performs the action required to capture meal image.	A valid analysis-session context is established or loaded.
2	Flutter / Django	Routes AI-related requests through the authenticated Django API rather than exposing the internal AI service directly.	Session ownership and service boundaries are preserved.
3	AI Service / VitaMate System	Produces or updates the provisional result relevant to capture meal image.	Confidence, component, mapping, or weight information remains provisional.
4	Registered User	Reviews, confirms, corrects, or completes the required AI-assisted information.	Only user-confirmed components and valid weights proceed toward finalization.
5	VitaMate System	Persists the reviewed result or exposes the documented fallback/error state.	Final meal creation is idempotent and does not use expired or non-owned sessions.

US-05-02 — Select Meal Image

Epic	EP-05 — AI-Assisted Meal Analysis	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to select meal image, so that I can use AI assistance while retaining control over the final meal record.		
Traceability	Source Story ID: US-05-02 | Source Epic: EP-05 | Appendix Story 55 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Select Meal Image' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Relevant inputs and preconditions shall be validated before the operation is accepted.
AC-3	Navigation shall resolve to the intended route or detail view while preserving valid session context.
AC-4	AI-derived information shall remain provisional until the user completes the required review or confirmation step.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Enters the AI meal-analysis workflow and performs the action required to select meal image.	A valid analysis-session context is established or loaded.
2	Flutter / Django	Routes AI-related requests through the authenticated Django API rather than exposing the internal AI service directly.	Session ownership and service boundaries are preserved.
3	AI Service / VitaMate System	Produces or updates the provisional result relevant to select meal image.	Confidence, component, mapping, or weight information remains provisional.
4	Registered User	Reviews, confirms, corrects, or completes the required AI-assisted information.	Only user-confirmed components and valid weights proceed toward finalization.
5	VitaMate System	Persists the reviewed result or exposes the documented fallback/error state.	Final meal creation is idempotent and does not use expired or non-owned sessions.

US-05-03 — Start AI Analysis

Epic	EP-05 — AI-Assisted Meal Analysis	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to start ai analysis, so that I can use AI assistance while retaining control over the final meal record.		
Traceability	Source Story ID: US-05-03 | Source Epic: EP-05 | Appendix Story 56 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Start AI Analysis' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	Failure states shall expose a recoverable retry or error path without reporting fabricated success.
AC-4	AI-derived information shall remain provisional until the user completes the required review or confirmation step.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Enters the AI meal-analysis workflow and performs the action required to start ai analysis.	A valid analysis-session context is established or loaded.
2	Flutter / Django	Routes AI-related requests through the authenticated Django API rather than exposing the internal AI service directly.	Session ownership and service boundaries are preserved.
3	AI Service / VitaMate System	Produces or updates the provisional result relevant to start ai analysis.	Confidence, component, mapping, or weight information remains provisional.
4	Registered User	Reviews, confirms, corrects, or completes the required AI-assisted information.	Only user-confirmed components and valid weights proceed toward finalization.
5	VitaMate System	Persists the reviewed result or exposes the documented fallback/error state.	Final meal creation is idempotent and does not use expired or non-owned sessions.

US-05-04 — Dish Suggestion

Epic	EP-05 — AI-Assisted Meal Analysis	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to use dish suggestion, so that I can use AI assistance while retaining control over the final meal record.		
Traceability	Source Story ID: US-05-04 | Source Epic: EP-05 | Appendix Story 57 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Dish Suggestion' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	AI-derived information shall remain provisional until the user completes the required review or confirmation step.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Enters the AI meal-analysis workflow and performs the action required to use dish suggestion.	A valid analysis-session context is established or loaded.
2	Flutter / Django	Routes AI-related requests through the authenticated Django API rather than exposing the internal AI service directly.	Session ownership and service boundaries are preserved.
3	AI Service / VitaMate System	Produces or updates the provisional result relevant to dish suggestion.	Confidence, component, mapping, or weight information remains provisional.
4	Registered User	Reviews, confirms, corrects, or completes the required AI-assisted information.	Only user-confirmed components and valid weights proceed toward finalization.
5	VitaMate System	Persists the reviewed result or exposes the documented fallback/error state.	Final meal creation is idempotent and does not use expired or non-owned sessions.

US-05-05 — Ingredient Segmentation

Epic	EP-05 — AI-Assisted Meal Analysis	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to use ingredient segmentation, so that I can use AI assistance while retaining control over the final meal record.		
Traceability	Source Story ID: US-05-05 | Source Epic: EP-05 | Appendix Story 58 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Ingredient Segmentation' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	AI-derived information shall remain provisional until the user completes the required review or confirmation step.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Enters the AI meal-analysis workflow and performs the action required to use ingredient segmentation.	A valid analysis-session context is established or loaded.
2	Flutter / Django	Routes AI-related requests through the authenticated Django API rather than exposing the internal AI service directly.	Session ownership and service boundaries are preserved.
3	AI Service / VitaMate System	Produces or updates the provisional result relevant to ingredient segmentation.	Confidence, component, mapping, or weight information remains provisional.
4	Registered User	Reviews, confirms, corrects, or completes the required AI-assisted information.	Only user-confirmed components and valid weights proceed toward finalization.
5	VitaMate System	Persists the reviewed result or exposes the documented fallback/error state.	Final meal creation is idempotent and does not use expired or non-owned sessions.

US-05-06 — Map Ingredient to Food Library

Epic	EP-05 — AI-Assisted Meal Analysis	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to map an AI-detected ingredient to an authorized food-catalog item, so that I can use AI assistance while retaining control over the final meal record.		
Traceability	Source Story ID: US-05-06 | Source Epic: EP-05 | Appendix Story 59 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Map Ingredient to Food Library' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Owner-scoped authorization shall prevent cross-user access to protected records.
AC-3	AI-derived information shall remain provisional until the user completes the required review or confirmation step.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Enters the AI meal-analysis workflow and performs the action required to map an AI-detected ingredient to an authorized food-catalog item.	A valid analysis-session context is established or loaded.
2	Flutter / Django	Routes AI-related requests through the authenticated Django API rather than exposing the internal AI service directly.	Session ownership and service boundaries are preserved.
3	AI Service / VitaMate System	Produces or updates the provisional result relevant to map ingredient to food library.	Confidence, component, mapping, or weight information remains provisional.
4	Registered User	Reviews, confirms, corrects, or completes the required AI-assisted information.	Only user-confirmed components and valid weights proceed toward finalization.
5	VitaMate System	Persists the reviewed result or exposes the documented fallback/error state.	Final meal creation is idempotent and does not use expired or non-owned sessions.

US-05-07 — Estimate Weight

Epic	EP-05 — AI-Assisted Meal Analysis	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to estimate weight, so that I can use AI assistance while retaining control over the final meal record.		
Traceability	Source Story ID: US-05-07 | Source Epic: EP-05 | Appendix Story 60 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Estimate Weight' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	AI-derived information shall remain provisional until the user completes the required review or confirmation step.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Enters the AI meal-analysis workflow and performs the action required to estimate weight.	A valid analysis-session context is established or loaded.
2	Flutter / Django	Routes AI-related requests through the authenticated Django API rather than exposing the internal AI service directly.	Session ownership and service boundaries are preserved.
3	AI Service / VitaMate System	Produces or updates the provisional result relevant to estimate weight.	Confidence, component, mapping, or weight information remains provisional.
4	Registered User	Reviews, confirms, corrects, or completes the required AI-assisted information.	Only user-confirmed components and valid weights proceed toward finalization.
5	VitaMate System	Persists the reviewed result or exposes the documented fallback/error state.	Final meal creation is idempotent and does not use expired or non-owned sessions.

US-05-08 — Edit Component Weights

Epic	EP-05 — AI-Assisted Meal Analysis	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to edit component weights, so that I can use AI assistance while retaining control over the final meal record.		
Traceability	Source Story ID: US-05-08 | Source Epic: EP-05 | Appendix Story 61 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Edit Component Weights' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Relevant inputs and preconditions shall be validated before the operation is accepted.
AC-3	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-4	AI-derived information shall remain provisional until the user completes the required review or confirmation step.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Enters the AI meal-analysis workflow and performs the action required to edit component weights.	A valid analysis-session context is established or loaded.
2	Flutter / Django	Routes AI-related requests through the authenticated Django API rather than exposing the internal AI service directly.	Session ownership and service boundaries are preserved.
3	AI Service / VitaMate System	Produces or updates the provisional result relevant to edit component weights.	Confidence, component, mapping, or weight information remains provisional.
4	Registered User	Reviews, confirms, corrects, or completes the required AI-assisted information.	Only user-confirmed components and valid weights proceed toward finalization.
5	VitaMate System	Persists the reviewed result or exposes the documented fallback/error state.	Final meal creation is idempotent and does not use expired or non-owned sessions.

US-05-09 — Complete Weight-Estimation Failure

Epic	EP-05 — AI-Assisted Meal Analysis	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to enter component weights manually when automatic weight estimation fails, so that I can use AI assistance while retaining control over the final meal record.		
Traceability	Source Story ID: US-05-09 | Source Epic: EP-05 | Appendix Story 62 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Complete Weight-Estimation Failure' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	When the automated path cannot provide a valid result, the documented manual fallback shall remain available or mandatory as specified.
AC-3	AI-derived information shall remain provisional until the user completes the required review or confirmation step.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Enters the AI meal-analysis workflow and performs the action required to enter component weights manually when automatic weight estimation fails.	A valid analysis-session context is established or loaded.
2	Flutter / Django	Routes AI-related requests through the authenticated Django API rather than exposing the internal AI service directly.	Session ownership and service boundaries are preserved.
3	AI Service / VitaMate System	Produces or updates the provisional result relevant to complete weight-estimation failure.	Confidence, component, mapping, or weight information remains provisional.
4	Registered User	Reviews, confirms, corrects, or completes the required AI-assisted information.	Only user-confirmed components and valid weights proceed toward finalization.
5	VitaMate System	Persists the reviewed result or exposes the documented fallback/error state.	Final meal creation is idempotent and does not use expired or non-owned sessions.

US-05-10 — Preview Nutritional Values

Epic	EP-05 — AI-Assisted Meal Analysis	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to use preview nutritional values, so that I can use AI assistance while retaining control over the final meal record.		
Traceability	Source Story ID: US-05-10 | Source Epic: EP-05 | Appendix Story 63 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Preview Nutritional Values' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	AI-derived information shall remain provisional until the user completes the required review or confirmation step.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Enters the AI meal-analysis workflow and performs the action required to use preview nutritional values.	A valid analysis-session context is established or loaded.
2	Flutter / Django	Routes AI-related requests through the authenticated Django API rather than exposing the internal AI service directly.	Session ownership and service boundaries are preserved.
3	AI Service / VitaMate System	Produces or updates the provisional result relevant to preview nutritional values.	Confidence, component, mapping, or weight information remains provisional.
4	Registered User	Reviews, confirms, corrects, or completes the required AI-assisted information.	Only user-confirmed components and valid weights proceed toward finalization.
5	VitaMate System	Persists the reviewed result or exposes the documented fallback/error state.	Final meal creation is idempotent and does not use expired or non-owned sessions.

US-05-11 — Idempotent Final Save

Epic	EP-05 — AI-Assisted Meal Analysis	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to finalize the reviewed AI meal once without duplicate meal creation, so that I can use AI assistance while retaining control over the final meal record.		
Traceability	Source Story ID: US-05-11 | Source Epic: EP-05 | Appendix Story 64 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Idempotent Final Save' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Duplicate records, rewards, plans, and side effects shall be prevented using idempotent or uniqueness-aware behavior.
AC-3	Failure states shall expose a recoverable retry or error path without reporting fabricated success.
AC-4	AI-derived information shall remain provisional until the user completes the required review or confirmation step.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Enters the AI meal-analysis workflow and performs the action required to finalize the reviewed AI meal once without duplicate meal creation.	A valid analysis-session context is established or loaded.
2	Flutter / Django	Routes AI-related requests through the authenticated Django API rather than exposing the internal AI service directly.	Session ownership and service boundaries are preserved.
3	AI Service / VitaMate System	Produces or updates the provisional result relevant to idempotent final save.	Confidence, component, mapping, or weight information remains provisional.
4	Registered User	Reviews, confirms, corrects, or completes the required AI-assisted information.	Only user-confirmed components and valid weights proceed toward finalization.
5	VitaMate System	Persists the reviewed result or exposes the documented fallback/error state.	Final meal creation is idempotent and does not use expired or non-owned sessions.

US-05-12 — Session Expiry and Privacy

Epic	EP-05 — AI-Assisted Meal Analysis	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to prevent expired or non-owned AI-analysis sessions from being modified, so that I can use AI assistance while retaining control over the final meal record.		
Traceability	Source Story ID: US-05-12 | Source Epic: EP-05 | Appendix Story 65 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Session Expiry and Privacy' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Owner-scoped authorization shall prevent cross-user access to protected records.
AC-3	AI-derived information shall remain provisional until the user completes the required review or confirmation step.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Enters the AI meal-analysis workflow and performs the action required to prevent expired or non-owned AI-analysis sessions from being modified.	A valid analysis-session context is established or loaded.
2	Flutter / Django	Routes AI-related requests through the authenticated Django API rather than exposing the internal AI service directly.	Session ownership and service boundaries are preserved.
3	AI Service / VitaMate System	Produces or updates the provisional result relevant to session expiry and privacy.	Confidence, component, mapping, or weight information remains provisional.
4	Registered User	Reviews, confirms, corrects, or completes the required AI-assisted information.	Only user-confirmed components and valid weights proceed toward finalization.
5	VitaMate System	Persists the reviewed result or exposes the documented fallback/error state.	Final meal creation is idempotent and does not use expired or non-owned sessions.

B.9 EP-06 — Vitamins, Minerals, and Focused Tracking

Focused micronutrient monitoring that can combine food snapshots, beverages, taken supplements, laboratory context, and effective health constraints.

Epic Attribute	Academic Description
Scope	Focused micronutrient monitoring that can combine food snapshots, beverages, taken supplements, laboratory context, and effective health constraints.
Story Count	9
Baseline Rule 1	Micronutrient intake is derived from confirmed meal, beverage, and taken-supplement data.
Baseline Rule 2	A medical target or laboratory result may generate an effective goal or upper/lower limit in the constraints engine.
Baseline Rule 3	Stopping focused tracking does not delete a linked supplement medication or erase historical records.

US-06-01 — Micronutrient Overview

Epic	EP-06 — Vitamins, Minerals, and Focused Tracking	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view micronutrient overview, so that I can monitor important vitamins and minerals against relevant targets and limits.		
Traceability	Source Story ID: US-06-01 | Source Epic: EP-06 | Appendix Story 66 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Micronutrient Overview' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-3	Compared and displayed values shall use the applicable units consistently.
AC-4	The requested information shall be presented from current authorized data without fabricating unavailable values.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens micronutrient tracking and selects micronutrient overview.	Current intake, goals, and supported nutrient context are loaded.
2	Registered User	Provides or reviews the information required to view micronutrient overview.	The application captures the target, laboratory, supplement, or filter input as applicable.
3	VitaMate System	Validates units, limits, ownership, and the relation between lower/target/upper values.	Invalid or contradictory limits are rejected.
4	VitaMate System	Updates the focused nutrient state and any applicable constraint or supplement linkage.	Confirmed intake and effective limits remain synchronized.
5	Registered User	Reviews the resulting nutrient status or warning.	The displayed state distinguishes low, acceptable, and high conditions using the effective rule.

US-06-02 — Filter Nutrient Status

Epic	EP-06 — Vitamins, Minerals, and Focused Tracking	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to filter nutrient status, so that I can monitor important vitamins and minerals against relevant targets and limits.		
Traceability	Source Story ID: US-06-02 | Source Epic: EP-06 | Appendix Story 67 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Filter Nutrient Status' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The visible result shall remain consistent with the selected locale, supported directionality, and accessibility semantics.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens micronutrient tracking and selects filter nutrient status.	Current intake, goals, and supported nutrient context are loaded.
2	Registered User	Provides or reviews the information required to filter nutrient status.	The application captures the target, laboratory, supplement, or filter input as applicable.
3	VitaMate System	Validates units, limits, ownership, and the relation between lower/target/upper values.	Invalid or contradictory limits are rejected.
4	VitaMate System	Updates the focused nutrient state and any applicable constraint or supplement linkage.	Confirmed intake and effective limits remain synchronized.
5	Registered User	Reviews the resulting nutrient status or warning.	The displayed state distinguishes low, acceptable, and high conditions using the effective rule.

US-06-03 — Start Focused Nutrient Tracking

Epic	EP-06 — Vitamins, Minerals, and Focused Tracking	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to start focused nutrient tracking, so that I can monitor important vitamins and minerals against relevant targets and limits.		
Traceability	Source Story ID: US-06-03 | Source Epic: EP-06 | Appendix Story 68 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Start Focused Nutrient Tracking' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	Navigation shall resolve to the intended route or detail view while preserving valid session context.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens micronutrient tracking and selects start focused nutrient tracking.	Current intake, goals, and supported nutrient context are loaded.
2	Registered User	Provides or reviews the information required to start focused nutrient tracking.	The application captures the target, laboratory, supplement, or filter input as applicable.
3	VitaMate System	Validates units, limits, ownership, and the relation between lower/target/upper values.	Invalid or contradictory limits are rejected.
4	VitaMate System	Updates the focused nutrient state and any applicable constraint or supplement linkage.	Confirmed intake and effective limits remain synchronized.
5	Registered User	Reviews the resulting nutrient status or warning.	The displayed state distinguishes low, acceptable, and high conditions using the effective rule.

US-06-04 — Enter Custom Limits

Epic	EP-06 — Vitamins, Minerals, and Focused Tracking	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to enter custom limits, so that I can monitor important vitamins and minerals against relevant targets and limits.		
Traceability	Source Story ID: US-06-04 | Source Epic: EP-06 | Appendix Story 69 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Enter Custom Limits' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Relevant inputs and preconditions shall be validated before the operation is accepted.
AC-3	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-4	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-5	Compared and displayed values shall use the applicable units consistently.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens micronutrient tracking and selects enter custom limits.	Current intake, goals, and supported nutrient context are loaded.
2	Registered User	Provides or reviews the information required to enter custom limits.	The application captures the target, laboratory, supplement, or filter input as applicable.
3	VitaMate System	Validates units, limits, ownership, and the relation between lower/target/upper values.	Invalid or contradictory limits are rejected.
4	VitaMate System	Updates the focused nutrient state and any applicable constraint or supplement linkage.	Confirmed intake and effective limits remain synchronized.
5	Registered User	Reviews the resulting nutrient status or warning.	The displayed state distinguishes low, acceptable, and high conditions using the effective rule.

US-06-05 — Add Laboratory Result

Epic	EP-06 — Vitamins, Minerals, and Focused Tracking	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to add laboratory result, so that I can monitor important vitamins and minerals against relevant targets and limits.		
Traceability	Source Story ID: US-06-05 | Source Epic: EP-06 | Appendix Story 70 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Add Laboratory Result' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens micronutrient tracking and selects add laboratory result.	Current intake, goals, and supported nutrient context are loaded.
2	Registered User	Provides or reviews the information required to add laboratory result.	The application captures the target, laboratory, supplement, or filter input as applicable.
3	VitaMate System	Validates units, limits, ownership, and the relation between lower/target/upper values.	Invalid or contradictory limits are rejected.
4	VitaMate System	Updates the focused nutrient state and any applicable constraint or supplement linkage.	Confirmed intake and effective limits remain synchronized.
5	Registered User	Reviews the resulting nutrient status or warning.	The displayed state distinguishes low, acceptable, and high conditions using the effective rule.

US-06-06 — Create Supplement Reminder

Epic	EP-06 — Vitamins, Minerals, and Focused Tracking	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to create supplement reminder, so that I can monitor important vitamins and minerals against relevant targets and limits.		
Traceability	Source Story ID: US-06-06 | Source Epic: EP-06 | Appendix Story 71 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Create Supplement Reminder' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Notification-related changes shall remain synchronized with current preferences, schedule, priority, and device state.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens micronutrient tracking and selects create supplement reminder.	Current intake, goals, and supported nutrient context are loaded.
2	Registered User	Provides or reviews the information required to create supplement reminder.	The application captures the target, laboratory, supplement, or filter input as applicable.
3	VitaMate System	Validates units, limits, ownership, and the relation between lower/target/upper values.	Invalid or contradictory limits are rejected.
4	VitaMate System	Updates the focused nutrient state and any applicable constraint or supplement linkage.	Confirmed intake and effective limits remain synchronized.
5	Registered User	Reviews the resulting nutrient status or warning.	The displayed state distinguishes low, acceptable, and high conditions using the effective rule.

US-06-07 — Count Taken Supplement Dose

Epic	EP-06 — Vitamins, Minerals, and Focused Tracking	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to include confirmed taken supplement doses in micronutrient intake, so that I can monitor important vitamins and minerals against relevant targets and limits.		
Traceability	Source Story ID: US-06-07 | Source Epic: EP-06 | Appendix Story 72 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Count Taken Supplement Dose' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Duplicate records, rewards, plans, and side effects shall be prevented using idempotent or uniqueness-aware behavior.
AC-3	Explicitly excluded records shall not contribute to the affected metric, completion rule, or mission.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens micronutrient tracking and selects count taken supplement dose.	Current intake, goals, and supported nutrient context are loaded.
2	Registered User	Provides or reviews the information required to include confirmed taken supplement doses in micronutrient intake.	The application captures the target, laboratory, supplement, or filter input as applicable.
3	VitaMate System	Validates units, limits, ownership, and the relation between lower/target/upper values.	Invalid or contradictory limits are rejected.
4	VitaMate System	Updates the focused nutrient state and any applicable constraint or supplement linkage.	Confirmed intake and effective limits remain synchronized.
5	Registered User	Reviews the resulting nutrient status or warning.	The displayed state distinguishes low, acceptable, and high conditions using the effective rule.

US-06-08 — Stop Focused Tracking

Epic	EP-06 — Vitamins, Minerals, and Focused Tracking	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to stop focused tracking, so that I can monitor important vitamins and minerals against relevant targets and limits.		
Traceability	Source Story ID: US-06-08 | Source Epic: EP-06 | Appendix Story 73 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Stop Focused Tracking' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-3	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens micronutrient tracking and selects stop focused tracking.	Current intake, goals, and supported nutrient context are loaded.
2	Registered User	Provides or reviews the information required to stop focused tracking.	The application captures the target, laboratory, supplement, or filter input as applicable.
3	VitaMate System	Validates units, limits, ownership, and the relation between lower/target/upper values.	Invalid or contradictory limits are rejected.
4	VitaMate System	Updates the focused nutrient state and any applicable constraint or supplement linkage.	Confirmed intake and effective limits remain synchronized.
5	Registered User	Reviews the resulting nutrient status or warning.	The displayed state distinguishes low, acceptable, and high conditions using the effective rule.

US-06-09 — Upper-Limit Warning

Epic	EP-06 — Vitamins, Minerals, and Focused Tracking	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to receive a warning when a monitored nutrient exceeds its effective upper limit, so that I can monitor important vitamins and minerals against relevant targets and limits.		
Traceability	Source Story ID: US-06-09 | Source Epic: EP-06 | Appendix Story 74 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Upper-Limit Warning' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Health-related output shall follow the validated safety and priority rules and shall not make unsupported diagnostic claims.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens micronutrient tracking and selects upper-limit warning.	Current intake, goals, and supported nutrient context are loaded.
2	Registered User	Provides or reviews the information required to receive a warning when a monitored nutrient exceeds its effective upper limit.	The application captures the target, laboratory, supplement, or filter input as applicable.
3	VitaMate System	Validates units, limits, ownership, and the relation between lower/target/upper values.	Invalid or contradictory limits are rejected.
4	VitaMate System	Updates the focused nutrient state and any applicable constraint or supplement linkage.	Confirmed intake and effective limits remain synchronized.
5	Registered User	Reviews the resulting nutrient status or warning.	The displayed state distinguishes low, acceptable, and high conditions using the effective rule.

B.10 EP-07 — Hydration and Beverages

Hydration logging that distinguishes effective hydration contribution from raw beverage volume and maintains bidirectional coordination with nutrition and habits.

Epic Attribute	Academic Description
Scope	Hydration logging that distinguishes effective hydration contribution from raw beverage volume and maintains bidirectional coordination with nutrition and habits.
Story Count	10
Baseline Rule 1	The effective hydration target may be influenced by activity, user profile, manual goals, and health constraints.
Baseline Rule 2	A beverage may create linked WaterLog and MealLog records, and edits must synchronize the relationship.
Baseline Rule 3	Caffeinated beverages may contribute to caffeine-habit tracking and sleep-coach context according to time and quantity.

US-07-01 — Hydration Summary

Epic	EP-07 — Hydration and Beverages	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view hydration summary, so that I can maintain an accurate hydration record and coordinated beverage data.		
Traceability	Source Story ID: US-07-01 | Source Epic: EP-07 | Appendix Story 75 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Hydration Summary' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Hydration and initiates hydration summary.	The effective hydration target and current-day context are available.
2	Registered User	Provides or reviews the beverage information required to view hydration summary.	Volume, beverage type, time, or reminder settings are captured.
3	VitaMate System	Validates the record and calculates the applicable hydration contribution.	Raw volume is not automatically treated as full hydration when a beverage factor applies.
4	VitaMate System	Synchronizes linked nutrition, caffeine-habit, notification, or motivation state when the source story requires it.	Cross-domain projections occur once and remain consistent.
5	Registered User	Reviews the updated hydration state or history.	Current intake, goal, remaining amount, and related effects reflect the confirmed record.

US-07-02 — Quick Water Logging

Epic	EP-07 — Hydration and Beverages	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to use quick water logging, so that I can maintain an accurate hydration record and coordinated beverage data.		
Traceability	Source Story ID: US-07-02 | Source Epic: EP-07 | Appendix Story 76 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Quick Water Logging' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Relevant inputs and preconditions shall be validated before the operation is accepted.
AC-3	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-4	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-5	Any motivation effect shall follow the defined point, mission, streak, badge, or level rules and shall not be duplicated.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Hydration and initiates quick water logging.	The effective hydration target and current-day context are available.
2	Registered User	Provides or reviews the beverage information required to use quick water logging.	Volume, beverage type, time, or reminder settings are captured.
3	VitaMate System	Validates the record and calculates the applicable hydration contribution.	Raw volume is not automatically treated as full hydration when a beverage factor applies.
4	VitaMate System	Synchronizes linked nutrition, caffeine-habit, notification, or motivation state when the source story requires it.	Cross-domain projections occur once and remain consistent.
5	Registered User	Reviews the updated hydration state or history.	Current intake, goal, remaining amount, and related effects reflect the confirmed record.

US-07-03 — Log Beverage

Epic	EP-07 — Hydration and Beverages	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to log beverage, so that I can maintain an accurate hydration record and coordinated beverage data.		
Traceability	Source Story ID: US-07-03 | Source Epic: EP-07 | Appendix Story 77 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Log Beverage' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Hydration and initiates log beverage.	The effective hydration target and current-day context are available.
2	Registered User	Provides or reviews the beverage information required to log beverage.	Volume, beverage type, time, or reminder settings are captured.
3	VitaMate System	Validates the record and calculates the applicable hydration contribution.	Raw volume is not automatically treated as full hydration when a beverage factor applies.
4	VitaMate System	Synchronizes linked nutrition, caffeine-habit, notification, or motivation state when the source story requires it.	Cross-domain projections occur once and remain consistent.
5	Registered User	Reviews the updated hydration state or history.	Current intake, goal, remaining amount, and related effects reflect the confirmed record.

US-07-04 — Create Custom Beverage

Epic	EP-07 — Hydration and Beverages	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to create custom beverage, so that I can maintain an accurate hydration record and coordinated beverage data.		
Traceability	Source Story ID: US-07-04 | Source Epic: EP-07 | Appendix Story 78 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Create Custom Beverage' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	Owner-scoped authorization shall prevent cross-user access to protected records.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Hydration and initiates create custom beverage.	The effective hydration target and current-day context are available.
2	Registered User	Provides or reviews the beverage information required to create custom beverage.	Volume, beverage type, time, or reminder settings are captured.
3	VitaMate System	Validates the record and calculates the applicable hydration contribution.	Raw volume is not automatically treated as full hydration when a beverage factor applies.
4	VitaMate System	Synchronizes linked nutrition, caffeine-habit, notification, or motivation state when the source story requires it.	Cross-domain projections occur once and remain consistent.
5	Registered User	Reviews the updated hydration state or history.	Current intake, goal, remaining amount, and related effects reflect the confirmed record.

US-07-05 — Hydration History

Epic	EP-07 — Hydration and Beverages	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view hydration history, so that I can maintain an accurate hydration record and coordinated beverage data.		
Traceability	Source Story ID: US-07-05 | Source Epic: EP-07 | Appendix Story 79 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Hydration History' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Hydration and initiates hydration history.	The effective hydration target and current-day context are available.
2	Registered User	Provides or reviews the beverage information required to view hydration history.	Volume, beverage type, time, or reminder settings are captured.
3	VitaMate System	Validates the record and calculates the applicable hydration contribution.	Raw volume is not automatically treated as full hydration when a beverage factor applies.
4	VitaMate System	Synchronizes linked nutrition, caffeine-habit, notification, or motivation state when the source story requires it.	Cross-domain projections occur once and remain consistent.
5	Registered User	Reviews the updated hydration state or history.	Current intake, goal, remaining amount, and related effects reflect the confirmed record.

US-07-06 — Edit Hydration Record

Epic	EP-07 — Hydration and Beverages	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to edit hydration record, so that I can maintain an accurate hydration record and coordinated beverage data.		
Traceability	Source Story ID: US-07-06 | Source Epic: EP-07 | Appendix Story 80 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Edit Hydration Record' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-3	The related domain record shall remain consistent with linked trackers and downstream summaries.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Hydration and initiates edit hydration record.	The effective hydration target and current-day context are available.
2	Registered User	Provides or reviews the beverage information required to edit hydration record.	Volume, beverage type, time, or reminder settings are captured.
3	VitaMate System	Validates the record and calculates the applicable hydration contribution.	Raw volume is not automatically treated as full hydration when a beverage factor applies.
4	VitaMate System	Synchronizes linked nutrition, caffeine-habit, notification, or motivation state when the source story requires it.	Cross-domain projections occur once and remain consistent.
5	Registered User	Reviews the updated hydration state or history.	Current intake, goal, remaining amount, and related effects reflect the confirmed record.

US-07-07 — Delete Hydration Record

Epic	EP-07 — Hydration and Beverages	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to delete hydration record, so that I can maintain an accurate hydration record and coordinated beverage data.		
Traceability	Source Story ID: US-07-07 | Source Epic: EP-07 | Appendix Story 81 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Delete Hydration Record' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-3	Any motivation effect shall follow the defined point, mission, streak, badge, or level rules and shall not be duplicated.
AC-4	The state-changing action shall preserve required historical information and linked consistency according to the source story.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Hydration and initiates delete hydration record.	The effective hydration target and current-day context are available.
2	Registered User	Provides or reviews the beverage information required to delete hydration record.	Volume, beverage type, time, or reminder settings are captured.
3	VitaMate System	Validates the record and calculates the applicable hydration contribution.	Raw volume is not automatically treated as full hydration when a beverage factor applies.
4	VitaMate System	Synchronizes linked nutrition, caffeine-habit, notification, or motivation state when the source story requires it.	Cross-domain projections occur once and remain consistent.
5	Registered User	Reviews the updated hydration state or history.	Current intake, goal, remaining amount, and related effects reflect the confirmed record.

US-07-08 — Set Manual Water Goal

Epic	EP-07 — Hydration and Beverages	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to set manual water goal, so that I can maintain an accurate hydration record and coordinated beverage data.		
Traceability	Source Story ID: US-07-08 | Source Epic: EP-07 | Appendix Story 82 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Set Manual Water Goal' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	When the automated path cannot provide a valid result, the documented manual fallback shall remain available or mandatory as specified.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Hydration and initiates set manual water goal.	The effective hydration target and current-day context are available.
2	Registered User	Provides or reviews the beverage information required to set manual water goal.	Volume, beverage type, time, or reminder settings are captured.
3	VitaMate System	Validates the record and calculates the applicable hydration contribution.	Raw volume is not automatically treated as full hydration when a beverage factor applies.
4	VitaMate System	Synchronizes linked nutrition, caffeine-habit, notification, or motivation state when the source story requires it.	Cross-domain projections occur once and remain consistent.
5	Registered User	Reviews the updated hydration state or history.	Current intake, goal, remaining amount, and related effects reflect the confirmed record.

US-07-09 — Water Reminders

Epic	EP-07 — Hydration and Beverages	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to manage water reminders, so that I can maintain an accurate hydration record and coordinated beverage data.		
Traceability	Source Story ID: US-07-09 | Source Epic: EP-07 | Appendix Story 83 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Water Reminders' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	Notification behavior shall respect configured quiet hours except where a validated critical-priority rule allows an override.
AC-4	Notification-related changes shall remain synchronized with current preferences, schedule, priority, and device state.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Hydration and initiates water reminders.	The effective hydration target and current-day context are available.
2	Registered User	Provides or reviews the beverage information required to manage water reminders.	Volume, beverage type, time, or reminder settings are captured.
3	VitaMate System	Validates the record and calculates the applicable hydration contribution.	Raw volume is not automatically treated as full hydration when a beverage factor applies.
4	VitaMate System	Synchronizes linked nutrition, caffeine-habit, notification, or motivation state when the source story requires it.	Cross-domain projections occur once and remain consistent.
5	Registered User	Reviews the updated hydration state or history.	Current intake, goal, remaining amount, and related effects reflect the confirmed record.

US-07-10 — Prevent Mission Mixing

Epic	EP-07 — Hydration and Beverages	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to keep hydration rewards separate from meal-completion logic, so that I can maintain an accurate hydration record and coordinated beverage data.		
Traceability	Source Story ID: US-07-10 | Source Epic: EP-07 | Appendix Story 84 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Prevent Mission Mixing' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any motivation effect shall follow the defined point, mission, streak, badge, or level rules and shall not be duplicated.
AC-3	Explicitly excluded records shall not contribute to the affected metric, completion rule, or mission.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Hydration and initiates prevent mission mixing.	The effective hydration target and current-day context are available.
2	Registered User	Provides or reviews the beverage information required to keep hydration rewards separate from meal-completion logic.	Volume, beverage type, time, or reminder settings are captured.
3	VitaMate System	Validates the record and calculates the applicable hydration contribution.	Raw volume is not automatically treated as full hydration when a beverage factor applies.
4	VitaMate System	Synchronizes linked nutrition, caffeine-habit, notification, or motivation state when the source story requires it.	Cross-domain projections occur once and remain consistent.
5	Registered User	Reviews the updated hydration state or history.	Current intake, goal, remaining amount, and related effects reflect the confirmed record.

B.11 EP-08 — Activity, Workouts, and Steps

Unified movement tracking that combines manual activity, live sessions, and automatic step data without duplicate counting.

Epic Attribute	Academic Description
Scope	Unified movement tracking that combines manual activity, live sessions, and automatic step data without duplicate counting.
Story Count	13
Baseline Rule 1	Steps are one component of activity and do not replace activity or workout requirements.
Baseline Rule 2	StepLog is maintained as one upserted record per local day and may exceed the target value.
Baseline Rule 3	Finalizing a live session is the authoritative path for creating its ActivityLog and awarding its points once.

US-08-01 — Activity Summary

Epic	EP-08 — Activity, Workouts, and Steps	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view activity summary, so that I can track movement and workouts consistently without duplicate counting.		
Traceability	Source Story ID: US-08-01 | Source Epic: EP-08 | Appendix Story 85 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Activity Summary' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Activity and initiates activity summary.	The current movement/session context and required device capability are identified.
2	Registered User	Provides permission, selects an exercise, or performs the session action required to view activity summary.	The application records only the permitted and relevant interaction.
3	VitaMate System	Validates session state, duration, device permission, and duplicate-counting rules.	Invalid transitions or unavailable capabilities are handled explicitly.
4	VitaMate System	Updates StepLog, live-session state, or ActivityLog according to the authoritative flow.	Steps and workouts do not create duplicate activity or reward effects.
5	Registered User	Reviews the updated activity result, session state, or reminder setting.	Displayed progress may exceed the target numerically while visual progress remains well-formed.

US-08-02 — Step-Sensor Permission

Epic	EP-08 — Activity, Workouts, and Steps	**Primary Actor**	Android User
Priority	High	**Implementation Status**	Implemented - Device/Permission Dependent
User Story	As a Android user, I want to grant or manage the permission required for automatic step tracking, so that I can track movement and workouts consistently without duplicate counting.		
Traceability	Source Story ID: US-08-02 | Source Epic: EP-08 | Appendix Story 86 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Step-Sensor Permission' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Required device capabilities or permissions shall be checked, and unavailable dependencies shall be communicated clearly.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Android User	Opens Activity and initiates step-sensor permission.	The current movement/session context and required device capability are identified.
2	Android User	Provides permission, selects an exercise, or performs the session action required to grant or manage the permission required for automatic step tracking.	The application records only the permitted and relevant interaction.
3	VitaMate System	Validates session state, duration, device permission, and duplicate-counting rules.	Invalid transitions or unavailable capabilities are handled explicitly.
4	VitaMate System	Updates StepLog, live-session state, or ActivityLog according to the authoritative flow.	Steps and workouts do not create duplicate activity or reward effects.
5	Android User	Reviews the updated activity result, session state, or reminder setting.	Displayed progress may exceed the target numerically while visual progress remains well-formed.

Implementation Note: The core flow exists, but complete acceptance depends on the device, permission, external service, or operational limitation identified by the source story.

US-08-03 — Automatic Step Synchronization

Epic	EP-08 — Activity, Workouts, and Steps	**Primary Actor**	Android User
Priority	High	**Implementation Status**	Implemented - Device/Permission Dependent
User Story	As a Android user, I want to synchronize device step counts automatically, so that I can track movement and workouts consistently without duplicate counting.		
Traceability	Source Story ID: US-08-03 | Source Epic: EP-08 | Appendix Story 87 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Automatic Step Synchronization' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-3	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Android User	Opens Activity and initiates automatic step synchronization.	The current movement/session context and required device capability are identified.
2	Android User	Provides permission, selects an exercise, or performs the session action required to synchronize device step counts automatically.	The application records only the permitted and relevant interaction.
3	VitaMate System	Validates session state, duration, device permission, and duplicate-counting rules.	Invalid transitions or unavailable capabilities are handled explicitly.
4	VitaMate System	Updates StepLog, live-session state, or ActivityLog according to the authoritative flow.	Steps and workouts do not create duplicate activity or reward effects.
5	Android User	Reviews the updated activity result, session state, or reminder setting.	Displayed progress may exceed the target numerically while visual progress remains well-formed.

Implementation Note: The core flow exists, but complete acceptance depends on the device, permission, external service, or operational limitation identified by the source story.

US-08-04 — Display Progress Above Goal

Epic	EP-08 — Activity, Workouts, and Steps	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to display progress above goal, so that I can track movement and workouts consistently without duplicate counting.		
Traceability	Source Story ID: US-08-04 | Source Epic: EP-08 | Appendix Story 88 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Display Progress Above Goal' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any motivation effect shall follow the defined point, mission, streak, badge, or level rules and shall not be duplicated.
AC-3	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Activity and initiates display progress above goal.	The current movement/session context and required device capability are identified.
2	Registered User	Provides permission, selects an exercise, or performs the session action required to display progress above goal.	The application records only the permitted and relevant interaction.
3	VitaMate System	Validates session state, duration, device permission, and duplicate-counting rules.	Invalid transitions or unavailable capabilities are handled explicitly.
4	VitaMate System	Updates StepLog, live-session state, or ActivityLog according to the authoritative flow.	Steps and workouts do not create duplicate activity or reward effects.
5	Registered User	Reviews the updated activity result, session state, or reminder setting.	Displayed progress may exceed the target numerically while visual progress remains well-formed.

US-08-05 — Distance and Calories Burned

Epic	EP-08 — Activity, Workouts, and Steps	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view distance and calorie estimates derived from valid activity data, so that I can track movement and workouts consistently without duplicate counting.		
Traceability	Source Story ID: US-08-05 | Source Epic: EP-08 | Appendix Story 89 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Distance and Calories Burned' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Compared and displayed values shall use the applicable units consistently.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Activity and initiates distance and calories burned.	The current movement/session context and required device capability are identified.
2	Registered User	Provides permission, selects an exercise, or performs the session action required to view distance and calorie estimates derived from valid activity data.	The application records only the permitted and relevant interaction.
3	VitaMate System	Validates session state, duration, device permission, and duplicate-counting rules.	Invalid transitions or unavailable capabilities are handled explicitly.
4	VitaMate System	Updates StepLog, live-session state, or ActivityLog according to the authoritative flow.	Steps and workouts do not create duplicate activity or reward effects.
5	Registered User	Reviews the updated activity result, session state, or reminder setting.	Displayed progress may exceed the target numerically while visual progress remains well-formed.

US-08-06 — Exercise Library

Epic	EP-08 — Activity, Workouts, and Steps	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to use exercise library, so that I can track movement and workouts consistently without duplicate counting.		
Traceability	Source Story ID: US-08-06 | Source Epic: EP-08 | Appendix Story 90 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Exercise Library' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	The visible result shall remain consistent with the selected locale, supported directionality, and accessibility semantics.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Activity and initiates exercise library.	The current movement/session context and required device capability are identified.
2	Registered User	Provides permission, selects an exercise, or performs the session action required to use exercise library.	The application records only the permitted and relevant interaction.
3	VitaMate System	Validates session state, duration, device permission, and duplicate-counting rules.	Invalid transitions or unavailable capabilities are handled explicitly.
4	VitaMate System	Updates StepLog, live-session state, or ActivityLog according to the authoritative flow.	Steps and workouts do not create duplicate activity or reward effects.
5	Registered User	Reviews the updated activity result, session state, or reminder setting.	Displayed progress may exceed the target numerically while visual progress remains well-formed.

US-08-07 — Set Up Live Session

Epic	EP-08 — Activity, Workouts, and Steps	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to set up live session, so that I can track movement and workouts consistently without duplicate counting.		
Traceability	Source Story ID: US-08-07 | Source Epic: EP-08 | Appendix Story 91 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Set Up Live Session' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Relevant inputs and preconditions shall be validated before the operation is accepted.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Activity and initiates set up live session.	The current movement/session context and required device capability are identified.
2	Registered User	Provides permission, selects an exercise, or performs the session action required to set up live session.	The application records only the permitted and relevant interaction.
3	VitaMate System	Validates session state, duration, device permission, and duplicate-counting rules.	Invalid transitions or unavailable capabilities are handled explicitly.
4	VitaMate System	Updates StepLog, live-session state, or ActivityLog according to the authoritative flow.	Steps and workouts do not create duplicate activity or reward effects.
5	Registered User	Reviews the updated activity result, session state, or reminder setting.	Displayed progress may exceed the target numerically while visual progress remains well-formed.

US-08-08 — Live Session Timer

Epic	EP-08 — Activity, Workouts, and Steps	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to use live session timer, so that I can track movement and workouts consistently without duplicate counting.		
Traceability	Source Story ID: US-08-08 | Source Epic: EP-08 | Appendix Story 92 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Live Session Timer' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-3	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Activity and initiates live session timer.	The current movement/session context and required device capability are identified.
2	Registered User	Provides permission, selects an exercise, or performs the session action required to use live session timer.	The application records only the permitted and relevant interaction.
3	VitaMate System	Validates session state, duration, device permission, and duplicate-counting rules.	Invalid transitions or unavailable capabilities are handled explicitly.
4	VitaMate System	Updates StepLog, live-session state, or ActivityLog according to the authoritative flow.	Steps and workouts do not create duplicate activity or reward effects.
5	Registered User	Reviews the updated activity result, session state, or reminder setting.	Displayed progress may exceed the target numerically while visual progress remains well-formed.

US-08-09 — Pause and Resume Session

Epic	EP-08 — Activity, Workouts, and Steps	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to pause and resume session, so that I can track movement and workouts consistently without duplicate counting.		
Traceability	Source Story ID: US-08-09 | Source Epic: EP-08 | Appendix Story 93 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Pause and Resume Session' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	The state-changing action shall preserve required historical information and linked consistency according to the source story.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Activity and initiates pause and resume session.	The current movement/session context and required device capability are identified.
2	Registered User	Provides permission, selects an exercise, or performs the session action required to pause and resume session.	The application records only the permitted and relevant interaction.
3	VitaMate System	Validates session state, duration, device permission, and duplicate-counting rules.	Invalid transitions or unavailable capabilities are handled explicitly.
4	VitaMate System	Updates StepLog, live-session state, or ActivityLog according to the authoritative flow.	Steps and workouts do not create duplicate activity or reward effects.
5	Registered User	Reviews the updated activity result, session state, or reminder setting.	Displayed progress may exceed the target numerically while visual progress remains well-formed.

US-08-10 — Edit Live Session

Epic	EP-08 — Activity, Workouts, and Steps	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to edit live session, so that I can track movement and workouts consistently without duplicate counting.		
Traceability	Source Story ID: US-08-10 | Source Epic: EP-08 | Appendix Story 94 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Edit Live Session' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	The prohibited record or side effect identified by the source story shall not be created.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Activity and initiates edit live session.	The current movement/session context and required device capability are identified.
2	Registered User	Provides permission, selects an exercise, or performs the session action required to edit live session.	The application records only the permitted and relevant interaction.
3	VitaMate System	Validates session state, duration, device permission, and duplicate-counting rules.	Invalid transitions or unavailable capabilities are handled explicitly.
4	VitaMate System	Updates StepLog, live-session state, or ActivityLog according to the authoritative flow.	Steps and workouts do not create duplicate activity or reward effects.
5	Registered User	Reviews the updated activity result, session state, or reminder setting.	Displayed progress may exceed the target numerically while visual progress remains well-formed.

US-08-11 — Complete Full or Partial Session

Epic	EP-08 — Activity, Workouts, and Steps	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to finish a live activity session and record the actual completed work, so that I can track movement and workouts consistently without duplicate counting.		
Traceability	Source Story ID: US-08-11 | Source Epic: EP-08 | Appendix Story 95 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Complete Full or Partial Session' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Duplicate records, rewards, plans, and side effects shall be prevented using idempotent or uniqueness-aware behavior.
AC-3	Any motivation effect shall follow the defined point, mission, streak, badge, or level rules and shall not be duplicated.
AC-4	The related domain record shall remain consistent with linked trackers and downstream summaries.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Activity and initiates complete full or partial session.	The current movement/session context and required device capability are identified.
2	Registered User	Provides permission, selects an exercise, or performs the session action required to finish a live activity session and record the actual completed work.	The application records only the permitted and relevant interaction.
3	VitaMate System	Validates session state, duration, device permission, and duplicate-counting rules.	Invalid transitions or unavailable capabilities are handled explicitly.
4	VitaMate System	Updates StepLog, live-session state, or ActivityLog according to the authoritative flow.	Steps and workouts do not create duplicate activity or reward effects.
5	Registered User	Reviews the updated activity result, session state, or reminder setting.	Displayed progress may exceed the target numerically while visual progress remains well-formed.

US-08-12 — Cancel Session

Epic	EP-08 — Activity, Workouts, and Steps	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to cancel session, so that I can track movement and workouts consistently without duplicate counting.		
Traceability	Source Story ID: US-08-12 | Source Epic: EP-08 | Appendix Story 96 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Cancel Session' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any motivation effect shall follow the defined point, mission, streak, badge, or level rules and shall not be duplicated.
AC-3	The related domain record shall remain consistent with linked trackers and downstream summaries.
AC-4	The prohibited record or side effect identified by the source story shall not be created.
AC-5	The prohibited reward identified by the source story shall not be granted.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Activity and initiates cancel session.	The current movement/session context and required device capability are identified.
2	Registered User	Provides permission, selects an exercise, or performs the session action required to cancel session.	The application records only the permitted and relevant interaction.
3	VitaMate System	Validates session state, duration, device permission, and duplicate-counting rules.	Invalid transitions or unavailable capabilities are handled explicitly.
4	VitaMate System	Updates StepLog, live-session state, or ActivityLog according to the authoritative flow.	Steps and workouts do not create duplicate activity or reward effects.
5	Registered User	Reviews the updated activity result, session state, or reminder setting.	Displayed progress may exceed the target numerically while visual progress remains well-formed.

US-08-13 — Activity and Inactivity Reminders

Epic	EP-08 — Activity, Workouts, and Steps	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to manage activity and inactivity reminders, so that I can track movement and workouts consistently without duplicate counting.		
Traceability	Source Story ID: US-08-13 | Source Epic: EP-08 | Appendix Story 97 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Activity and Inactivity Reminders' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Notification behavior shall respect configured quiet hours except where a validated critical-priority rule allows an override.
AC-3	Notification-related changes shall remain synchronized with current preferences, schedule, priority, and device state.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Activity and initiates activity and inactivity reminders.	The current movement/session context and required device capability are identified.
2	Registered User	Provides permission, selects an exercise, or performs the session action required to manage activity and inactivity reminders.	The application records only the permitted and relevant interaction.
3	VitaMate System	Validates session state, duration, device permission, and duplicate-counting rules.	Invalid transitions or unavailable capabilities are handled explicitly.
4	VitaMate System	Updates StepLog, live-session state, or ActivityLog according to the authoritative flow.	Steps and workouts do not create duplicate activity or reward effects.
5	Registered User	Reviews the updated activity result, session state, or reminder setting.	Displayed progress may exceed the target numerically while visual progress remains well-formed.

B.12 EP-09 — Sleep and Sleep Coach

Sleep recording and planning that uses behavior and caffeine context as explanatory factors rather than diagnostic conclusions.

Epic Attribute	Academic Description
Scope	Sleep recording and planning that uses behavior and caffeine context as explanatory factors rather than diagnostic conclusions.
Story Count	9
Baseline Rule 1	Sleep duration is derived from start and end times and supports intervals crossing midnight.
Baseline Rule 2	Late caffeine and habit information may explain a sleep-plan suggestion but is not used as a diagnosis.
Baseline Rule 3	Morning feedback completes the sleep plan and creates or updates the corresponding SleepLog consistently.

US-09-01 — Log Sleep

Epic	EP-09 — Sleep and Sleep Coach	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to log sleep, so that I can monitor sleep and follow a practical sleep routine.		
Traceability	Source Story ID: US-09-01 | Source Epic: EP-09 | Appendix Story 98 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Log Sleep' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Relevant inputs and preconditions shall be validated before the operation is accepted.
AC-3	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-4	Time-sensitive values shall preserve the validated UTC/local-date or timezone behavior.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Sleep and selects log sleep.	Current sleep target, plan, history, and relevant context are loaded.
2	Registered User	Provides or reviews the information required to log sleep.	Sleep times, wake target, feedback, or plan action are captured.
3	VitaMate System	Validates time ordering, midnight crossing, and applicable plan state.	A coherent sleep interval or plan is derived.
4	VitaMate System	Updates SleepLog, plan state, contextual factors, and notification scheduling as required.	Late caffeine or habits are explanatory factors rather than diagnostic conclusions.
5	Registered User	Reviews the resulting duration, plan, feedback state, or history.	The displayed result uses recorded data and does not invent missing days.

US-09-02 — Sleep Summary

Epic	EP-09 — Sleep and Sleep Coach	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view sleep summary, so that I can monitor sleep and follow a practical sleep routine.		
Traceability	Source Story ID: US-09-02 | Source Epic: EP-09 | Appendix Story 99 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Sleep Summary' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-3	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Sleep and selects sleep summary.	Current sleep target, plan, history, and relevant context are loaded.
2	Registered User	Provides or reviews the information required to view sleep summary.	Sleep times, wake target, feedback, or plan action are captured.
3	VitaMate System	Validates time ordering, midnight crossing, and applicable plan state.	A coherent sleep interval or plan is derived.
4	VitaMate System	Updates SleepLog, plan state, contextual factors, and notification scheduling as required.	Late caffeine or habits are explanatory factors rather than diagnostic conclusions.
5	Registered User	Reviews the resulting duration, plan, feedback state, or history.	The displayed result uses recorded data and does not invent missing days.

US-09-03 — Set Sleep Goal and Wake Time

Epic	EP-09 — Sleep and Sleep Coach	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to set sleep goal and wake time, so that I can monitor sleep and follow a practical sleep routine.		
Traceability	Source Story ID: US-09-03 | Source Epic: EP-09 | Appendix Story 100 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Set Sleep Goal and Wake Time' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Sleep and selects set sleep goal and wake time.	Current sleep target, plan, history, and relevant context are loaded.
2	Registered User	Provides or reviews the information required to set sleep goal and wake time.	Sleep times, wake target, feedback, or plan action are captured.
3	VitaMate System	Validates time ordering, midnight crossing, and applicable plan state.	A coherent sleep interval or plan is derived.
4	VitaMate System	Updates SleepLog, plan state, contextual factors, and notification scheduling as required.	Late caffeine or habits are explanatory factors rather than diagnostic conclusions.
5	Registered User	Reviews the resulting duration, plan, feedback state, or history.	The displayed result uses recorded data and does not invent missing days.

US-09-04 — Create Sleep Plan

Epic	EP-09 — Sleep and Sleep Coach	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to create sleep plan, so that I can monitor sleep and follow a practical sleep routine.		
Traceability	Source Story ID: US-09-04 | Source Epic: EP-09 | Appendix Story 101 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Create Sleep Plan' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Sleep and selects create sleep plan.	Current sleep target, plan, history, and relevant context are loaded.
2	Registered User	Provides or reviews the information required to create sleep plan.	Sleep times, wake target, feedback, or plan action are captured.
3	VitaMate System	Validates time ordering, midnight crossing, and applicable plan state.	A coherent sleep interval or plan is derived.
4	VitaMate System	Updates SleepLog, plan state, contextual factors, and notification scheduling as required.	Late caffeine or habits are explanatory factors rather than diagnostic conclusions.
5	Registered User	Reviews the resulting duration, plan, feedback state, or history.	The displayed result uses recorded data and does not invent missing days.

US-09-05 — Explain Contributing Factors

Epic	EP-09 — Sleep and Sleep Coach	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to use explain contributing factors, so that I can monitor sleep and follow a practical sleep routine.		
Traceability	Source Story ID: US-09-05 | Source Epic: EP-09 | Appendix Story 102 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Explain Contributing Factors' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-3	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Sleep and selects explain contributing factors.	Current sleep target, plan, history, and relevant context are loaded.
2	Registered User	Provides or reviews the information required to use explain contributing factors.	Sleep times, wake target, feedback, or plan action are captured.
3	VitaMate System	Validates time ordering, midnight crossing, and applicable plan state.	A coherent sleep interval or plan is derived.
4	VitaMate System	Updates SleepLog, plan state, contextual factors, and notification scheduling as required.	Late caffeine or habits are explanatory factors rather than diagnostic conclusions.
5	Registered User	Reviews the resulting duration, plan, feedback state, or history.	The displayed result uses recorded data and does not invent missing days.

US-09-06 — Morning Feedback

Epic	EP-09 — Sleep and Sleep Coach	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to use morning feedback, so that I can monitor sleep and follow a practical sleep routine.		
Traceability	Source Story ID: US-09-06 | Source Epic: EP-09 | Appendix Story 103 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Morning Feedback' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-4	The related domain record shall remain consistent with linked trackers and downstream summaries.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Sleep and selects morning feedback.	Current sleep target, plan, history, and relevant context are loaded.
2	Registered User	Provides or reviews the information required to use morning feedback.	Sleep times, wake target, feedback, or plan action are captured.
3	VitaMate System	Validates time ordering, midnight crossing, and applicable plan state.	A coherent sleep interval or plan is derived.
4	VitaMate System	Updates SleepLog, plan state, contextual factors, and notification scheduling as required.	Late caffeine or habits are explanatory factors rather than diagnostic conclusions.
5	Registered User	Reviews the resulting duration, plan, feedback state, or history.	The displayed result uses recorded data and does not invent missing days.

US-09-07 — Cancel Sleep Plan

Epic	EP-09 — Sleep and Sleep Coach	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to cancel sleep plan, so that I can monitor sleep and follow a practical sleep routine.		
Traceability	Source Story ID: US-09-07 | Source Epic: EP-09 | Appendix Story 104 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Cancel Sleep Plan' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The prohibited record or side effect identified by the source story shall not be created.
AC-3	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-4	The state-changing action shall preserve required historical information and linked consistency according to the source story.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Sleep and selects cancel sleep plan.	Current sleep target, plan, history, and relevant context are loaded.
2	Registered User	Provides or reviews the information required to cancel sleep plan.	Sleep times, wake target, feedback, or plan action are captured.
3	VitaMate System	Validates time ordering, midnight crossing, and applicable plan state.	A coherent sleep interval or plan is derived.
4	VitaMate System	Updates SleepLog, plan state, contextual factors, and notification scheduling as required.	Late caffeine or habits are explanatory factors rather than diagnostic conclusions.
5	Registered User	Reviews the resulting duration, plan, feedback state, or history.	The displayed result uses recorded data and does not invent missing days.

US-09-08 — Sleep History

Epic	EP-09 — Sleep and Sleep Coach	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view sleep history, so that I can monitor sleep and follow a practical sleep routine.		
Traceability	Source Story ID: US-09-08 | Source Epic: EP-09 | Appendix Story 105 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Sleep History' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Sleep and selects sleep history.	Current sleep target, plan, history, and relevant context are loaded.
2	Registered User	Provides or reviews the information required to view sleep history.	Sleep times, wake target, feedback, or plan action are captured.
3	VitaMate System	Validates time ordering, midnight crossing, and applicable plan state.	A coherent sleep interval or plan is derived.
4	VitaMate System	Updates SleepLog, plan state, contextual factors, and notification scheduling as required.	Late caffeine or habits are explanatory factors rather than diagnostic conclusions.
5	Registered User	Reviews the resulting duration, plan, feedback state, or history.	The displayed result uses recorded data and does not invent missing days.

US-09-09 — Sleep Reminders

Epic	EP-09 — Sleep and Sleep Coach	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to manage sleep reminders, so that I can monitor sleep and follow a practical sleep routine.		
Traceability	Source Story ID: US-09-09 | Source Epic: EP-09 | Appendix Story 106 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Sleep Reminders' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-3	Required device capabilities or permissions shall be checked, and unavailable dependencies shall be communicated clearly.
AC-4	Notification-related changes shall remain synchronized with current preferences, schedule, priority, and device state.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Sleep and selects sleep reminders.	Current sleep target, plan, history, and relevant context are loaded.
2	Registered User	Provides or reviews the information required to manage sleep reminders.	Sleep times, wake target, feedback, or plan action are captured.
3	VitaMate System	Validates time ordering, midnight crossing, and applicable plan state.	A coherent sleep interval or plan is derived.
4	VitaMate System	Updates SleepLog, plan state, contextual factors, and notification scheduling as required.	Late caffeine or habits are explanatory factors rather than diagnostic conclusions.
5	Registered User	Reviews the resulting duration, plan, feedback state, or history.	The displayed result uses recorded data and does not invent missing days.

B.13 EP-10 — Medication and Adherence

A unified medication-plan model for manual medications, condition-linked medications, and supplements, with actionable dose states.

Epic Attribute	Academic Description
Scope	A unified medication-plan model for manual medications, condition-linked medications, and supplements, with actionable dose states.
Story Count	15
Baseline Rule 1	ConditionMedication is the authoritative plan model for all supported medication sources.
Baseline Rule 2	A late dose remains actionable and is recorded as taken late with the applicable partial reward.
Baseline Rule 3	Taken, missed, and skipped are terminal dose states; snoozed is temporary and overdue remains actionable.

US-10-01 — Medication Overview

Epic	EP-10 — Medication and Adherence	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view medication overview, so that I can manage medication schedules and adherence safely.		
Traceability	Source Story ID: US-10-01 | Source Epic: EP-10 | Appendix Story 107 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Medication Overview' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-3	State transitions involving delay or overdue status shall remain explicit and traceable.
AC-4	The requested information shall be presented from current authorized data without fabricating unavailable values.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Medication and initiates medication overview.	The authoritative medication plan and current dose state are loaded.
2	Registered User	Provides or selects the information required to view medication overview.	The application captures schedule, dose action, or medication details as applicable.
3	VitaMate System	Validates the medication plan, timing, ownership, and current dose transition.	Invalid or duplicate dose transitions are rejected.
4	VitaMate System	Updates the unified medication plan, concrete dose state, adherence summary, motivation effect, and notification plan when applicable.	Dose status remains traceable and consistent.
5	Registered User	Receives the updated dose or adherence result.	Late, snoozed, skipped, missed, PRN, and taken states remain distinguishable.

US-10-02 — Add Medication

Epic	EP-10 — Medication and Adherence	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to add medication, so that I can manage medication schedules and adherence safely.		
Traceability	Source Story ID: US-10-02 | Source Epic: EP-10 | Appendix Story 108 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Add Medication' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Relevant inputs and preconditions shall be validated before the operation is accepted.
AC-3	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-4	When the automated path cannot provide a valid result, the documented manual fallback shall remain available or mandatory as specified.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Medication and initiates add medication.	The authoritative medication plan and current dose state are loaded.
2	Registered User	Provides or selects the information required to add medication.	The application captures schedule, dose action, or medication details as applicable.
3	VitaMate System	Validates the medication plan, timing, ownership, and current dose transition.	Invalid or duplicate dose transitions are rejected.
4	VitaMate System	Updates the unified medication plan, concrete dose state, adherence summary, motivation effect, and notification plan when applicable.	Dose status remains traceable and consistent.
5	Registered User	Receives the updated dose or adherence result.	Late, snoozed, skipped, missed, PRN, and taken states remain distinguishable.

US-10-03 — Set Daily Schedule

Epic	EP-10 — Medication and Adherence	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to set daily schedule, so that I can manage medication schedules and adherence safely.		
Traceability	Source Story ID: US-10-03 | Source Epic: EP-10 | Appendix Story 109 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Set Daily Schedule' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Relevant inputs and preconditions shall be validated before the operation is accepted.
AC-3	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-4	Duplicate records, rewards, plans, and side effects shall be prevented using idempotent or uniqueness-aware behavior.
AC-5	Time-sensitive values shall preserve the validated UTC/local-date or timezone behavior.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Medication and initiates set daily schedule.	The authoritative medication plan and current dose state are loaded.
2	Registered User	Provides or selects the information required to set daily schedule.	The application captures schedule, dose action, or medication details as applicable.
3	VitaMate System	Validates the medication plan, timing, ownership, and current dose transition.	Invalid or duplicate dose transitions are rejected.
4	VitaMate System	Updates the unified medication plan, concrete dose state, adherence summary, motivation effect, and notification plan when applicable.	Dose status remains traceable and consistent.
5	Registered User	Receives the updated dose or adherence result.	Late, snoozed, skipped, missed, PRN, and taken states remain distinguishable.

US-10-04 — Select Specific Weekdays

Epic	EP-10 — Medication and Adherence	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to select specific weekdays, so that I can manage medication schedules and adherence safely.		
Traceability	Source Story ID: US-10-04 | Source Epic: EP-10 | Appendix Story 110 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Select Specific Weekdays' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Medication and initiates select specific weekdays.	The authoritative medication plan and current dose state are loaded.
2	Registered User	Provides or selects the information required to select specific weekdays.	The application captures schedule, dose action, or medication details as applicable.
3	VitaMate System	Validates the medication plan, timing, ownership, and current dose transition.	Invalid or duplicate dose transitions are rejected.
4	VitaMate System	Updates the unified medication plan, concrete dose state, adherence summary, motivation effect, and notification plan when applicable.	Dose status remains traceable and consistent.
5	Registered User	Receives the updated dose or adherence result.	Late, snoozed, skipped, missed, PRN, and taken states remain distinguishable.

US-10-05 — Schedule Every N Hours

Epic	EP-10 — Medication and Adherence	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to schedule every n hours, so that I can manage medication schedules and adherence safely.		
Traceability	Source Story ID: US-10-05 | Source Epic: EP-10 | Appendix Story 111 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Schedule Every N Hours' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Relevant inputs and preconditions shall be validated before the operation is accepted.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Medication and initiates schedule every n hours.	The authoritative medication plan and current dose state are loaded.
2	Registered User	Provides or selects the information required to schedule every n hours.	The application captures schedule, dose action, or medication details as applicable.
3	VitaMate System	Validates the medication plan, timing, ownership, and current dose transition.	Invalid or duplicate dose transitions are rejected.
4	VitaMate System	Updates the unified medication plan, concrete dose state, adherence summary, motivation effect, and notification plan when applicable.	Dose status remains traceable and consistent.
5	Registered User	Receives the updated dose or adherence result.	Late, snoozed, skipped, missed, PRN, and taken states remain distinguishable.

US-10-06 — PRN Medication

Epic	EP-10 — Medication and Adherence	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to manage medication taken on an as-needed basis without a fixed daily schedule, so that I can manage medication schedules and adherence safely.		
Traceability	Source Story ID: US-10-06 | Source Epic: EP-10 | Appendix Story 112 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'PRN Medication' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-3	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Medication and initiates prn medication.	The authoritative medication plan and current dose state are loaded.
2	Registered User	Provides or selects the information required to manage medication taken on an as-needed basis without a fixed daily schedule.	The application captures schedule, dose action, or medication details as applicable.
3	VitaMate System	Validates the medication plan, timing, ownership, and current dose transition.	Invalid or duplicate dose transitions are rejected.
4	VitaMate System	Updates the unified medication plan, concrete dose state, adherence summary, motivation effect, and notification plan when applicable.	Dose status remains traceable and consistent.
5	Registered User	Receives the updated dose or adherence result.	Late, snoozed, skipped, missed, PRN, and taken states remain distinguishable.

US-10-07 — Today's Dose Plan

Epic	EP-10 — Medication and Adherence	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to use today's dose plan, so that I can manage medication schedules and adherence safely.		
Traceability	Source Story ID: US-10-07 | Source Epic: EP-10 | Appendix Story 113 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Today's Dose Plan' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Owner-scoped authorization shall prevent cross-user access to protected records.
AC-3	State transitions involving delay or overdue status shall remain explicit and traceable.
AC-4	The requested information shall be presented from current authorized data without fabricating unavailable values.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Medication and initiates today's dose plan.	The authoritative medication plan and current dose state are loaded.
2	Registered User	Provides or selects the information required to use today's dose plan.	The application captures schedule, dose action, or medication details as applicable.
3	VitaMate System	Validates the medication plan, timing, ownership, and current dose transition.	Invalid or duplicate dose transitions are rejected.
4	VitaMate System	Updates the unified medication plan, concrete dose state, adherence summary, motivation effect, and notification plan when applicable.	Dose status remains traceable and consistent.
5	Registered User	Receives the updated dose or adherence result.	Late, snoozed, skipped, missed, PRN, and taken states remain distinguishable.

US-10-08 — Take Dose On Time

Epic	EP-10 — Medication and Adherence	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to take dose on time, so that I can manage medication schedules and adherence safely.		
Traceability	Source Story ID: US-10-08 | Source Epic: EP-10 | Appendix Story 114 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Take Dose On Time' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-4	Duplicate records, rewards, plans, and side effects shall be prevented using idempotent or uniqueness-aware behavior.
AC-5	Any motivation effect shall follow the defined point, mission, streak, badge, or level rules and shall not be duplicated.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Medication and initiates take dose on time.	The authoritative medication plan and current dose state are loaded.
2	Registered User	Provides or selects the information required to take dose on time.	The application captures schedule, dose action, or medication details as applicable.
3	VitaMate System	Validates the medication plan, timing, ownership, and current dose transition.	Invalid or duplicate dose transitions are rejected.
4	VitaMate System	Updates the unified medication plan, concrete dose state, adherence summary, motivation effect, and notification plan when applicable.	Dose status remains traceable and consistent.
5	Registered User	Receives the updated dose or adherence result.	Late, snoozed, skipped, missed, PRN, and taken states remain distinguishable.

US-10-09 — Take Dose Late

Epic	EP-10 — Medication and Adherence	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to record a dose that is taken after its planned time, so that I can manage medication schedules and adherence safely.		
Traceability	Source Story ID: US-10-09 | Source Epic: EP-10 | Appendix Story 115 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Take Dose Late' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	Any motivation effect shall follow the defined point, mission, streak, badge, or level rules and shall not be duplicated.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Medication and initiates take dose late.	The authoritative medication plan and current dose state are loaded.
2	Registered User	Provides or selects the information required to record a dose that is taken after its planned time.	The application captures schedule, dose action, or medication details as applicable.
3	VitaMate System	Validates the medication plan, timing, ownership, and current dose transition.	Invalid or duplicate dose transitions are rejected.
4	VitaMate System	Updates the unified medication plan, concrete dose state, adherence summary, motivation effect, and notification plan when applicable.	Dose status remains traceable and consistent.
5	Registered User	Receives the updated dose or adherence result.	Late, snoozed, skipped, missed, PRN, and taken states remain distinguishable.

US-10-10 — Snooze Dose

Epic	EP-10 — Medication and Adherence	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to snooze dose, so that I can manage medication schedules and adherence safely.		
Traceability	Source Story ID: US-10-10 | Source Epic: EP-10 | Appendix Story 116 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Snooze Dose' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	State transitions involving delay or overdue status shall remain explicit and traceable.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Medication and initiates snooze dose.	The authoritative medication plan and current dose state are loaded.
2	Registered User	Provides or selects the information required to snooze dose.	The application captures schedule, dose action, or medication details as applicable.
3	VitaMate System	Validates the medication plan, timing, ownership, and current dose transition.	Invalid or duplicate dose transitions are rejected.
4	VitaMate System	Updates the unified medication plan, concrete dose state, adherence summary, motivation effect, and notification plan when applicable.	Dose status remains traceable and consistent.
5	Registered User	Receives the updated dose or adherence result.	Late, snoozed, skipped, missed, PRN, and taken states remain distinguishable.

US-10-11 — Skip or Miss Dose

Epic	EP-10 — Medication and Adherence	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to record a dose as skipped or missed, so that I can manage medication schedules and adherence safely.		
Traceability	Source Story ID: US-10-11 | Source Epic: EP-10 | Appendix Story 117 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Skip or Miss Dose' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-3	Any motivation effect shall follow the defined point, mission, streak, badge, or level rules and shall not be duplicated.
AC-4	The prohibited reward identified by the source story shall not be granted.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Medication and initiates skip or miss dose.	The authoritative medication plan and current dose state are loaded.
2	Registered User	Provides or selects the information required to record a dose as skipped or missed.	The application captures schedule, dose action, or medication details as applicable.
3	VitaMate System	Validates the medication plan, timing, ownership, and current dose transition.	Invalid or duplicate dose transitions are rejected.
4	VitaMate System	Updates the unified medication plan, concrete dose state, adherence summary, motivation effect, and notification plan when applicable.	Dose status remains traceable and consistent.
5	Registered User	Receives the updated dose or adherence result.	Late, snoozed, skipped, missed, PRN, and taken states remain distinguishable.

US-10-12 — Dose Success Feedback

Epic	EP-10 — Medication and Adherence	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to use dose success feedback, so that I can manage medication schedules and adherence safely.		
Traceability	Source Story ID: US-10-12 | Source Epic: EP-10 | Appendix Story 118 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Dose Success Feedback' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-3	Any motivation effect shall follow the defined point, mission, streak, badge, or level rules and shall not be duplicated.
AC-4	The interface shall preserve readable layout and shall not obscure or overflow required content.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Medication and initiates dose success feedback.	The authoritative medication plan and current dose state are loaded.
2	Registered User	Provides or selects the information required to use dose success feedback.	The application captures schedule, dose action, or medication details as applicable.
3	VitaMate System	Validates the medication plan, timing, ownership, and current dose transition.	Invalid or duplicate dose transitions are rejected.
4	VitaMate System	Updates the unified medication plan, concrete dose state, adherence summary, motivation effect, and notification plan when applicable.	Dose status remains traceable and consistent.
5	Registered User	Receives the updated dose or adherence result.	Late, snoozed, skipped, missed, PRN, and taken states remain distinguishable.

US-10-13 — Dose History

Epic	EP-10 — Medication and Adherence	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view dose history, so that I can manage medication schedules and adherence safely.		
Traceability	Source Story ID: US-10-13 | Source Epic: EP-10 | Appendix Story 119 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Dose History' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Medication and initiates dose history.	The authoritative medication plan and current dose state are loaded.
2	Registered User	Provides or selects the information required to view dose history.	The application captures schedule, dose action, or medication details as applicable.
3	VitaMate System	Validates the medication plan, timing, ownership, and current dose transition.	Invalid or duplicate dose transitions are rejected.
4	VitaMate System	Updates the unified medication plan, concrete dose state, adherence summary, motivation effect, and notification plan when applicable.	Dose status remains traceable and consistent.
5	Registered User	Receives the updated dose or adherence result.	Late, snoozed, skipped, missed, PRN, and taken states remain distinguishable.

US-10-14 — Disable Medication Plan

Epic	EP-10 — Medication and Adherence	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to disable medication plan, so that I can manage medication schedules and adherence safely.		
Traceability	Source Story ID: US-10-14 | Source Epic: EP-10 | Appendix Story 120 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Disable Medication Plan' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-4	The state-changing action shall preserve required historical information and linked consistency according to the source story.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Medication and initiates disable medication plan.	The authoritative medication plan and current dose state are loaded.
2	Registered User	Provides or selects the information required to disable medication plan.	The application captures schedule, dose action, or medication details as applicable.
3	VitaMate System	Validates the medication plan, timing, ownership, and current dose transition.	Invalid or duplicate dose transitions are rejected.
4	VitaMate System	Updates the unified medication plan, concrete dose state, adherence summary, motivation effect, and notification plan when applicable.	Dose status remains traceable and consistent.
5	Registered User	Receives the updated dose or adherence result.	Late, snoozed, skipped, missed, PRN, and taken states remain distinguishable.

US-10-15 — Link Medication to Condition or Supplement

Epic	EP-10 — Medication and Adherence	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to link medication to condition or supplement, so that I can manage medication schedules and adherence safely.		
Traceability	Source Story ID: US-10-15 | Source Epic: EP-10 | Appendix Story 121 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Link Medication to Condition or Supplement' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	The prohibited record or side effect identified by the source story shall not be created.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Medication and initiates link medication to condition or supplement.	The authoritative medication plan and current dose state are loaded.
2	Registered User	Provides or selects the information required to link medication to condition or supplement.	The application captures schedule, dose action, or medication details as applicable.
3	VitaMate System	Validates the medication plan, timing, ownership, and current dose transition.	Invalid or duplicate dose transitions are rejected.
4	VitaMate System	Updates the unified medication plan, concrete dose state, adherence summary, motivation effect, and notification plan when applicable.	Dose status remains traceable and consistent.
5	Registered User	Receives the updated dose or adherence result.	Late, snoozed, skipped, missed, PRN, and taken states remain distinguishable.

B.14 EP-11 — Unhealthy Habits and Bidirectional Integration

Reduction plans for smoking, caffeine, and fast food using explicit check-ins and cross-tracker projections without duplicate logging.

Epic Attribute	Academic Description
Scope	Reduction plans for smoking, caffeine, and fast food using explicit check-ins and cross-tracker projections without duplicate logging.
Story Count	11
Baseline Rule 1	Absence of a record is not treated as success; a successful day requires an explicit check-in or confirmed evidence.
Baseline Rule 2	Coffee and fast-food records may be projected from hydration or nutrition without duplicate entry.
Baseline Rule 3	Exceeding the active limit records a relapse state without an arbitrary default point penalty.

US-11-01 — Habit Cards

Epic	EP-11 — Unhealthy Habits and Bidirectional Integration	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view habit cards, so that I can monitor and reduce unhealthy habits using confirmed behavior data.		
Traceability	Source Story ID: US-11-01 | Source Epic: EP-11 | Appendix Story 122 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Habit Cards' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The visible result shall remain consistent with the selected locale, supported directionality, and accessibility semantics.
AC-3	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Unhealthy Habits and selects habit cards.	The active habit plan, baseline, and current-day evidence are loaded.
2	Registered User	Provides or reviews the information required to view habit cards.	The system captures direct check-in data or uses an eligible linked tracker event.
3	VitaMate System	Validates the source event and prevents duplicate projection from nutrition or hydration.	Missing data alone is not interpreted as successful abstinence.
4	VitaMate System	Updates habit progress, relapse/on-track state, streak, motivation, and reminder state as required.	The result reflects confirmed behavior rather than assumptions.
5	Registered User	Reviews the current habit status or related sleep context.	Cross-domain effects remain synchronized without double entry.

US-11-02 — Set Up Personalized Reduction Plan

Epic	EP-11 — Unhealthy Habits and Bidirectional Integration	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to set up personalized reduction plan, so that I can monitor and reduce unhealthy habits using confirmed behavior data.		
Traceability	Source Story ID: US-11-02 | Source Epic: EP-11 | Appendix Story 123 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Set Up Personalized Reduction Plan' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Unhealthy Habits and selects set up personalized reduction plan.	The active habit plan, baseline, and current-day evidence are loaded.
2	Registered User	Provides or reviews the information required to set up personalized reduction plan.	The system captures direct check-in data or uses an eligible linked tracker event.
3	VitaMate System	Validates the source event and prevents duplicate projection from nutrition or hydration.	Missing data alone is not interpreted as successful abstinence.
4	VitaMate System	Updates habit progress, relapse/on-track state, streak, motivation, and reminder state as required.	The result reflects confirmed behavior rather than assumptions.
5	Registered User	Reviews the current habit status or related sleep context.	Cross-domain effects remain synchronized without double entry.

US-11-03 — Log Habit Occurrence

Epic	EP-11 — Unhealthy Habits and Bidirectional Integration	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to log habit occurrence, so that I can monitor and reduce unhealthy habits using confirmed behavior data.		
Traceability	Source Story ID: US-11-03 | Source Epic: EP-11 | Appendix Story 124 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Log Habit Occurrence' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-4	Duplicate records, rewards, plans, and side effects shall be prevented using idempotent or uniqueness-aware behavior.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Unhealthy Habits and selects log habit occurrence.	The active habit plan, baseline, and current-day evidence are loaded.
2	Registered User	Provides or reviews the information required to log habit occurrence.	The system captures direct check-in data or uses an eligible linked tracker event.
3	VitaMate System	Validates the source event and prevents duplicate projection from nutrition or hydration.	Missing data alone is not interpreted as successful abstinence.
4	VitaMate System	Updates habit progress, relapse/on-track state, streak, motivation, and reminder state as required.	The result reflects confirmed behavior rather than assumptions.
5	Registered User	Reviews the current habit status or related sleep context.	Cross-domain effects remain synchronized without double entry.

US-11-04 — Confirm Abstinence

Epic	EP-11 — Unhealthy Habits and Bidirectional Integration	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to confirm abstinence, so that I can monitor and reduce unhealthy habits using confirmed behavior data.		
Traceability	Source Story ID: US-11-04 | Source Epic: EP-11 | Appendix Story 125 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Confirm Abstinence' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-3	Any motivation effect shall follow the defined point, mission, streak, badge, or level rules and shall not be duplicated.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Unhealthy Habits and selects confirm abstinence.	The active habit plan, baseline, and current-day evidence are loaded.
2	Registered User	Provides or reviews the information required to confirm abstinence.	The system captures direct check-in data or uses an eligible linked tracker event.
3	VitaMate System	Validates the source event and prevents duplicate projection from nutrition or hydration.	Missing data alone is not interpreted as successful abstinence.
4	VitaMate System	Updates habit progress, relapse/on-track state, streak, motivation, and reminder state as required.	The result reflects confirmed behavior rather than assumptions.
5	Registered User	Reviews the current habit status or related sleep context.	Cross-domain effects remain synchronized without double entry.

US-11-05 — Project Fast-Food Entry

Epic	EP-11 — Unhealthy Habits and Bidirectional Integration	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to project a fast-food nutrition entry into the unhealthy-habit tracker, so that I can monitor and reduce unhealthy habits using confirmed behavior data.		
Traceability	Source Story ID: US-11-05 | Source Epic: EP-11 | Appendix Story 126 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Project Fast-Food Entry' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Duplicate records, rewards, plans, and side effects shall be prevented using idempotent or uniqueness-aware behavior.
AC-3	Any motivation effect shall follow the defined point, mission, streak, badge, or level rules and shall not be duplicated.
AC-4	The prohibited reward identified by the source story shall not be granted.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Unhealthy Habits and selects project fast-food entry.	The active habit plan, baseline, and current-day evidence are loaded.
2	Registered User	Provides or reviews the information required to project a fast-food nutrition entry into the unhealthy-habit tracker.	The system captures direct check-in data or uses an eligible linked tracker event.
3	VitaMate System	Validates the source event and prevents duplicate projection from nutrition or hydration.	Missing data alone is not interpreted as successful abstinence.
4	VitaMate System	Updates habit progress, relapse/on-track state, streak, motivation, and reminder state as required.	The result reflects confirmed behavior rather than assumptions.
5	Registered User	Reviews the current habit status or related sleep context.	Cross-domain effects remain synchronized without double entry.

US-11-06 — Project Caffeine Intake

Epic	EP-11 — Unhealthy Habits and Bidirectional Integration	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to project caffeine from relevant beverage or food records into habit tracking, so that I can monitor and reduce unhealthy habits using confirmed behavior data.		
Traceability	Source Story ID: US-11-06 | Source Epic: EP-11 | Appendix Story 127 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Project Caffeine Intake' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The related domain record shall remain consistent with linked trackers and downstream summaries.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Unhealthy Habits and selects project caffeine intake.	The active habit plan, baseline, and current-day evidence are loaded.
2	Registered User	Provides or reviews the information required to project caffeine from relevant beverage or food records into habit tracking.	The system captures direct check-in data or uses an eligible linked tracker event.
3	VitaMate System	Validates the source event and prevents duplicate projection from nutrition or hydration.	Missing data alone is not interpreted as successful abstinence.
4	VitaMate System	Updates habit progress, relapse/on-track state, streak, motivation, and reminder state as required.	The result reflects confirmed behavior rather than assumptions.
5	Registered User	Reviews the current habit status or related sleep context.	Cross-domain effects remain synchronized without double entry.

US-11-07 — Backfill Existing Daily Records

Epic	EP-11 — Unhealthy Habits and Bidirectional Integration	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to backfill eligible records from the current day when a habit plan is first activated, so that I can monitor and reduce unhealthy habits using confirmed behavior data.		
Traceability	Source Story ID: US-11-07 | Source Epic: EP-11 | Appendix Story 128 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Backfill Existing Daily Records' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Duplicate records, rewards, plans, and side effects shall be prevented using idempotent or uniqueness-aware behavior.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Unhealthy Habits and selects backfill existing daily records.	The active habit plan, baseline, and current-day evidence are loaded.
2	Registered User	Provides or reviews the information required to backfill eligible records from the current day when a habit plan is first activated.	The system captures direct check-in data or uses an eligible linked tracker event.
3	VitaMate System	Validates the source event and prevents duplicate projection from nutrition or hydration.	Missing data alone is not interpreted as successful abstinence.
4	VitaMate System	Updates habit progress, relapse/on-track state, streak, motivation, and reminder state as required.	The result reflects confirmed behavior rather than assumptions.
5	Registered User	Reviews the current habit status or related sleep context.	Cross-domain effects remain synchronized without double entry.

US-11-08 — Habit Progress

Epic	EP-11 — Unhealthy Habits and Bidirectional Integration	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view habit progress, so that I can monitor and reduce unhealthy habits using confirmed behavior data.		
Traceability	Source Story ID: US-11-08 | Source Epic: EP-11 | Appendix Story 129 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Habit Progress' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Unhealthy Habits and selects habit progress.	The active habit plan, baseline, and current-day evidence are loaded.
2	Registered User	Provides or reviews the information required to view habit progress.	The system captures direct check-in data or uses an eligible linked tracker event.
3	VitaMate System	Validates the source event and prevents duplicate projection from nutrition or hydration.	Missing data alone is not interpreted as successful abstinence.
4	VitaMate System	Updates habit progress, relapse/on-track state, streak, motivation, and reminder state as required.	The result reflects confirmed behavior rather than assumptions.
5	Registered User	Reviews the current habit status or related sleep context.	Cross-domain effects remain synchronized without double entry.

US-11-09 — Pause and Resume Habit Plan

Epic	EP-11 — Unhealthy Habits and Bidirectional Integration	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to pause and resume habit plan, so that I can monitor and reduce unhealthy habits using confirmed behavior data.		
Traceability	Source Story ID: US-11-09 | Source Epic: EP-11 | Appendix Story 130 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Pause and Resume Habit Plan' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-4	The state-changing action shall preserve required historical information and linked consistency according to the source story.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Unhealthy Habits and selects pause and resume habit plan.	The active habit plan, baseline, and current-day evidence are loaded.
2	Registered User	Provides or reviews the information required to pause and resume habit plan.	The system captures direct check-in data or uses an eligible linked tracker event.
3	VitaMate System	Validates the source event and prevents duplicate projection from nutrition or hydration.	Missing data alone is not interpreted as successful abstinence.
4	VitaMate System	Updates habit progress, relapse/on-track state, streak, motivation, and reminder state as required.	The result reflects confirmed behavior rather than assumptions.
5	Registered User	Reviews the current habit status or related sleep context.	Cross-domain effects remain synchronized without double entry.

US-11-10 — Integrate with Sleep Coach

Epic	EP-11 — Unhealthy Habits and Bidirectional Integration	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to integrate with sleep coach, so that I can monitor and reduce unhealthy habits using confirmed behavior data.		
Traceability	Source Story ID: US-11-10 | Source Epic: EP-11 | Appendix Story 131 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Integrate with Sleep Coach' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Duplicate records, rewards, plans, and side effects shall be prevented using idempotent or uniqueness-aware behavior.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Unhealthy Habits and selects integrate with sleep coach.	The active habit plan, baseline, and current-day evidence are loaded.
2	Registered User	Provides or reviews the information required to integrate with sleep coach.	The system captures direct check-in data or uses an eligible linked tracker event.
3	VitaMate System	Validates the source event and prevents duplicate projection from nutrition or hydration.	Missing data alone is not interpreted as successful abstinence.
4	VitaMate System	Updates habit progress, relapse/on-track state, streak, motivation, and reminder state as required.	The result reflects confirmed behavior rather than assumptions.
5	Registered User	Reviews the current habit status or related sleep context.	Cross-domain effects remain synchronized without double entry.

US-11-11 — Habit Reminder

Epic	EP-11 — Unhealthy Habits and Bidirectional Integration	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to manage habit reminder, so that I can monitor and reduce unhealthy habits using confirmed behavior data.		
Traceability	Source Story ID: US-11-11 | Source Epic: EP-11 | Appendix Story 132 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Habit Reminder' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Notification-related changes shall remain synchronized with current preferences, schedule, priority, and device state.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Unhealthy Habits and selects habit reminder.	The active habit plan, baseline, and current-day evidence are loaded.
2	Registered User	Provides or reviews the information required to manage habit reminder.	The system captures direct check-in data or uses an eligible linked tracker event.
3	VitaMate System	Validates the source event and prevents duplicate projection from nutrition or hydration.	Missing data alone is not interpreted as successful abstinence.
4	VitaMate System	Updates habit progress, relapse/on-track state, streak, motivation, and reminder state as required.	The result reflects confirmed behavior rather than assumptions.
5	Registered User	Reviews the current habit status or related sleep context.	Cross-domain effects remain synchronized without double entry.

B.15 EP-12 — Chronic Conditions, Readings, and Health Constraints

Management of supported chronic conditions, readings, goals, alerts, and constraints that safely influence related trackers.

Epic Attribute	Academic Description
Scope	Management of supported chronic conditions, readings, goals, alerts, and constraints that safely influence related trackers.
Story Count	14
Baseline Rule 1	Currently supported conditions are diabetes, hypertension, and hyperlipidemia.
Baseline Rule 2	The constraints engine resolves profile, manual-goal, chronic-condition, and micronutrient rules using the safer applicable value.
Baseline Rule 3	Health text remains supportive and non-diagnostic; a critical alert is not generated from moderate guidance alone.

US-12-01 — Supported Conditions Catalog

Epic	EP-12 — Chronic Conditions, Readings, and Health Constraints	**Primary Actor**	User with a Chronic Condition
Priority	High	**Implementation Status**	Implemented
User Story	As a user with a chronic condition, I want to view supported conditions catalog, so that I can manage chronic-condition information and health constraints safely.		
Traceability	Source Story ID: US-12-01 | Source Epic: EP-12 | Appendix Story 133 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Supported Conditions Catalog' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The visible result shall remain consistent with the selected locale, supported directionality, and accessibility semantics.
AC-3	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	User with a Chronic Condition	Opens the chronic-condition area and selects supported conditions catalog.	The supported condition, reading, goal, medication, and constraint context is loaded.
2	User with a Chronic Condition	Provides or reviews the condition information required to view supported conditions catalog.	The application captures only supported condition types and valid reading fields.
3	VitaMate System	Validates units, ownership, duplication, and current reading context.	Unsupported or unauthorized condition data are rejected.
4	VitaMate System	Persists the valid condition/read information and recomputes applicable constraints and unified health state.	Related nutrition, hydration, activity, alert, or medication state reflects the effective rules.
5	User with a Chronic Condition	Reviews the resulting condition status, goals, guidance, history, or alert.	Health information remains supportive, current, owner-scoped, and non-diagnostic.

US-12-02 — Add Chronic Condition

Epic	EP-12 — Chronic Conditions, Readings, and Health Constraints	**Primary Actor**	User with a Chronic Condition
Priority	High	**Implementation Status**	Implemented
User Story	As a user with a chronic condition, I want to add chronic condition, so that I can manage chronic-condition information and health constraints safely.		
Traceability	Source Story ID: US-12-02 | Source Epic: EP-12 | Appendix Story 134 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Add Chronic Condition' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-3	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	User with a Chronic Condition	Opens the chronic-condition area and selects add chronic condition.	The supported condition, reading, goal, medication, and constraint context is loaded.
2	User with a Chronic Condition	Provides or reviews the condition information required to add chronic condition.	The application captures only supported condition types and valid reading fields.
3	VitaMate System	Validates units, ownership, duplication, and current reading context.	Unsupported or unauthorized condition data are rejected.
4	VitaMate System	Persists the valid condition/read information and recomputes applicable constraints and unified health state.	Related nutrition, hydration, activity, alert, or medication state reflects the effective rules.
5	User with a Chronic Condition	Reviews the resulting condition status, goals, guidance, history, or alert.	Health information remains supportive, current, owner-scoped, and non-diagnostic.

US-12-03 — Edit or Disable Condition

Epic	EP-12 — Chronic Conditions, Readings, and Health Constraints	**Primary Actor**	User with a Chronic Condition
Priority	High	**Implementation Status**	Implemented
User Story	As a user with a chronic condition, I want to edit or disable condition, so that I can manage chronic-condition information and health constraints safely.		
Traceability	Source Story ID: US-12-03 | Source Epic: EP-12 | Appendix Story 135 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Edit or Disable Condition' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	The state-changing action shall preserve required historical information and linked consistency according to the source story.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	User with a Chronic Condition	Opens the chronic-condition area and selects edit or disable condition.	The supported condition, reading, goal, medication, and constraint context is loaded.
2	User with a Chronic Condition	Provides or reviews the condition information required to edit or disable condition.	The application captures only supported condition types and valid reading fields.
3	VitaMate System	Validates units, ownership, duplication, and current reading context.	Unsupported or unauthorized condition data are rejected.
4	VitaMate System	Persists the valid condition/read information and recomputes applicable constraints and unified health state.	Related nutrition, hydration, activity, alert, or medication state reflects the effective rules.
5	User with a Chronic Condition	Reviews the resulting condition status, goals, guidance, history, or alert.	Health information remains supportive, current, owner-scoped, and non-diagnostic.

US-12-04 — Record Glucose Reading

Epic	EP-12 — Chronic Conditions, Readings, and Health Constraints	**Primary Actor**	User with a Chronic Condition
Priority	High	**Implementation Status**	Implemented
User Story	As a user with a chronic condition, I want to record glucose reading, so that I can manage chronic-condition information and health constraints safely.		
Traceability	Source Story ID: US-12-04 | Source Epic: EP-12 | Appendix Story 136 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Record Glucose Reading' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Relevant inputs and preconditions shall be validated before the operation is accepted.
AC-3	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-4	Compared and displayed values shall use the applicable units consistently.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	User with a Chronic Condition	Opens the chronic-condition area and selects record glucose reading.	The supported condition, reading, goal, medication, and constraint context is loaded.
2	User with a Chronic Condition	Provides or reviews the condition information required to record glucose reading.	The application captures only supported condition types and valid reading fields.
3	VitaMate System	Validates units, ownership, duplication, and current reading context.	Unsupported or unauthorized condition data are rejected.
4	VitaMate System	Persists the valid condition/read information and recomputes applicable constraints and unified health state.	Related nutrition, hydration, activity, alert, or medication state reflects the effective rules.
5	User with a Chronic Condition	Reviews the resulting condition status, goals, guidance, history, or alert.	Health information remains supportive, current, owner-scoped, and non-diagnostic.

US-12-05 — Record Blood Pressure

Epic	EP-12 — Chronic Conditions, Readings, and Health Constraints	**Primary Actor**	User with a Chronic Condition
Priority	High	**Implementation Status**	Implemented
User Story	As a user with a chronic condition, I want to record blood pressure, so that I can manage chronic-condition information and health constraints safely.		
Traceability	Source Story ID: US-12-05 | Source Epic: EP-12 | Appendix Story 137 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Record Blood Pressure' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	User with a Chronic Condition	Opens the chronic-condition area and selects record blood pressure.	The supported condition, reading, goal, medication, and constraint context is loaded.
2	User with a Chronic Condition	Provides or reviews the condition information required to record blood pressure.	The application captures only supported condition types and valid reading fields.
3	VitaMate System	Validates units, ownership, duplication, and current reading context.	Unsupported or unauthorized condition data are rejected.
4	VitaMate System	Persists the valid condition/read information and recomputes applicable constraints and unified health state.	Related nutrition, hydration, activity, alert, or medication state reflects the effective rules.
5	User with a Chronic Condition	Reviews the resulting condition status, goals, guidance, history, or alert.	Health information remains supportive, current, owner-scoped, and non-diagnostic.

US-12-06 — Track Lipid Readings

Epic	EP-12 — Chronic Conditions, Readings, and Health Constraints	**Primary Actor**	User with a Chronic Condition
Priority	High	**Implementation Status**	Implemented
User Story	As a user with a chronic condition, I want to track lipid readings, so that I can manage chronic-condition information and health constraints safely.		
Traceability	Source Story ID: US-12-06 | Source Epic: EP-12 | Appendix Story 138 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Track Lipid Readings' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	User with a Chronic Condition	Opens the chronic-condition area and selects track lipid readings.	The supported condition, reading, goal, medication, and constraint context is loaded.
2	User with a Chronic Condition	Provides or reviews the condition information required to track lipid readings.	The application captures only supported condition types and valid reading fields.
3	VitaMate System	Validates units, ownership, duplication, and current reading context.	Unsupported or unauthorized condition data are rejected.
4	VitaMate System	Persists the valid condition/read information and recomputes applicable constraints and unified health state.	Related nutrition, hydration, activity, alert, or medication state reflects the effective rules.
5	User with a Chronic Condition	Reviews the resulting condition status, goals, guidance, history, or alert.	Health information remains supportive, current, owner-scoped, and non-diagnostic.

US-12-07 — Condition Details

Epic	EP-12 — Chronic Conditions, Readings, and Health Constraints	**Primary Actor**	User with a Chronic Condition
Priority	High	**Implementation Status**	Implemented
User Story	As a user with a chronic condition, I want to view condition details, so that I can manage chronic-condition information and health constraints safely.		
Traceability	Source Story ID: US-12-07 | Source Epic: EP-12 | Appendix Story 139 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Condition Details' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	User with a Chronic Condition	Opens the chronic-condition area and selects condition details.	The supported condition, reading, goal, medication, and constraint context is loaded.
2	User with a Chronic Condition	Provides or reviews the condition information required to view condition details.	The application captures only supported condition types and valid reading fields.
3	VitaMate System	Validates units, ownership, duplication, and current reading context.	Unsupported or unauthorized condition data are rejected.
4	VitaMate System	Persists the valid condition/read information and recomputes applicable constraints and unified health state.	Related nutrition, hydration, activity, alert, or medication state reflects the effective rules.
5	User with a Chronic Condition	Reviews the resulting condition status, goals, guidance, history, or alert.	Health information remains supportive, current, owner-scoped, and non-diagnostic.

US-12-08 — Goals and Limits

Epic	EP-12 — Chronic Conditions, Readings, and Health Constraints	**Primary Actor**	User with a Chronic Condition
Priority	High	**Implementation Status**	Implemented
User Story	As a user with a chronic condition, I want to view goals and limits, so that I can manage chronic-condition information and health constraints safely.		
Traceability	Source Story ID: US-12-08 | Source Epic: EP-12 | Appendix Story 140 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Goals and Limits' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Compared and displayed values shall use the applicable units consistently.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	User with a Chronic Condition	Opens the chronic-condition area and selects goals and limits.	The supported condition, reading, goal, medication, and constraint context is loaded.
2	User with a Chronic Condition	Provides or reviews the condition information required to view goals and limits.	The application captures only supported condition types and valid reading fields.
3	VitaMate System	Validates units, ownership, duplication, and current reading context.	Unsupported or unauthorized condition data are rejected.
4	VitaMate System	Persists the valid condition/read information and recomputes applicable constraints and unified health state.	Related nutrition, hydration, activity, alert, or medication state reflects the effective rules.
5	User with a Chronic Condition	Reviews the resulting condition status, goals, guidance, history, or alert.	Health information remains supportive, current, owner-scoped, and non-diagnostic.

US-12-09 — Health Guidance

Epic	EP-12 — Chronic Conditions, Readings, and Health Constraints	**Primary Actor**	User with a Chronic Condition
Priority	High	**Implementation Status**	Implemented
User Story	As a user with a chronic condition, I want to view localized health guidance linked to the relevant reason, so that I can manage chronic-condition information and health constraints safely.		
Traceability	Source Story ID: US-12-09 | Source Epic: EP-12 | Appendix Story 141 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Health Guidance' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The visible result shall remain consistent with the selected locale, supported directionality, and accessibility semantics.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	User with a Chronic Condition	Opens the chronic-condition area and selects health guidance.	The supported condition, reading, goal, medication, and constraint context is loaded.
2	User with a Chronic Condition	Provides or reviews the condition information required to view localized health guidance linked to the relevant reason.	The application captures only supported condition types and valid reading fields.
3	VitaMate System	Validates units, ownership, duplication, and current reading context.	Unsupported or unauthorized condition data are rejected.
4	VitaMate System	Persists the valid condition/read information and recomputes applicable constraints and unified health state.	Related nutrition, hydration, activity, alert, or medication state reflects the effective rules.
5	User with a Chronic Condition	Reviews the resulting condition status, goals, guidance, history, or alert.	Health information remains supportive, current, owner-scoped, and non-diagnostic.

US-12-10 — Health Alerts

Epic	EP-12 — Chronic Conditions, Readings, and Health Constraints	**Primary Actor**	User with a Chronic Condition
Priority	High	**Implementation Status**	Implemented
User Story	As a user with a chronic condition, I want to receive an alert when a current risk indicator requires attention, so that I can manage chronic-condition information and health constraints safely.		
Traceability	Source Story ID: US-12-10 | Source Epic: EP-12 | Appendix Story 142 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Health Alerts' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Relevant inputs and preconditions shall be validated before the operation is accepted.
AC-3	Duplicate records, rewards, plans, and side effects shall be prevented using idempotent or uniqueness-aware behavior.
AC-4	Navigation shall resolve to the intended route or detail view while preserving valid session context.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	User with a Chronic Condition	Opens the chronic-condition area and selects health alerts.	The supported condition, reading, goal, medication, and constraint context is loaded.
2	User with a Chronic Condition	Provides or reviews the condition information required to receive an alert when a current risk indicator requires attention.	The application captures only supported condition types and valid reading fields.
3	VitaMate System	Validates units, ownership, duplication, and current reading context.	Unsupported or unauthorized condition data are rejected.
4	VitaMate System	Persists the valid condition/read information and recomputes applicable constraints and unified health state.	Related nutrition, hydration, activity, alert, or medication state reflects the effective rules.
5	User with a Chronic Condition	Reviews the resulting condition status, goals, guidance, history, or alert.	Health information remains supportive, current, owner-scoped, and non-diagnostic.

US-12-11 — Condition-Linked Medication

Epic	EP-12 — Chronic Conditions, Readings, and Health Constraints	**Primary Actor**	User with a Chronic Condition
Priority	High	**Implementation Status**	Implemented
User Story	As a user with a chronic condition, I want to use condition-linked medication, so that I can manage chronic-condition information and health constraints safely.		
Traceability	Source Story ID: US-12-11 | Source Epic: EP-12 | Appendix Story 143 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Condition-Linked Medication' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-3	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	User with a Chronic Condition	Opens the chronic-condition area and selects condition-linked medication.	The supported condition, reading, goal, medication, and constraint context is loaded.
2	User with a Chronic Condition	Provides or reviews the condition information required to use condition-linked medication.	The application captures only supported condition types and valid reading fields.
3	VitaMate System	Validates units, ownership, duplication, and current reading context.	Unsupported or unauthorized condition data are rejected.
4	VitaMate System	Persists the valid condition/read information and recomputes applicable constraints and unified health state.	Related nutrition, hydration, activity, alert, or medication state reflects the effective rules.
5	User with a Chronic Condition	Reviews the resulting condition status, goals, guidance, history, or alert.	Health information remains supportive, current, owner-scoped, and non-diagnostic.

US-12-12 — Condition Effect on Trackers

Epic	EP-12 — Chronic Conditions, Readings, and Health Constraints	**Primary Actor**	User with a Chronic Condition
Priority	High	**Implementation Status**	Implemented
User Story	As a user with a chronic condition, I want to see tracker targets adjusted by active health constraints, so that I can manage chronic-condition information and health constraints safely.		
Traceability	Source Story ID: US-12-12 | Source Epic: EP-12 | Appendix Story 144 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Condition Effect on Trackers' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	User with a Chronic Condition	Opens the chronic-condition area and selects condition effect on trackers.	The supported condition, reading, goal, medication, and constraint context is loaded.
2	User with a Chronic Condition	Provides or reviews the condition information required to see tracker targets adjusted by active health constraints.	The application captures only supported condition types and valid reading fields.
3	VitaMate System	Validates units, ownership, duplication, and current reading context.	Unsupported or unauthorized condition data are rejected.
4	VitaMate System	Persists the valid condition/read information and recomputes applicable constraints and unified health state.	Related nutrition, hydration, activity, alert, or medication state reflects the effective rules.
5	User with a Chronic Condition	Reviews the resulting condition status, goals, guidance, history, or alert.	Health information remains supportive, current, owner-scoped, and non-diagnostic.

US-12-13 — Condition History and Audit Trail

Epic	EP-12 — Chronic Conditions, Readings, and Health Constraints	**Primary Actor**	User with a Chronic Condition
Priority	High	**Implementation Status**	Implemented
User Story	As a user with a chronic condition, I want to view condition history and audit trail, so that I can manage chronic-condition information and health constraints safely.		
Traceability	Source Story ID: US-12-13 | Source Epic: EP-12 | Appendix Story 145 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Condition History and Audit Trail' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Owner-scoped authorization shall prevent cross-user access to protected records.
AC-3	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	User with a Chronic Condition	Opens the chronic-condition area and selects condition history and audit trail.	The supported condition, reading, goal, medication, and constraint context is loaded.
2	User with a Chronic Condition	Provides or reviews the condition information required to view condition history and audit trail.	The application captures only supported condition types and valid reading fields.
3	VitaMate System	Validates units, ownership, duplication, and current reading context.	Unsupported or unauthorized condition data are rejected.
4	VitaMate System	Persists the valid condition/read information and recomputes applicable constraints and unified health state.	Related nutrition, hydration, activity, alert, or medication state reflects the effective rules.
5	User with a Chronic Condition	Reviews the resulting condition status, goals, guidance, history, or alert.	Health information remains supportive, current, owner-scoped, and non-diagnostic.

US-12-14 — Ownership Protection

Epic	EP-12 — Chronic Conditions, Readings, and Health Constraints	**Primary Actor**	User with a Chronic Condition
Priority	High	**Implementation Status**	Implemented
User Story	As a user with a chronic condition, I want to protect chronic-condition records from cross-user access, so that I can manage chronic-condition information and health constraints safely.		
Traceability	Source Story ID: US-12-14 | Source Epic: EP-12 | Appendix Story 146 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Ownership Protection' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Owner-scoped authorization shall prevent cross-user access to protected records.
AC-3	Any motivation effect shall follow the defined point, mission, streak, badge, or level rules and shall not be duplicated.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	User with a Chronic Condition	Opens the chronic-condition area and selects ownership protection.	The supported condition, reading, goal, medication, and constraint context is loaded.
2	User with a Chronic Condition	Provides or reviews the condition information required to protect chronic-condition records from cross-user access.	The application captures only supported condition types and valid reading fields.
3	VitaMate System	Validates units, ownership, duplication, and current reading context.	Unsupported or unauthorized condition data are rejected.
4	VitaMate System	Persists the valid condition/read information and recomputes applicable constraints and unified health state.	Related nutrition, hydration, activity, alert, or medication state reflects the effective rules.
5	User with a Chronic Condition	Reviews the resulting condition status, goals, guidance, history, or alert.	Health information remains supportive, current, owner-scoped, and non-diagnostic.

B.16 EP-13 — Unified Health State, Progress, and History

A single consistent daily health-state view across relevant trackers and constraints, separate from motivational XP.

Epic Attribute	Academic Description
Scope	A single consistent daily health-state view across relevant trackers and constraints, separate from motivational XP.
Story Count	10
Baseline Rule 1	UnifiedHealthState is a materialized read model recomputed after successful write transactions.
Baseline Rule 2	Daily health progress is separate from XP and requires genuine nutrition entries when nutrition is applicable.
Baseline Rule 3	Activity completion follows the actual effective requirements and does not automatically treat step count as a complete workout.

US-13-01 — Unified Daily Progress

Epic	EP-13 — Unified Health State, Progress, and History	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view unified daily progress, so that I can review a consistent daily health state across applicable trackers.		
Traceability	Source Story ID: US-13-01 | Source Epic: EP-13 | Appendix Story 147 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Unified Daily Progress' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any motivation effect shall follow the defined point, mission, streak, badge, or level rules and shall not be duplicated.
AC-3	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Home, Progress, History, or the relevant tracker to use unified daily progress.	The materialized UnifiedHealthState and supporting snapshots are loaded.
2	VitaMate System	Determines which trackers and constraints are applicable for the requested day.	Not-applicable trackers are excluded from the daily-progress denominator.
3	VitaMate System	Applies the calculation or presentation rule required for unified daily progress.	Nutrition, activity, and other tracker requirements use the same authoritative rules across views.
4	Registered User	Reviews the resulting daily or historical health state.	Health progress is clearly separated from motivation XP and points.
5	VitaMate System	Maintains read-only behavior unless a controlled missing-snapshot bootstrap is required.	Viewing progress does not award points or create missions.

US-13-02 — Require Nutrition Completion

Epic	EP-13 — Unified Health State, Progress, and History	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to require genuine nutrition entries before daily health progress can be considered complete, so that I can review a consistent daily health state across applicable trackers.		
Traceability	Source Story ID: US-13-02 | Source Epic: EP-13 | Appendix Story 148 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Require Nutrition Completion' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Explicitly excluded records shall not contribute to the affected metric, completion rule, or mission.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Home, Progress, History, or the relevant tracker to use require nutrition completion.	The materialized UnifiedHealthState and supporting snapshots are loaded.
2	VitaMate System	Determines which trackers and constraints are applicable for the requested day.	Not-applicable trackers are excluded from the daily-progress denominator.
3	VitaMate System	Applies the calculation or presentation rule required for require nutrition completion.	Nutrition, activity, and other tracker requirements use the same authoritative rules across views.
4	Registered User	Reviews the resulting daily or historical health state.	Health progress is clearly separated from motivation XP and points.
5	VitaMate System	Maintains read-only behavior unless a controlled missing-snapshot bootstrap is required.	Viewing progress does not award points or create missions.

US-13-03 — Not-Applicable Tracker Handling

Epic	EP-13 — Unified Health State, Progress, and History	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to exclude non-applicable trackers from the daily progress calculation, so that I can review a consistent daily health state across applicable trackers.		
Traceability	Source Story ID: US-13-03 | Source Epic: EP-13 | Appendix Story 149 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Not-Applicable Tracker Handling' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-3	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Home, Progress, History, or the relevant tracker to use not-applicable tracker handling.	The materialized UnifiedHealthState and supporting snapshots are loaded.
2	VitaMate System	Determines which trackers and constraints are applicable for the requested day.	Not-applicable trackers are excluded from the daily-progress denominator.
3	VitaMate System	Applies the calculation or presentation rule required for not-applicable tracker handling.	Nutrition, activity, and other tracker requirements use the same authoritative rules across views.
4	Registered User	Reviews the resulting daily or historical health state.	Health progress is clearly separated from motivation XP and points.
5	VitaMate System	Maintains read-only behavior unless a controlled missing-snapshot bootstrap is required.	Viewing progress does not award points or create missions.

US-13-04 — Resolve Constraint Conflict

Epic	EP-13 — Unified Health State, Progress, and History	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to receive the safer effective target when multiple constraints conflict, so that I can review a consistent daily health state across applicable trackers.		
Traceability	Source Story ID: US-13-04 | Source Epic: EP-13 | Appendix Story 150 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Resolve Constraint Conflict' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	Compared and displayed values shall use the applicable units consistently.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Home, Progress, History, or the relevant tracker to use resolve constraint conflict.	The materialized UnifiedHealthState and supporting snapshots are loaded.
2	VitaMate System	Determines which trackers and constraints are applicable for the requested day.	Not-applicable trackers are excluded from the daily-progress denominator.
3	VitaMate System	Applies the calculation or presentation rule required for resolve constraint conflict.	Nutrition, activity, and other tracker requirements use the same authoritative rules across views.
4	Registered User	Reviews the resulting daily or historical health state.	Health progress is clearly separated from motivation XP and points.
5	VitaMate System	Maintains read-only behavior unless a controlled missing-snapshot bootstrap is required.	Viewing progress does not award points or create missions.

US-13-05 — Constraint Validity

Epic	EP-13 — Unified Health State, Progress, and History	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to apply only health constraints that are valid for the relevant time period, so that I can review a consistent daily health state across applicable trackers.		
Traceability	Source Story ID: US-13-05 | Source Epic: EP-13 | Appendix Story 151 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Constraint Validity' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-3	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Home, Progress, History, or the relevant tracker to use constraint validity.	The materialized UnifiedHealthState and supporting snapshots are loaded.
2	VitaMate System	Determines which trackers and constraints are applicable for the requested day.	Not-applicable trackers are excluded from the daily-progress denominator.
3	VitaMate System	Applies the calculation or presentation rule required for constraint validity.	Nutrition, activity, and other tracker requirements use the same authoritative rules across views.
4	Registered User	Reviews the resulting daily or historical health state.	Health progress is clearly separated from motivation XP and points.
5	VitaMate System	Maintains read-only behavior unless a controlled missing-snapshot bootstrap is required.	Viewing progress does not award points or create missions.

US-13-06 — Progress Overview

Epic	EP-13 — Unified Health State, Progress, and History	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view progress overview, so that I can review a consistent daily health state across applicable trackers.		
Traceability	Source Story ID: US-13-06 | Source Epic: EP-13 | Appendix Story 152 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Progress Overview' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	Navigation shall resolve to the intended route or detail view while preserving valid session context.
AC-4	The requested information shall be presented from current authorized data without fabricating unavailable values.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Home, Progress, History, or the relevant tracker to use progress overview.	The materialized UnifiedHealthState and supporting snapshots are loaded.
2	VitaMate System	Determines which trackers and constraints are applicable for the requested day.	Not-applicable trackers are excluded from the daily-progress denominator.
3	VitaMate System	Applies the calculation or presentation rule required for progress overview.	Nutrition, activity, and other tracker requirements use the same authoritative rules across views.
4	Registered User	Reviews the resulting daily or historical health state.	Health progress is clearly separated from motivation XP and points.
5	VitaMate System	Maintains read-only behavior unless a controlled missing-snapshot bootstrap is required.	Viewing progress does not award points or create missions.

US-13-07 — Tracker Details

Epic	EP-13 — Unified Health State, Progress, and History	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view tracker details, so that I can review a consistent daily health state across applicable trackers.		
Traceability	Source Story ID: US-13-07 | Source Epic: EP-13 | Appendix Story 153 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Tracker Details' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Duplicate records, rewards, plans, and side effects shall be prevented using idempotent or uniqueness-aware behavior.
AC-3	The visible result shall remain consistent with the selected locale, supported directionality, and accessibility semantics.
AC-4	The requested information shall be presented from current authorized data without fabricating unavailable values.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Home, Progress, History, or the relevant tracker to use tracker details.	The materialized UnifiedHealthState and supporting snapshots are loaded.
2	VitaMate System	Determines which trackers and constraints are applicable for the requested day.	Not-applicable trackers are excluded from the daily-progress denominator.
3	VitaMate System	Applies the calculation or presentation rule required for tracker details.	Nutrition, activity, and other tracker requirements use the same authoritative rules across views.
4	Registered User	Reviews the resulting daily or historical health state.	Health progress is clearly separated from motivation XP and points.
5	VitaMate System	Maintains read-only behavior unless a controlled missing-snapshot bootstrap is required.	Viewing progress does not award points or create missions.

US-13-08 — Historical Progress

Epic	EP-13 — Unified Health State, Progress, and History	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view historical progress, so that I can review a consistent daily health state across applicable trackers.		
Traceability	Source Story ID: US-13-08 | Source Epic: EP-13 | Appendix Story 154 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Historical Progress' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Home, Progress, History, or the relevant tracker to use historical progress.	The materialized UnifiedHealthState and supporting snapshots are loaded.
2	VitaMate System	Determines which trackers and constraints are applicable for the requested day.	Not-applicable trackers are excluded from the daily-progress denominator.
3	VitaMate System	Applies the calculation or presentation rule required for historical progress.	Nutrition, activity, and other tracker requirements use the same authoritative rules across views.
4	Registered User	Reviews the resulting daily or historical health state.	Health progress is clearly separated from motivation XP and points.
5	VitaMate System	Maintains read-only behavior unless a controlled missing-snapshot bootstrap is required.	Viewing progress does not award points or create missions.

US-13-09 — Post-Write Consistency

Epic	EP-13 — Unified Health State, Progress, and History	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to see updated health state after a successful write operation, so that I can review a consistent daily health state across applicable trackers.		
Traceability	Source Story ID: US-13-09 | Source Epic: EP-13 | Appendix Story 155 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Post-Write Consistency' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Home, Progress, History, or the relevant tracker to use post-write consistency.	The materialized UnifiedHealthState and supporting snapshots are loaded.
2	VitaMate System	Determines which trackers and constraints are applicable for the requested day.	Not-applicable trackers are excluded from the daily-progress denominator.
3	VitaMate System	Applies the calculation or presentation rule required for post-write consistency.	Nutrition, activity, and other tracker requirements use the same authoritative rules across views.
4	Registered User	Reviews the resulting daily or historical health state.	Health progress is clearly separated from motivation XP and points.
5	VitaMate System	Maintains read-only behavior unless a controlled missing-snapshot bootstrap is required.	Viewing progress does not award points or create missions.

US-13-10 — Side-Effect-Free Read

Epic	EP-13 — Unified Health State, Progress, and History	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view health information without creating motivation side effects, so that I can review a consistent daily health state across applicable trackers.		
Traceability	Source Story ID: US-13-10 | Source Epic: EP-13 | Appendix Story 156 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Side-Effect-Free Read' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	Any motivation effect shall follow the defined point, mission, streak, badge, or level rules and shall not be duplicated.
AC-4	The prohibited reward identified by the source story shall not be granted.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Opens Home, Progress, History, or the relevant tracker to use side-effect-free read.	The materialized UnifiedHealthState and supporting snapshots are loaded.
2	VitaMate System	Determines which trackers and constraints are applicable for the requested day.	Not-applicable trackers are excluded from the daily-progress denominator.
3	VitaMate System	Applies the calculation or presentation rule required for side-effect-free read.	Nutrition, activity, and other tracker requirements use the same authoritative rules across views.
4	Registered User	Reviews the resulting daily or historical health state.	Health progress is clearly separated from motivation XP and points.
5	VitaMate System	Maintains read-only behavior unless a controlled missing-snapshot bootstrap is required.	Viewing progress does not award points or create missions.

B.17 EP-14 — Points, Missions, Badges, and Motivation

A motivation subsystem that rewards verified user actions through an auditable point ledger, missions, streaks, badges, levels, and contextual feedback.

Epic Attribute	Academic Description
Scope	A motivation subsystem that rewards verified user actions through an auditable point ledger, missions, streaks, badges, levels, and contextual feedback.
Story Count	12
Baseline Rule 1	PointsTransaction is the source of truth for point changes and supported deductions.
Baseline Rule 2	Rewards and achievement events are processed idempotently to avoid duplicate credit.
Baseline Rule 3	Motivational feedback must not override health-critical alerts or alter the health-progress percentage.

US-14-01 — Earn Points

Epic	EP-14 — Points, Missions, Badges, and Motivation	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to earn points, so that I can receive fair motivational feedback for verified health behavior.		
Traceability	Source Story ID: US-14-01 | Source Epic: EP-14 | Appendix Story 157 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Earn Points' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-3	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	VitaMate System	Receives a verified health action or motivation condition relevant to earn points.	The source event is eligible for motivation processing.
2	Motivation Service	Evaluates the point, mission, streak, badge, level, or celebration rule associated with the event.	Only allowed rewards or deductions are considered.
3	Motivation Service	Writes the authoritative point transaction or achievement state using idempotent processing.	The same logical event cannot be rewarded or deducted twice.
4	Registered User	Views or receives the result needed to earn points.	Feedback is proportionate to the achievement and does not override health-critical information.
5	VitaMate System	Updates the Motivation Feed and relevant summary views.	Motivation remains separate from the health-progress percentage.

US-14-02 — Controlled Point Deduction

Epic	EP-14 — Points, Missions, Badges, and Motivation	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to receive point deductions only when an explicit rule allows them, so that I can receive fair motivational feedback for verified health behavior.		
Traceability	Source Story ID: US-14-02 | Source Epic: EP-14 | Appendix Story 158 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Controlled Point Deduction' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	VitaMate System	Receives a verified health action or motivation condition relevant to controlled point deduction.	The source event is eligible for motivation processing.
2	Motivation Service	Evaluates the point, mission, streak, badge, level, or celebration rule associated with the event.	Only allowed rewards or deductions are considered.
3	Motivation Service	Writes the authoritative point transaction or achievement state using idempotent processing.	The same logical event cannot be rewarded or deducted twice.
4	Registered User	Views or receives the result needed to receive point deductions only when an explicit rule allows them.	Feedback is proportionate to the achievement and does not override health-critical information.
5	VitaMate System	Updates the Motivation Feed and relevant summary views.	Motivation remains separate from the health-progress percentage.

US-14-03 — Level Progress

Epic	EP-14 — Points, Missions, Badges, and Motivation	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view level progress, so that I can receive fair motivational feedback for verified health behavior.		
Traceability	Source Story ID: US-14-03 | Source Epic: EP-14 | Appendix Story 159 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Level Progress' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Duplicate records, rewards, plans, and side effects shall be prevented using idempotent or uniqueness-aware behavior.
AC-3	Any motivation effect shall follow the defined point, mission, streak, badge, or level rules and shall not be duplicated.
AC-4	The requested information shall be presented from current authorized data without fabricating unavailable values.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	VitaMate System	Receives a verified health action or motivation condition relevant to level progress.	The source event is eligible for motivation processing.
2	Motivation Service	Evaluates the point, mission, streak, badge, level, or celebration rule associated with the event.	Only allowed rewards or deductions are considered.
3	Motivation Service	Writes the authoritative point transaction or achievement state using idempotent processing.	The same logical event cannot be rewarded or deducted twice.
4	Registered User	Views or receives the result needed to view level progress.	Feedback is proportionate to the achievement and does not override health-critical information.
5	VitaMate System	Updates the Motivation Feed and relevant summary views.	Motivation remains separate from the health-progress percentage.

US-14-04 — Streak

Epic	EP-14 — Points, Missions, Badges, and Motivation	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view streak, so that I can receive fair motivational feedback for verified health behavior.		
Traceability	Source Story ID: US-14-04 | Source Epic: EP-14 | Appendix Story 160 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Streak' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-3	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	VitaMate System	Receives a verified health action or motivation condition relevant to streak.	The source event is eligible for motivation processing.
2	Motivation Service	Evaluates the point, mission, streak, badge, level, or celebration rule associated with the event.	Only allowed rewards or deductions are considered.
3	Motivation Service	Writes the authoritative point transaction or achievement state using idempotent processing.	The same logical event cannot be rewarded or deducted twice.
4	Registered User	Views or receives the result needed to view streak.	Feedback is proportionate to the achievement and does not override health-critical information.
5	VitaMate System	Updates the Motivation Feed and relevant summary views.	Motivation remains separate from the health-progress percentage.

US-14-05 — Daily Missions

Epic	EP-14 — Points, Missions, Badges, and Motivation	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view daily missions, so that I can receive fair motivational feedback for verified health behavior.		
Traceability	Source Story ID: US-14-05 | Source Epic: EP-14 | Appendix Story 161 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Daily Missions' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-3	Duplicate records, rewards, plans, and side effects shall be prevented using idempotent or uniqueness-aware behavior.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	VitaMate System	Receives a verified health action or motivation condition relevant to daily missions.	The source event is eligible for motivation processing.
2	Motivation Service	Evaluates the point, mission, streak, badge, level, or celebration rule associated with the event.	Only allowed rewards or deductions are considered.
3	Motivation Service	Writes the authoritative point transaction or achievement state using idempotent processing.	The same logical event cannot be rewarded or deducted twice.
4	Registered User	Views or receives the result needed to view daily missions.	Feedback is proportionate to the achievement and does not override health-critical information.
5	VitaMate System	Updates the Motivation Feed and relevant summary views.	Motivation remains separate from the health-progress percentage.

US-14-06 — Mission Completion

Epic	EP-14 — Points, Missions, Badges, and Motivation	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to use mission completion, so that I can receive fair motivational feedback for verified health behavior.		
Traceability	Source Story ID: US-14-06 | Source Epic: EP-14 | Appendix Story 162 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Mission Completion' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any motivation effect shall follow the defined point, mission, streak, badge, or level rules and shall not be duplicated.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	VitaMate System	Receives a verified health action or motivation condition relevant to mission completion.	The source event is eligible for motivation processing.
2	Motivation Service	Evaluates the point, mission, streak, badge, level, or celebration rule associated with the event.	Only allowed rewards or deductions are considered.
3	Motivation Service	Writes the authoritative point transaction or achievement state using idempotent processing.	The same logical event cannot be rewarded or deducted twice.
4	Registered User	Views or receives the result needed to use mission completion.	Feedback is proportionate to the achievement and does not override health-critical information.
5	VitaMate System	Updates the Motivation Feed and relevant summary views.	Motivation remains separate from the health-progress percentage.

US-14-07 — Badges

Epic	EP-14 — Points, Missions, Badges, and Motivation	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view badges, so that I can receive fair motivational feedback for verified health behavior.		
Traceability	Source Story ID: US-14-07 | Source Epic: EP-14 | Appendix Story 163 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Badges' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-3	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	VitaMate System	Receives a verified health action or motivation condition relevant to badges.	The source event is eligible for motivation processing.
2	Motivation Service	Evaluates the point, mission, streak, badge, level, or celebration rule associated with the event.	Only allowed rewards or deductions are considered.
3	Motivation Service	Writes the authoritative point transaction or achievement state using idempotent processing.	The same logical event cannot be rewarded or deducted twice.
4	Registered User	Views or receives the result needed to view badges.	Feedback is proportionate to the achievement and does not override health-critical information.
5	VitaMate System	Updates the Motivation Feed and relevant summary views.	Motivation remains separate from the health-progress percentage.

US-14-08 — Motivation Status Bar

Epic	EP-14 — Points, Missions, Badges, and Motivation	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view a compact motivation status bar, so that I can receive fair motivational feedback for verified health behavior.		
Traceability	Source Story ID: US-14-08 | Source Epic: EP-14 | Appendix Story 164 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Motivation Status Bar' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The visible result shall remain consistent with the selected locale, supported directionality, and accessibility semantics.
AC-3	The interface shall preserve readable layout and shall not obscure or overflow required content.
AC-4	Navigation shall resolve to the intended route or detail view while preserving valid session context.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	VitaMate System	Receives a verified health action or motivation condition relevant to motivation status bar.	The source event is eligible for motivation processing.
2	Motivation Service	Evaluates the point, mission, streak, badge, level, or celebration rule associated with the event.	Only allowed rewards or deductions are considered.
3	Motivation Service	Writes the authoritative point transaction or achievement state using idempotent processing.	The same logical event cannot be rewarded or deducted twice.
4	Registered User	Views or receives the result needed to view a compact motivation status bar.	Feedback is proportionate to the achievement and does not override health-critical information.
5	VitaMate System	Updates the Motivation Feed and relevant summary views.	Motivation remains separate from the health-progress percentage.

US-14-09 — Home Motivation Focus

Epic	EP-14 — Points, Missions, Badges, and Motivation	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view one prioritized motivation focus on the Home screen, so that I can receive fair motivational feedback for verified health behavior.		
Traceability	Source Story ID: US-14-09 | Source Epic: EP-14 | Appendix Story 165 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Home Motivation Focus' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Duplicate records, rewards, plans, and side effects shall be prevented using idempotent or uniqueness-aware behavior.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	VitaMate System	Receives a verified health action or motivation condition relevant to home motivation focus.	The source event is eligible for motivation processing.
2	Motivation Service	Evaluates the point, mission, streak, badge, level, or celebration rule associated with the event.	Only allowed rewards or deductions are considered.
3	Motivation Service	Writes the authoritative point transaction or achievement state using idempotent processing.	The same logical event cannot be rewarded or deducted twice.
4	Registered User	Views or receives the result needed to view one prioritized motivation focus on the Home screen.	Feedback is proportionate to the achievement and does not override health-critical information.
5	VitaMate System	Updates the Motivation Feed and relevant summary views.	Motivation remains separate from the health-progress percentage.

US-14-10 — Point Celebration

Epic	EP-14 — Points, Missions, Badges, and Motivation	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to receive a brief point celebration after an eligible reward, so that I can receive fair motivational feedback for verified health behavior.		
Traceability	Source Story ID: US-14-10 | Source Epic: EP-14 | Appendix Story 166 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Point Celebration' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-3	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	VitaMate System	Receives a verified health action or motivation condition relevant to point celebration.	The source event is eligible for motivation processing.
2	Motivation Service	Evaluates the point, mission, streak, badge, level, or celebration rule associated with the event.	Only allowed rewards or deductions are considered.
3	Motivation Service	Writes the authoritative point transaction or achievement state using idempotent processing.	The same logical event cannot be rewarded or deducted twice.
4	Registered User	Views or receives the result needed to receive a brief point celebration after an eligible reward.	Feedback is proportionate to the achievement and does not override health-critical information.
5	VitaMate System	Updates the Motivation Feed and relevant summary views.	Motivation remains separate from the health-progress percentage.

US-14-11 — Major Achievement Celebration

Epic	EP-14 — Points, Missions, Badges, and Motivation	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to receive a stronger celebration for significant achievements, so that I can receive fair motivational feedback for verified health behavior.		
Traceability	Source Story ID: US-14-11 | Source Epic: EP-14 | Appendix Story 167 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Major Achievement Celebration' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-3	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	VitaMate System	Receives a verified health action or motivation condition relevant to major achievement celebration.	The source event is eligible for motivation processing.
2	Motivation Service	Evaluates the point, mission, streak, badge, level, or celebration rule associated with the event.	Only allowed rewards or deductions are considered.
3	Motivation Service	Writes the authoritative point transaction or achievement state using idempotent processing.	The same logical event cannot be rewarded or deducted twice.
4	Registered User	Views or receives the result needed to receive a stronger celebration for significant achievements.	Feedback is proportionate to the achievement and does not override health-critical information.
5	VitaMate System	Updates the Motivation Feed and relevant summary views.	Motivation remains separate from the health-progress percentage.

US-14-12 — Points Ledger

Epic	EP-14 — Points, Missions, Badges, and Motivation	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view points ledger, so that I can receive fair motivational feedback for verified health behavior.		
Traceability	Source Story ID: US-14-12 | Source Epic: EP-14 | Appendix Story 168 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Points Ledger' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The visible result shall remain consistent with the selected locale, supported directionality, and accessibility semantics.
AC-3	The requested information shall be presented from current authorized data without fabricating unavailable values.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	VitaMate System	Receives a verified health action or motivation condition relevant to points ledger.	The source event is eligible for motivation processing.
2	Motivation Service	Evaluates the point, mission, streak, badge, level, or celebration rule associated with the event.	Only allowed rewards or deductions are considered.
3	Motivation Service	Writes the authoritative point transaction or achievement state using idempotent processing.	The same logical event cannot be rewarded or deducted twice.
4	Registered User	Views or receives the result needed to view points ledger.	Feedback is proportionate to the achievement and does not override health-critical information.
5	VitaMate System	Updates the Motivation Feed and relevant summary views.	Motivation remains separate from the health-progress percentage.

B.18 EP-15 — Notification Hub and In-App Alerts

Centralized notification decision-making in Django with synchronized local Android delivery, health priority, quiet hours, and localization.

Epic Attribute	Academic Description
Scope	Centralized notification decision-making in Django with synchronized local Android delivery, health priority, quiet hours, and localization.
Story Count	14
Baseline Rule 1	Django builds the notification plan, Flutter synchronizes it, and Android schedules it locally; FCM is not used in the current version.
Baseline Rule 2	One primary device performs local scheduling to prevent duplicate notifications across devices.
Baseline Rule 3	Priority is health/medication first, then routine, motivation, and celebration; only critical events may bypass quiet hours.

US-15-01 — Register Device

Epic	EP-15 — Notification Hub and In-App Alerts	**Primary Actor**	Android User / VitaMate System
Priority	High	**Implementation Status**	Implemented
User Story	As a VitaMate user, I want to register device, so that I can receive relevant notifications without unnecessary duplication or interruption.		
Traceability	Source Story ID: US-15-01 | Source Epic: EP-15 | Appendix Story 169 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Register Device' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-4	Notification-related changes shall remain synchronized with current preferences, schedule, priority, and device state.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Android User / VitaMate System	Configures or triggers the notification behavior required to register device.	Current device, locale, permission, preference, and scheduling context is available.
2	Django Notification Hub	Compiles or updates the authoritative notification plan using category, priority, quiet-hours, and source-domain rules.	The plan reflects the latest validated backend state.
3	Flutter Client	Synchronizes only the required plan changes to the primary device.	Unchanged entries are preserved and stale entries are removed when appropriate.
4	Android Local Delivery	Schedules, presents, or routes the notification according to local capabilities and foreground/background state.	Health-critical events retain the validated priority behavior.
5	Flutter / Django	Reports delivery or interaction outcome when the story requires reporting.	Acknowledgement and delivery transitions remain idempotent and traceable.

US-15-02 — Select Primary Device

Epic	EP-15 — Notification Hub and In-App Alerts	**Primary Actor**	Android User / VitaMate System
Priority	High	**Implementation Status**	Implemented
User Story	As a VitaMate user, I want to select primary device, so that I can receive relevant notifications without unnecessary duplication or interruption.		
Traceability	Source Story ID: US-15-02 | Source Epic: EP-15 | Appendix Story 170 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Select Primary Device' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Required device capabilities or permissions shall be checked, and unavailable dependencies shall be communicated clearly.
AC-3	Notification-related changes shall remain synchronized with current preferences, schedule, priority, and device state.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Android User / VitaMate System	Configures or triggers the notification behavior required to select primary device.	Current device, locale, permission, preference, and scheduling context is available.
2	Django Notification Hub	Compiles or updates the authoritative notification plan using category, priority, quiet-hours, and source-domain rules.	The plan reflects the latest validated backend state.
3	Flutter Client	Synchronizes only the required plan changes to the primary device.	Unchanged entries are preserved and stale entries are removed when appropriate.
4	Android Local Delivery	Schedules, presents, or routes the notification according to local capabilities and foreground/background state.	Health-critical events retain the validated priority behavior.
5	Flutter / Django	Reports delivery or interaction outcome when the story requires reporting.	Acknowledgement and delivery transitions remain idempotent and traceable.

US-15-03 — Centralized Notification Preferences

Epic	EP-15 — Notification Hub and In-App Alerts	**Primary Actor**	Android User / VitaMate System
Priority	High	**Implementation Status**	Implemented
User Story	As a VitaMate user, I want to use centralized notification preferences, so that I can receive relevant notifications without unnecessary duplication or interruption.		
Traceability	Source Story ID: US-15-03 | Source Epic: EP-15 | Appendix Story 171 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Centralized Notification Preferences' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	Notification-related changes shall remain synchronized with current preferences, schedule, priority, and device state.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Android User / VitaMate System	Configures or triggers the notification behavior required to use centralized notification preferences.	Current device, locale, permission, preference, and scheduling context is available.
2	Django Notification Hub	Compiles or updates the authoritative notification plan using category, priority, quiet-hours, and source-domain rules.	The plan reflects the latest validated backend state.
3	Flutter Client	Synchronizes only the required plan changes to the primary device.	Unchanged entries are preserved and stale entries are removed when appropriate.
4	Android Local Delivery	Schedules, presents, or routes the notification according to local capabilities and foreground/background state.	Health-critical events retain the validated priority behavior.
5	Flutter / Django	Reports delivery or interaction outcome when the story requires reporting.	Acknowledgement and delivery transitions remain idempotent and traceable.

US-15-04 — Quiet Hours

Epic	EP-15 — Notification Hub and In-App Alerts	**Primary Actor**	Android User / VitaMate System
Priority	High	**Implementation Status**	Implemented
User Story	As a VitaMate user, I want to define a quiet period for non-critical notifications, so that I can receive relevant notifications without unnecessary duplication or interruption.		
Traceability	Source Story ID: US-15-04 | Source Epic: EP-15 | Appendix Story 172 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Quiet Hours' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Time-sensitive values shall preserve the validated UTC/local-date or timezone behavior.
AC-3	State transitions involving delay or overdue status shall remain explicit and traceable.
AC-4	Notification-related changes shall remain synchronized with current preferences, schedule, priority, and device state.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Android User / VitaMate System	Configures or triggers the notification behavior required to define a quiet period for non-critical notifications.	Current device, locale, permission, preference, and scheduling context is available.
2	Django Notification Hub	Compiles or updates the authoritative notification plan using category, priority, quiet-hours, and source-domain rules.	The plan reflects the latest validated backend state.
3	Flutter Client	Synchronizes only the required plan changes to the primary device.	Unchanged entries are preserved and stale entries are removed when appropriate.
4	Android Local Delivery	Schedules, presents, or routes the notification according to local capabilities and foreground/background state.	Health-critical events retain the validated priority behavior.
5	Flutter / Django	Reports delivery or interaction outcome when the story requires reporting.	Acknowledgement and delivery transitions remain idempotent and traceable.

US-15-05 — Health and Medication Reminders

Epic	EP-15 — Notification Hub and In-App Alerts	**Primary Actor**	Android User / VitaMate System
Priority	High	**Implementation Status**	Implemented
User Story	As a VitaMate user, I want to manage health and medication reminders, so that I can receive relevant notifications without unnecessary duplication or interruption.		
Traceability	Source Story ID: US-15-05 | Source Epic: EP-15 | Appendix Story 173 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Health and Medication Reminders' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Relevant inputs and preconditions shall be validated before the operation is accepted.
AC-3	Navigation shall resolve to the intended route or detail view while preserving valid session context.
AC-4	Notification-related changes shall remain synchronized with current preferences, schedule, priority, and device state.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Android User / VitaMate System	Configures or triggers the notification behavior required to manage health and medication reminders.	Current device, locale, permission, preference, and scheduling context is available.
2	Django Notification Hub	Compiles or updates the authoritative notification plan using category, priority, quiet-hours, and source-domain rules.	The plan reflects the latest validated backend state.
3	Flutter Client	Synchronizes only the required plan changes to the primary device.	Unchanged entries are preserved and stale entries are removed when appropriate.
4	Android Local Delivery	Schedules, presents, or routes the notification according to local capabilities and foreground/background state.	Health-critical events retain the validated priority behavior.
5	Flutter / Django	Reports delivery or interaction outcome when the story requires reporting.	Acknowledgement and delivery transitions remain idempotent and traceable.

US-15-06 — Routine Reminders

Epic	EP-15 — Notification Hub and In-App Alerts	**Primary Actor**	Android User / VitaMate System
Priority	High	**Implementation Status**	Implemented
User Story	As a VitaMate user, I want to manage routine reminders, so that I can receive relevant notifications without unnecessary duplication or interruption.		
Traceability	Source Story ID: US-15-06 | Source Epic: EP-15 | Appendix Story 174 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Routine Reminders' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Notification-related changes shall remain synchronized with current preferences, schedule, priority, and device state.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Android User / VitaMate System	Configures or triggers the notification behavior required to manage routine reminders.	Current device, locale, permission, preference, and scheduling context is available.
2	Django Notification Hub	Compiles or updates the authoritative notification plan using category, priority, quiet-hours, and source-domain rules.	The plan reflects the latest validated backend state.
3	Flutter Client	Synchronizes only the required plan changes to the primary device.	Unchanged entries are preserved and stale entries are removed when appropriate.
4	Android Local Delivery	Schedules, presents, or routes the notification according to local capabilities and foreground/background state.	Health-critical events retain the validated priority behavior.
5	Flutter / Django	Reports delivery or interaction outcome when the story requires reporting.	Acknowledgement and delivery transitions remain idempotent and traceable.

US-15-07 — Controlled Motivation Notifications

Epic	EP-15 — Notification Hub and In-App Alerts	**Primary Actor**	Android User / VitaMate System
Priority	High	**Implementation Status**	Implemented
User Story	As a VitaMate user, I want to receive motivation reminders at a controlled frequency, so that I can receive relevant notifications without unnecessary duplication or interruption.		
Traceability	Source Story ID: US-15-07 | Source Epic: EP-15 | Appendix Story 175 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Controlled Motivation Notifications' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Notification-related changes shall remain synchronized with current preferences, schedule, priority, and device state.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Android User / VitaMate System	Configures or triggers the notification behavior required to receive motivation reminders at a controlled frequency.	Current device, locale, permission, preference, and scheduling context is available.
2	Django Notification Hub	Compiles or updates the authoritative notification plan using category, priority, quiet-hours, and source-domain rules.	The plan reflects the latest validated backend state.
3	Flutter Client	Synchronizes only the required plan changes to the primary device.	Unchanged entries are preserved and stale entries are removed when appropriate.
4	Android Local Delivery	Schedules, presents, or routes the notification according to local capabilities and foreground/background state.	Health-critical events retain the validated priority behavior.
5	Flutter / Django	Reports delivery or interaction outcome when the story requires reporting.	Acknowledgement and delivery transitions remain idempotent and traceable.

US-15-08 — Foreground Event Presentation

Epic	EP-15 — Notification Hub and In-App Alerts	**Primary Actor**	Android User / VitaMate System
Priority	High	**Implementation Status**	Implemented
User Story	As a VitaMate user, I want to receive appropriate in-app presentation while the application is in the foreground, so that I can receive relevant notifications without unnecessary duplication or interruption.		
Traceability	Source Story ID: US-15-08 | Source Epic: EP-15 | Appendix Story 176 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Foreground Event Presentation' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Notification-related changes shall remain synchronized with current preferences, schedule, priority, and device state.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Android User / VitaMate System	Configures or triggers the notification behavior required to receive appropriate in-app presentation while the application is in the foreground.	Current device, locale, permission, preference, and scheduling context is available.
2	Django Notification Hub	Compiles or updates the authoritative notification plan using category, priority, quiet-hours, and source-domain rules.	The plan reflects the latest validated backend state.
3	Flutter Client	Synchronizes only the required plan changes to the primary device.	Unchanged entries are preserved and stale entries are removed when appropriate.
4	Android Local Delivery	Schedules, presents, or routes the notification according to local capabilities and foreground/background state.	Health-critical events retain the validated priority behavior.
5	Flutter / Django	Reports delivery or interaction outcome when the story requires reporting.	Acknowledgement and delivery transitions remain idempotent and traceable.

US-15-09 — Deep Links

Epic	EP-15 — Notification Hub and In-App Alerts	**Primary Actor**	Android User / VitaMate System
Priority	High	**Implementation Status**	Implemented
User Story	As a VitaMate user, I want to open the correct application destination from a notification, so that I can receive relevant notifications without unnecessary duplication or interruption.		
Traceability	Source Story ID: US-15-09 | Source Epic: EP-15 | Appendix Story 177 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Deep Links' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Navigation shall resolve to the intended route or detail view while preserving valid session context.
AC-3	Notification-related changes shall remain synchronized with current preferences, schedule, priority, and device state.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Android User / VitaMate System	Configures or triggers the notification behavior required to open the correct application destination from a notification.	Current device, locale, permission, preference, and scheduling context is available.
2	Django Notification Hub	Compiles or updates the authoritative notification plan using category, priority, quiet-hours, and source-domain rules.	The plan reflects the latest validated backend state.
3	Flutter Client	Synchronizes only the required plan changes to the primary device.	Unchanged entries are preserved and stale entries are removed when appropriate.
4	Android Local Delivery	Schedules, presents, or routes the notification according to local capabilities and foreground/background state.	Health-critical events retain the validated priority behavior.
5	Flutter / Django	Reports delivery or interaction outcome when the story requires reporting.	Acknowledgement and delivery transitions remain idempotent and traceable.

US-15-10 — Notification Language

Epic	EP-15 — Notification Hub and In-App Alerts	**Primary Actor**	Android User / VitaMate System
Priority	High	**Implementation Status**	Implemented
User Story	As a VitaMate user, I want to receive notifications in the selected application language, so that I can receive relevant notifications without unnecessary duplication or interruption.		
Traceability	Source Story ID: US-15-10 | Source Epic: EP-15 | Appendix Story 178 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Notification Language' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The visible result shall remain consistent with the selected locale, supported directionality, and accessibility semantics.
AC-3	Time-sensitive values shall preserve the validated UTC/local-date or timezone behavior.
AC-4	Notification-related changes shall remain synchronized with current preferences, schedule, priority, and device state.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Android User / VitaMate System	Configures or triggers the notification behavior required to receive notifications in the selected application language.	Current device, locale, permission, preference, and scheduling context is available.
2	Django Notification Hub	Compiles or updates the authoritative notification plan using category, priority, quiet-hours, and source-domain rules.	The plan reflects the latest validated backend state.
3	Flutter Client	Synchronizes only the required plan changes to the primary device.	Unchanged entries are preserved and stale entries are removed when appropriate.
4	Android Local Delivery	Schedules, presents, or routes the notification according to local capabilities and foreground/background state.	Health-critical events retain the validated priority behavior.
5	Flutter / Django	Reports delivery or interaction outcome when the story requires reporting.	Acknowledgement and delivery transitions remain idempotent and traceable.

US-15-11 — Notification Sounds and Channels

Epic	EP-15 — Notification Hub and In-App Alerts	**Primary Actor**	Android User / VitaMate System
Priority	High	**Implementation Status**	Implemented
User Story	As a VitaMate user, I want to use notification sounds and channels appropriate to the event category, so that I can receive relevant notifications without unnecessary duplication or interruption.		
Traceability	Source Story ID: US-15-11 | Source Epic: EP-15 | Appendix Story 179 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Notification Sounds and Channels' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Notification-related changes shall remain synchronized with current preferences, schedule, priority, and device state.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Android User / VitaMate System	Configures or triggers the notification behavior required to use notification sounds and channels appropriate to the event category.	Current device, locale, permission, preference, and scheduling context is available.
2	Django Notification Hub	Compiles or updates the authoritative notification plan using category, priority, quiet-hours, and source-domain rules.	The plan reflects the latest validated backend state.
3	Flutter Client	Synchronizes only the required plan changes to the primary device.	Unchanged entries are preserved and stale entries are removed when appropriate.
4	Android Local Delivery	Schedules, presents, or routes the notification according to local capabilities and foreground/background state.	Health-critical events retain the validated priority behavior.
5	Flutter / Django	Reports delivery or interaction outcome when the story requires reporting.	Acknowledgement and delivery transitions remain idempotent and traceable.

US-15-12 — Permission Management

Epic	EP-15 — Notification Hub and In-App Alerts	**Primary Actor**	Android User / VitaMate System
Priority	High	**Implementation Status**	Implemented
User Story	As a VitaMate user, I want to understand and repair missing notification or alarm permissions, so that I can receive relevant notifications without unnecessary duplication or interruption.		
Traceability	Source Story ID: US-15-12 | Source Epic: EP-15 | Appendix Story 180 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Permission Management' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Any successful write shall persist the intended record or configuration using the appropriate domain model.
AC-3	Notification-related changes shall remain synchronized with current preferences, schedule, priority, and device state.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Android User / VitaMate System	Configures or triggers the notification behavior required to understand and repair missing notification or alarm permissions.	Current device, locale, permission, preference, and scheduling context is available.
2	Django Notification Hub	Compiles or updates the authoritative notification plan using category, priority, quiet-hours, and source-domain rules.	The plan reflects the latest validated backend state.
3	Flutter Client	Synchronizes only the required plan changes to the primary device.	Unchanged entries are preserved and stale entries are removed when appropriate.
4	Android Local Delivery	Schedules, presents, or routes the notification according to local capabilities and foreground/background state.	Health-critical events retain the validated priority behavior.
5	Flutter / Django	Reports delivery or interaction outcome when the story requires reporting.	Acknowledgement and delivery transitions remain idempotent and traceable.

US-15-13 — Differential Synchronization

Epic	EP-15 — Notification Hub and In-App Alerts	**Primary Actor**	Android User / VitaMate System
Priority	High	**Implementation Status**	Implemented
User Story	As a VitaMate user, I want to synchronize only notification-plan changes that are actually stale, so that I can receive relevant notifications without unnecessary duplication or interruption.		
Traceability	Source Story ID: US-15-13 | Source Epic: EP-15 | Appendix Story 181 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Differential Synchronization' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Duplicate records, rewards, plans, and side effects shall be prevented using idempotent or uniqueness-aware behavior.
AC-3	Notification-related changes shall remain synchronized with current preferences, schedule, priority, and device state.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Android User / VitaMate System	Configures or triggers the notification behavior required to synchronize only notification-plan changes that are actually stale.	Current device, locale, permission, preference, and scheduling context is available.
2	Django Notification Hub	Compiles or updates the authoritative notification plan using category, priority, quiet-hours, and source-domain rules.	The plan reflects the latest validated backend state.
3	Flutter Client	Synchronizes only the required plan changes to the primary device.	Unchanged entries are preserved and stale entries are removed when appropriate.
4	Android Local Delivery	Schedules, presents, or routes the notification according to local capabilities and foreground/background state.	Health-critical events retain the validated priority behavior.
5	Flutter / Django	Reports delivery or interaction outcome when the story requires reporting.	Acknowledgement and delivery transitions remain idempotent and traceable.

US-15-14 — Delivery Reports

Epic	EP-15 — Notification Hub and In-App Alerts	**Primary Actor**	Android User / VitaMate System
Priority	High	**Implementation Status**	Implemented
User Story	As a VitaMate user, I want to report notification delivery and interaction outcomes, so that I can receive relevant notifications without unnecessary duplication or interruption.		
Traceability	Source Story ID: US-15-14 | Source Epic: EP-15 | Appendix Story 182 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Delivery Reports' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-3	Duplicate records, rewards, plans, and side effects shall be prevented using idempotent or uniqueness-aware behavior.
AC-4	Notification-related changes shall remain synchronized with current preferences, schedule, priority, and device state.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Android User / VitaMate System	Configures or triggers the notification behavior required to report notification delivery and interaction outcomes.	Current device, locale, permission, preference, and scheduling context is available.
2	Django Notification Hub	Compiles or updates the authoritative notification plan using category, priority, quiet-hours, and source-domain rules.	The plan reflects the latest validated backend state.
3	Flutter Client	Synchronizes only the required plan changes to the primary device.	Unchanged entries are preserved and stale entries are removed when appropriate.
4	Android Local Delivery	Schedules, presents, or routes the notification according to local capabilities and foreground/background state.	Health-critical events retain the validated priority behavior.
5	Flutter / Django	Reports delivery or interaction outcome when the story requires reporting.	Acknowledgement and delivery transitions remain idempotent and traceable.

B.19 EP-16 — Language, Accessibility, Reliability, and Cross-Cutting Security

Cross-cutting requirements for bilingual operation, accessibility, responsive layouts, error recovery, data isolation, idempotency, and medical safety.

Epic Attribute	Academic Description
Scope	Cross-cutting requirements for bilingual operation, accessibility, responsive layouts, error recovery, data isolation, idempotency, and medical safety.
Story Count	12
Baseline Rule 1	Static interface text is localized, while supported dynamic backend content uses stable codes or localization mapping.
Baseline Rule 2	User-entered names, medication names, food names, and standard medical abbreviations remain unchanged when appropriate.
Baseline Rule 3	Protected health endpoints enforce ownership and isolate one user's records from another.

US-16-01 — Live Language Switching

Epic	EP-16 — Language, Accessibility, Reliability, and Cross-Cutting Security	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to use live language switching, so that I can use the application reliably, accessibly, securely, and in my selected language.		
Traceability	Source Story ID: US-16-01 | Source Epic: EP-16 | Appendix Story 183 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Live Language Switching' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-3	The visible result shall remain consistent with the selected locale, supported directionality, and accessibility semantics.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Uses the application in a context that requires live language switching.	The relevant locale, layout, network, session, ownership, or safety context is detected.
2	VitaMate System	Applies the cross-cutting rule needed to use live language switching.	The requested behavior is enforced consistently across affected screens or endpoints.
3	VitaMate System	Validates session, ownership, retry, device, or presentation conditions relevant to the source story.	Unsafe, unauthorized, duplicated, or misleading behavior is prevented.
4	Registered User	Continues the workflow or follows the presented recovery path.	The interface remains understandable and does not hide the actual failure state.
5	VitaMate System	Preserves consistent state across subsequent requests and views.	Localization, accessibility, reliability, security, and medical-safety constraints remain intact.

US-16-02 — Full RTL/LTR Support

Epic	EP-16 — Language, Accessibility, Reliability, and Cross-Cutting Security	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to use interfaces that correctly follow right-to-left or left-to-right direction, so that I can use the application reliably, accessibly, securely, and in my selected language.		
Traceability	Source Story ID: US-16-02 | Source Epic: EP-16 | Appendix Story 184 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Full RTL/LTR Support' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Dependent summaries, targets, notification plans, or read models shall be refreshed after a successful state change.
AC-3	The interface shall preserve readable layout and shall not obscure or overflow required content.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Uses the application in a context that requires full rtl/ltr support.	The relevant locale, layout, network, session, ownership, or safety context is detected.
2	VitaMate System	Applies the cross-cutting rule needed to use interfaces that correctly follow right-to-left or left-to-right direction.	The requested behavior is enforced consistently across affected screens or endpoints.
3	VitaMate System	Validates session, ownership, retry, device, or presentation conditions relevant to the source story.	Unsafe, unauthorized, duplicated, or misleading behavior is prevented.
4	Registered User	Continues the workflow or follows the presented recovery path.	The interface remains understandable and does not hide the actual failure state.
5	VitaMate System	Preserves consistent state across subsequent requests and views.	Localization, accessibility, reliability, security, and medical-safety constraints remain intact.

US-16-03 — Dynamic Content Localization

Epic	EP-16 — Language, Accessibility, Reliability, and Cross-Cutting Security	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view dynamic backend content in the selected language, so that I can use the application reliably, accessibly, securely, and in my selected language.		
Traceability	Source Story ID: US-16-03 | Source Epic: EP-16 | Appendix Story 185 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Dynamic Content Localization' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The visible result shall remain consistent with the selected locale, supported directionality, and accessibility semantics.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Uses the application in a context that requires dynamic content localization.	The relevant locale, layout, network, session, ownership, or safety context is detected.
2	VitaMate System	Applies the cross-cutting rule needed to view dynamic backend content in the selected language.	The requested behavior is enforced consistently across affected screens or endpoints.
3	VitaMate System	Validates session, ownership, retry, device, or presentation conditions relevant to the source story.	Unsafe, unauthorized, duplicated, or misleading behavior is prevented.
4	Registered User	Continues the workflow or follows the presented recovery path.	The interface remains understandable and does not hide the actual failure state.
5	VitaMate System	Preserves consistent state across subsequent requests and views.	Localization, accessibility, reliability, security, and medical-safety constraints remain intact.

US-16-04 — Localized Dates, Numbers, and Units

Epic	EP-16 — Language, Accessibility, Reliability, and Cross-Cutting Security	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to view dates, numbers, and units in an understandable localized format, so that I can use the application reliably, accessibly, securely, and in my selected language.		
Traceability	Source Story ID: US-16-04 | Source Epic: EP-16 | Appendix Story 186 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Localized Dates, Numbers, and Units' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The visible result shall remain consistent with the selected locale, supported directionality, and accessibility semantics.
AC-3	Health-related output shall follow the validated safety and priority rules and shall not make unsupported diagnostic claims.
AC-4	Compared and displayed values shall use the applicable units consistently.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Uses the application in a context that requires localized dates, numbers, and units.	The relevant locale, layout, network, session, ownership, or safety context is detected.
2	VitaMate System	Applies the cross-cutting rule needed to view dates, numbers, and units in an understandable localized format.	The requested behavior is enforced consistently across affected screens or endpoints.
3	VitaMate System	Validates session, ownership, retry, device, or presentation conditions relevant to the source story.	Unsafe, unauthorized, duplicated, or misleading behavior is prevented.
4	Registered User	Continues the workflow or follows the presented recovery path.	The interface remains understandable and does not hide the actual failure state.
5	VitaMate System	Preserves consistent state across subsequent requests and views.	Localization, accessibility, reliability, security, and medical-safety constraints remain intact.

US-16-05 — Accessibility Semantics

Epic	EP-16 — Language, Accessibility, Reliability, and Cross-Cutting Security	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to use accessible semantic labels for key controls and status information, so that I can use the application reliably, accessibly, securely, and in my selected language.		
Traceability	Source Story ID: US-16-05 | Source Epic: EP-16 | Appendix Story 187 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Accessibility Semantics' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The visible result shall remain consistent with the selected locale, supported directionality, and accessibility semantics.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Uses the application in a context that requires accessibility semantics.	The relevant locale, layout, network, session, ownership, or safety context is detected.
2	VitaMate System	Applies the cross-cutting rule needed to use accessible semantic labels for key controls and status information.	The requested behavior is enforced consistently across affected screens or endpoints.
3	VitaMate System	Validates session, ownership, retry, device, or presentation conditions relevant to the source story.	Unsafe, unauthorized, duplicated, or misleading behavior is prevented.
4	Registered User	Continues the workflow or follows the presented recovery path.	The interface remains understandable and does not hide the actual failure state.
5	VitaMate System	Preserves consistent state across subsequent requests and views.	Localization, accessibility, reliability, security, and medical-safety constraints remain intact.

US-16-06 — Responsive Screen Sizes

Epic	EP-16 — Language, Accessibility, Reliability, and Cross-Cutting Security	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to use the interface without layout overflow across supported phone sizes, so that I can use the application reliably, accessibly, securely, and in my selected language.		
Traceability	Source Story ID: US-16-06 | Source Epic: EP-16 | Appendix Story 188 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Responsive Screen Sizes' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-3	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Uses the application in a context that requires responsive screen sizes.	The relevant locale, layout, network, session, ownership, or safety context is detected.
2	VitaMate System	Applies the cross-cutting rule needed to use the interface without layout overflow across supported phone sizes.	The requested behavior is enforced consistently across affected screens or endpoints.
3	VitaMate System	Validates session, ownership, retry, device, or presentation conditions relevant to the source story.	Unsafe, unauthorized, duplicated, or misleading behavior is prevented.
4	Registered User	Continues the workflow or follows the presented recovery path.	The interface remains understandable and does not hide the actual failure state.
5	VitaMate System	Preserves consistent state across subsequent requests and views.	Localization, accessibility, reliability, security, and medical-safety constraints remain intact.

US-16-07 — Understandable Network Errors

Epic	EP-16 — Language, Accessibility, Reliability, and Cross-Cutting Security	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to use understandable network errors, so that I can use the application reliably, accessibly, securely, and in my selected language.		
Traceability	Source Story ID: US-16-07 | Source Epic: EP-16 | Appendix Story 189 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Understandable Network Errors' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Failure states shall expose a recoverable retry or error path without reporting fabricated success.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Uses the application in a context that requires understandable network errors.	The relevant locale, layout, network, session, ownership, or safety context is detected.
2	VitaMate System	Applies the cross-cutting rule needed to use understandable network errors.	The requested behavior is enforced consistently across affected screens or endpoints.
3	VitaMate System	Validates session, ownership, retry, device, or presentation conditions relevant to the source story.	Unsafe, unauthorized, duplicated, or misleading behavior is prevented.
4	Registered User	Continues the workflow or follows the presented recovery path.	The interface remains understandable and does not hide the actual failure state.
5	VitaMate System	Preserves consistent state across subsequent requests and views.	Localization, accessibility, reliability, security, and medical-safety constraints remain intact.

US-16-08 — Local Android Connectivity

Epic	EP-16 — Language, Accessibility, Reliability, and Cross-Cutting Security	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to use local android connectivity, so that I can use the application reliably, accessibly, securely, and in my selected language.		
Traceability	Source Story ID: US-16-08 | Source Epic: EP-16 | Appendix Story 190 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Local Android Connectivity' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-3	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Uses the application in a context that requires local android connectivity.	The relevant locale, layout, network, session, ownership, or safety context is detected.
2	VitaMate System	Applies the cross-cutting rule needed to use local android connectivity.	The requested behavior is enforced consistently across affected screens or endpoints.
3	VitaMate System	Validates session, ownership, retry, device, or presentation conditions relevant to the source story.	Unsafe, unauthorized, duplicated, or misleading behavior is prevented.
4	Registered User	Continues the workflow or follows the presented recovery path.	The interface remains understandable and does not hide the actual failure state.
5	VitaMate System	Preserves consistent state across subsequent requests and views.	Localization, accessibility, reliability, security, and medical-safety constraints remain intact.

US-16-09 — HTTP 401 Handling

Epic	EP-16 — Language, Accessibility, Reliability, and Cross-Cutting Security	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to recover gracefully from expired or invalid authentication, so that I can use the application reliably, accessibly, securely, and in my selected language.		
Traceability	Source Story ID: US-16-09 | Source Epic: EP-16 | Appendix Story 191 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'HTTP 401 Handling' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Duplicate records, rewards, plans, and side effects shall be prevented using idempotent or uniqueness-aware behavior.
AC-3	Authentication and session recovery shall follow the defined token/refresh behavior without exposing protected data.
AC-4	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Uses the application in a context that requires http 401 handling.	The relevant locale, layout, network, session, ownership, or safety context is detected.
2	VitaMate System	Applies the cross-cutting rule needed to recover gracefully from expired or invalid authentication.	The requested behavior is enforced consistently across affected screens or endpoints.
3	VitaMate System	Validates session, ownership, retry, device, or presentation conditions relevant to the source story.	Unsafe, unauthorized, duplicated, or misleading behavior is prevented.
4	Registered User	Continues the workflow or follows the presented recovery path.	The interface remains understandable and does not hide the actual failure state.
5	VitaMate System	Preserves consistent state across subsequent requests and views.	Localization, accessibility, reliability, security, and medical-safety constraints remain intact.

US-16-10 — User Data Isolation

Epic	EP-16 — Language, Accessibility, Reliability, and Cross-Cutting Security	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to keep personal health data isolated from other users, so that I can use the application reliably, accessibly, securely, and in my selected language.		
Traceability	Source Story ID: US-16-10 | Source Epic: EP-16 | Appendix Story 192 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'User Data Isolation' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Owner-scoped authorization shall prevent cross-user access to protected records.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Uses the application in a context that requires user data isolation.	The relevant locale, layout, network, session, ownership, or safety context is detected.
2	VitaMate System	Applies the cross-cutting rule needed to keep personal health data isolated from other users.	The requested behavior is enforced consistently across affected screens or endpoints.
3	VitaMate System	Validates session, ownership, retry, device, or presentation conditions relevant to the source story.	Unsafe, unauthorized, duplicated, or misleading behavior is prevented.
4	Registered User	Continues the workflow or follows the presented recovery path.	The interface remains understandable and does not hide the actual failure state.
5	VitaMate System	Preserves consistent state across subsequent requests and views.	Localization, accessibility, reliability, security, and medical-safety constraints remain intact.

US-16-11 — Idempotent Retry Operations

Epic	EP-16 — Language, Accessibility, Reliability, and Cross-Cutting Security	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to retry operations without creating duplicate side effects, so that I can use the application reliably, accessibly, securely, and in my selected language.		
Traceability	Source Story ID: US-16-11 | Source Epic: EP-16 | Appendix Story 193 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Idempotent Retry Operations' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Duplicate records, rewards, plans, and side effects shall be prevented using idempotent or uniqueness-aware behavior.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Uses the application in a context that requires idempotent retry operations.	The relevant locale, layout, network, session, ownership, or safety context is detected.
2	VitaMate System	Applies the cross-cutting rule needed to retry operations without creating duplicate side effects.	The requested behavior is enforced consistently across affected screens or endpoints.
3	VitaMate System	Validates session, ownership, retry, device, or presentation conditions relevant to the source story.	Unsafe, unauthorized, duplicated, or misleading behavior is prevented.
4	Registered User	Continues the workflow or follows the presented recovery path.	The interface remains understandable and does not hide the actual failure state.
5	VitaMate System	Preserves consistent state across subsequent requests and views.	Localization, accessibility, reliability, security, and medical-safety constraints remain intact.

US-16-12 — Medical Safety

Epic	EP-16 — Language, Accessibility, Reliability, and Cross-Cutting Security	**Primary Actor**	Registered User
Priority	High	**Implementation Status**	Implemented
User Story	As a registered user, I want to receive non-diagnostic health guidance and appropriate safety messaging, so that I can use the application reliably, accessibly, securely, and in my selected language.		
Traceability	Source Story ID: US-16-12 | Source Epic: EP-16 | Appendix Story 194 of 194		

Acceptance Criteria

AC ID	Testable Acceptance Criterion
AC-1	The 'Medical Safety' capability shall behave consistently with the validated source user story and its implementation status.
AC-2	Health-related output shall follow the validated safety and priority rules and shall not make unsupported diagnostic claims.
AC-3	The interface shall present a clear success, empty, validation, or failure state appropriate to the operation.
AC-4	The resulting application state shall remain consistent with related VitaMate modules and subsequent views.

Operational Flow

Step	Actor / Component	Action / System Behavior	Expected Result
1	Registered User	Uses the application in a context that requires medical safety.	The relevant locale, layout, network, session, ownership, or safety context is detected.
2	VitaMate System	Applies the cross-cutting rule needed to receive non-diagnostic health guidance and appropriate safety messaging.	The requested behavior is enforced consistently across affected screens or endpoints.
3	VitaMate System	Validates session, ownership, retry, device, or presentation conditions relevant to the source story.	Unsafe, unauthorized, duplicated, or misleading behavior is prevented.
4	Registered User	Continues the workflow or follows the presented recovery path.	The interface remains understandable and does not hide the actual failure state.
5	VitaMate System	Preserves consistent state across subsequent requests and views.	Localization, accessibility, reliability, security, and medical-safety constraints remain intact.

B.20 Appendix Use and Traceability Note

This appendix should be used together with the Requirements Traceability Matrix and the testing appendix. The story identifiers remain unchanged so that each requirement can be traced to the corresponding epic, implementation evidence, and test case. If an API contract, business rule, or implementation status changes, the affected user story, RTM entry, and acceptance evidence should be updated together.

The appendix intentionally distinguishes implemented behavior from device-dependent, partially implemented, and backlog items. This distinction preserves academic accuracy and prevents incomplete functionality from being presented as a verified final capability.